Group A: Early progression

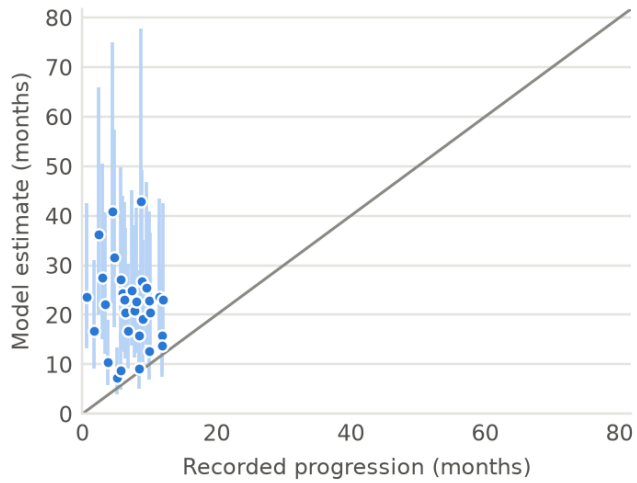
PrimeKG Patient Reasoning · Precision Oncology Research

Group A · 32 patients

Report date 31 Oct 2026

Group A · Early progression recorded progression-free survival: ≤ 12 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
32	**77**	**7.1 mo**	**12%**
3.0% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 0.7–12.0 mo	estimate in this window · median |error| 14 mo



Most frequent path end-points in this group	Patients
breast neoplasm	3
breast carcinoma	3
Formation of the cornified envelope	2
breast cancer	2
Stimuli-sensing channels	2
mammary neoplasms, animal	2
PIP3 activates AKT signaling	2
hereditary breast carcinoma	2

Logic formulas firing most often in this group	Mean weight	Patients
NStage(p, N0) ⇒ Progress(p)	-0.183	11
Stage(p, II) ⇒ Progress(p)	-0.097	7
TStage(p, T4) ⇒ Progress(p)	+0.120	4
Histology(p, other) ⇒ Progress(p)	+0.063	4
Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p)	-0.025	3
Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p)	-0.141	3

Most common driver alterations

TP53 mutated (41%), PIK3CA mutated (31%), ERBB2 amplified (22%), PTEN deleted (16%), MYC amplified (16%), CDH1 mutated (12%), PTEN mutated (12%), GATA3 mutated (12%), CCND1 amplified (12%), FGFR1 amplified (9%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-D8-A73W	79	III	LumA	0.7 mo	24 mo	✗	76	A_progression_within_12m/TCGA-D8-A73W.pdf
TCGA-E2-A152	56	I	Her2	1.8 mo	17 mo	✗	42	A_progression_within_12m/TCGA-E2-A152.pdf
TCGA-E2-A10A	41	II	LumA	2.5 mo	36 mo	✗	32	A_progression_within_12m/TCGA-E2-A10A.pdf
TCGA-E9-A1NF	60	II	LumB	3.0 mo	27 mo	✗	40	A_progression_within_12m/TCGA-E9-A1NF.pdf
TCGA-A2-A04P	36	III	Basal	3.4 mo	22 mo	✗	85	A_progression_within_12m/TCGA-A2-A04P.pdf
TCGA-UU-A93S	63	IV	Her2	3.8 mo	10 mo	✓	99	A_progression_within_12m/TCGA-UU-A93S.pdf
TCGA-C8-A137	34	II	Her2	4.5 mo	41 mo	✗	64	A_progression_within_12m/TCGA-C8-A137.pdf
TCGA-AO-A0JA	36	III	LumA	4.7 mo	32 mo	✗	69	A_progression_within_12m/TCGA-AO-A0JA.pdf
TCGA-E2-A1LK	84	III	Basal	5.1 mo	7 mo	✓	91	A_progression_within_12m/TCGA-E2-A1LK.pdf
TCGA-A7-A3RF	79	II	LumA	5.7 mo	27 mo	✗	9	A_progression_within_12m/TCGA-A7-A3RF.pdf
TCGA-AC-A23H	90	II	Her2	5.7 mo	9 mo	✓	82	A_progression_within_12m/TCGA-AC-A23H.pdf
TCGA-Z7-A8R5	61	III	LumA	6.0 mo	24 mo	✗	67	A_progression_within_12m/TCGA-Z7-A8R5.pdf
TCGA-E2-A14Z	64	I	LumA	6.2 mo	23 mo	✗	4	A_progression_within_12m/TCGA-E2-A14Z.pdf
TCGA-LL-A73Z	55	IV	Normal	6.3 mo	23 mo	✗	99	A_progression_within_12m/TCGA-LL-A73Z.pdf
TCGA-E2-A10B	67	II	LumA	6.4 mo	20 mo	✗	17	A_progression_within_12m/TCGA-E2-A10B.pdf
TCGA-EW-A1P8	58	III	Basal	6.8 mo	17 mo	✗	84	A_progression_within_12m/TCGA-EW-A1P8.pdf
TCGA-BH-A18T	70	II	Basal	7.4 mo	25 mo	✗	58	A_progression_within_12m/TCGA-BH-A18T.pdf
TCGA-AC-A2QJ	48	III	missing	7.8 mo	21 mo	✗	95	A_progression_within_12m/TCGA-AC-A2QJ.pdf
TCGA-A8-A091	61	II	LumA	8.0 mo	23 mo	✗	6	A_progression_within_12m/TCGA-A8-A091.pdf
TCGA-A2-A0T2	66	IV	Basal	8.4 mo	9 mo	✓	100	A_progression_within_12m/TCGA-A2-A0T2.pdf
TCGA-LL-A5YM	88	III	LumB	8.5 mo	16 mo	✗	96	A_progression_within_12m/TCGA-LL-A5YM.pdf
TCGA-AO-A126	39	II	LumA	8.8 mo	43 mo	✗	27	A_progression_within_12m/TCGA-AO-A126.pdf
TCGA-AR-A1AR	50	III	Basal	8.8 mo	27 mo	✗	92	A_progression_within_12m/TCGA-AR-A1AR.pdf
TCGA-OL-A97C	67	II	Normal	9.0 mo	19 mo	✗	40	A_progression_within_12m/TCGA-OL-A97C.pdf
TCGA-A7-A26H	72	II	LumA	9.6 mo	25 mo	✗	22	A_progression_within_12m/TCGA-A7-A26H.pdf
TCGA-AR-A5QQ	68	III	Basal	9.9 mo	23 mo	✗	90	A_progression_within_12m/TCGA-AR-A5QQ.pdf
TCGA-D8-A1Y1	80	III	LumB	9.9 mo	13 mo	✗	78	A_progression_within_12m/TCGA-D8-A1Y1.pdf
TCGA-E9-A243	52	II	Basal	10.0 mo	20 mo	✗	58	A_progression_within_12m/TCGA-E9-A243.pdf
TCGA-AC-A62V	58	IV	missing	11.4 mo	24 mo	✗	98	A_progression_within_12m/TCGA-AC-A62V.pdf
TCGA-E2-A1LE	71	III	missing	11.9 mo	16 mo	✗	95	A_progression_within_12m/TCGA-E2-A1LE.pdf
TCGA-B6-A0I9	62	IV	Her2	11.9 mo	14 mo	✗	100	A_progression_within_12m/TCGA-B6-A0I9.pdf
TCGA-BH-A1EV	45	III	Her2	12.0 mo	23 mo	✗	98	A_progression_within_12m/TCGA-BH-A1EV.pdf

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

Sample: **TCGA-D8-A73W-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.3** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

76th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 26%

Model-estimated progression time

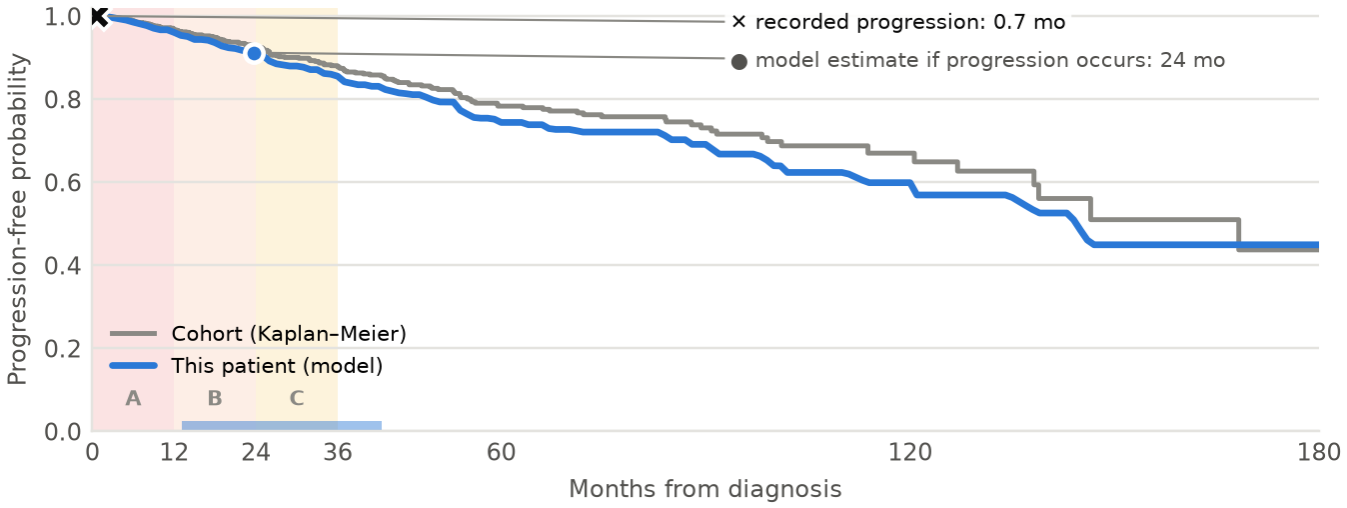
≈ 24 months

Window 12–24 months · IQR 13–42 mo
if progression occurs

Recorded progression

0.7 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 23 mo



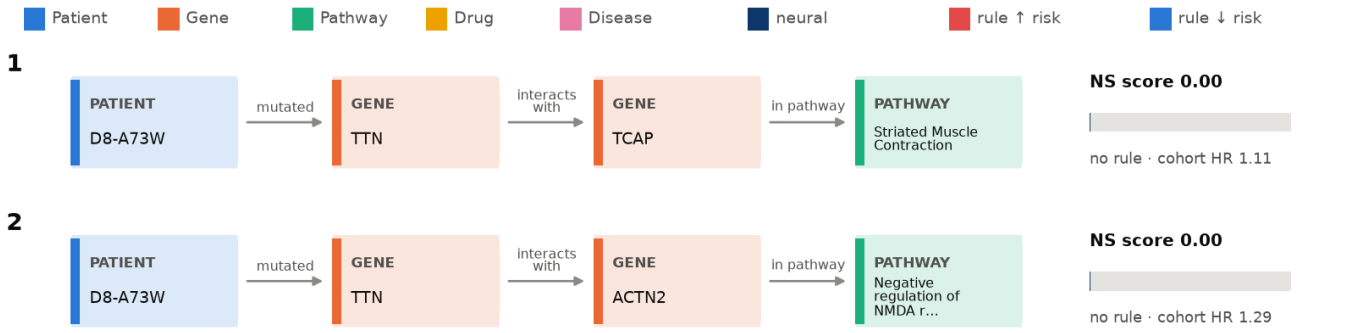
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **76th risk percentile** (high tier). The strongest learned formula that holds for them (*Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.010 per SD). The highest-scoring knowledge-graph path runs from **TTN** to **Striated Muscle Contraction**. Progression was recorded at 0.7 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

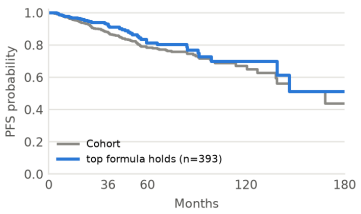


None of the 25 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-D8-A73W** mutated → **TTN** interacts with → **TCAP** in pathway → **Striated Muscle Contraction**	0	No learned formula fires on this path; its score is neural only. cohort: HR 1.11, n = 334 carriers	**0.00**
2	**TCGA-D8-A73W** mutated → **TTN** interacts with → **ACTN2** in pathway → **Negative regulation of NMDA receptor-mediated neuronal transmission**	0	No learned formula fires on this path; its score is neural only. cohort: HR 1.29, n = 108 carriers	**0.00**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

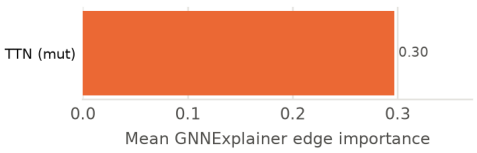
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TTN	Mutation	p.E18412A	2	0.30



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3KD	47	III	LumA	0.70	no progression, followed 39.6 mo
TCGA-AC-A3W6	90	III	missing	0.68	no progression, followed 19.8 mo
TCGA-AC-A2QI	76	III	missing	0.67	no progression, followed 19.3 mo
TCGA-BH-A1FR	73	III	LumA	0.66	progressed at 54.0 mo (group D)
TCGA-LQ-A4E4	73	III	missing	0.65	progressed at 22.4 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time

estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A152 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**56**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

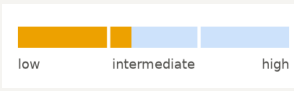
Sample: **TCGA-E2-A152-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **4.13** · MSIsensor score: **0.43** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

42nd

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

≈ 17 months

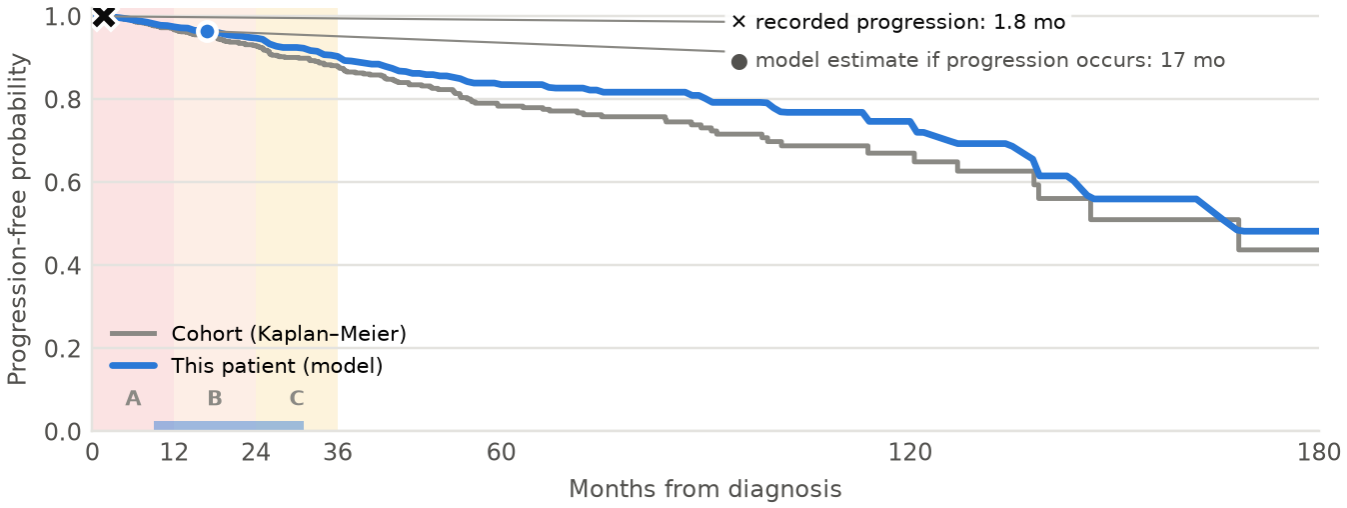
Window 12–24 months · IQR 9–31 mo if progression occurs

Recorded progression

1.8 months

Group A · Early progression

✗ recorded window differs · |error| 15 mo



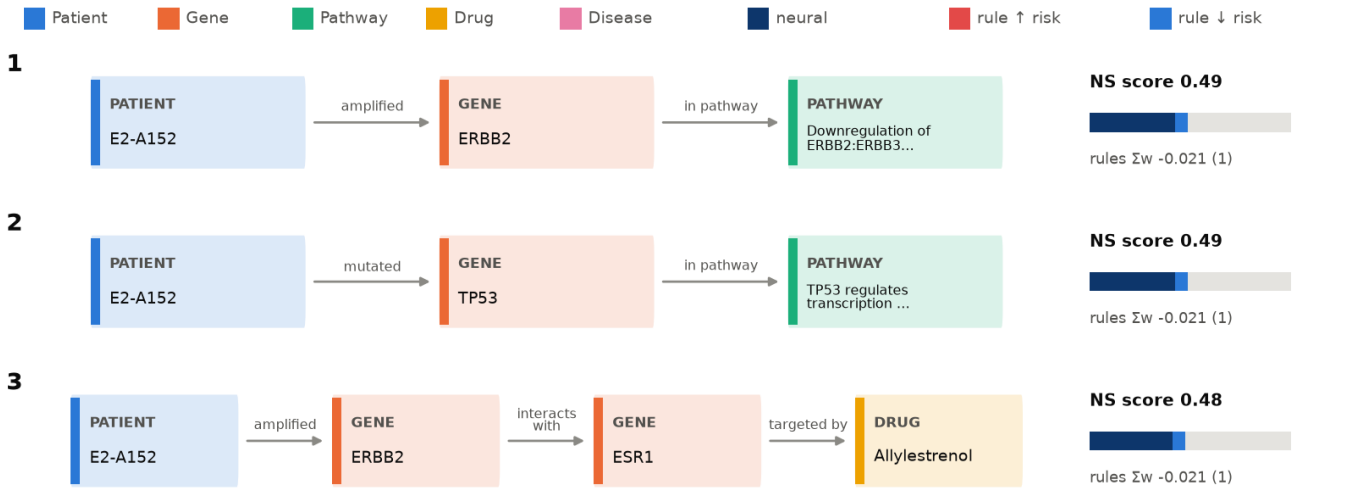
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 56-year-old patient has a Her2 tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **42nd risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.199 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **Downregulation of ERBB2:ERBB3 signaling**; the learned formulas it instantiates point to slower progression (Σw -0.021). Progression was recorded at 1.8 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

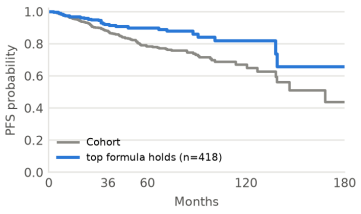


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E2-A152** amplified → **ERBB2** in pathway → **Downregulation of ERBB2:ERBB3 signaling**	71	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 0.93, n = 217 carriers	**0.49**
2	**TCGA-E2-A152** mutated → **TP53** in pathway → **TP53 regulates transcription of additional cell cycle genes whose exact role in the p53 pathway remain uncertain**	71	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 1.05, n = 284 carriers	**0.49**
3	**TCGA-E2-A152** amplified → **ERBB2** interacts with → **ESR1** targeted by → **Allylestrenol**	69	↓ **slower progression** · Σw -0.021 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) cohort: HR 2.93, n = 28 carriers	**0.48**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

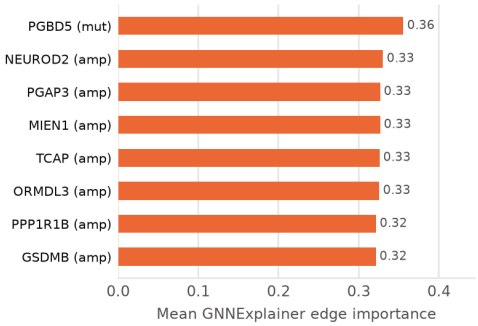
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.199 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	413 pts
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts
-0.021 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Signaling by ERBB2') ∧ Altered(p, g) ∧ InPathway(g, 'DNA Damage/Telomere Stress Induced Senescence') ∧ Altered(p, g) ∧ InPathway(g, 'Ub-specific processing proteases') ⇒ Progress(p) [KG]	RL-proposed	73 pts
-0.021 ↓ risk	NStage(p, N0) ∧ TStage(p, T1) ⇒ Progress(p)	mined pair	139 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Stage(p, I) ⇒ Progress(p)	mined pair	138 pts
-0.000 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Pregnenolone biosynthesis') ⇒ Progress(p) [KG]	mined pair	69 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.K132E	57	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
PIK3CA · driver	Mutation	p.X764_splice	70	0.32
PGBD5	Mutation	p.*42*	0	0.36
NEUROD2	High-level amplification	GISTIC ±2	0	0.33
PGAP3	High-level amplification	GISTIC ±2	0	0.33
MIEN1	High-level amplification	GISTIC ±2	0	0.33
TCAP	High-level amplification	GISTIC ±2	1	0.33
ORMDL3	High-level amplification	GISTIC ±2	2	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 49 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A1AX	64	I	LumA	0.84	no progression, followed 86.4 mo
TCGA-B6-A0RH	51	II	Her2	0.82	progressed at 71.2 mo (group D)
TCGA-D8-A1J9	48	I	LumB	0.82	no progression, followed 17.5 mo
TCGA-A2-A3XZ	46	I	Her2	0.81	no progression, followed 50.4 mo
TCGA-E2-A1IN	60	I	LumA	0.80	no progression, followed 22.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A10A · Group A

Report date 31 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**41**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**—**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

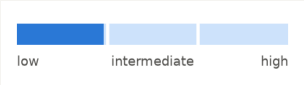
Sample: **TCGA-E2-A10A-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.73** · MSIsensor score: **0.26** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

32nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

≈ 36 months

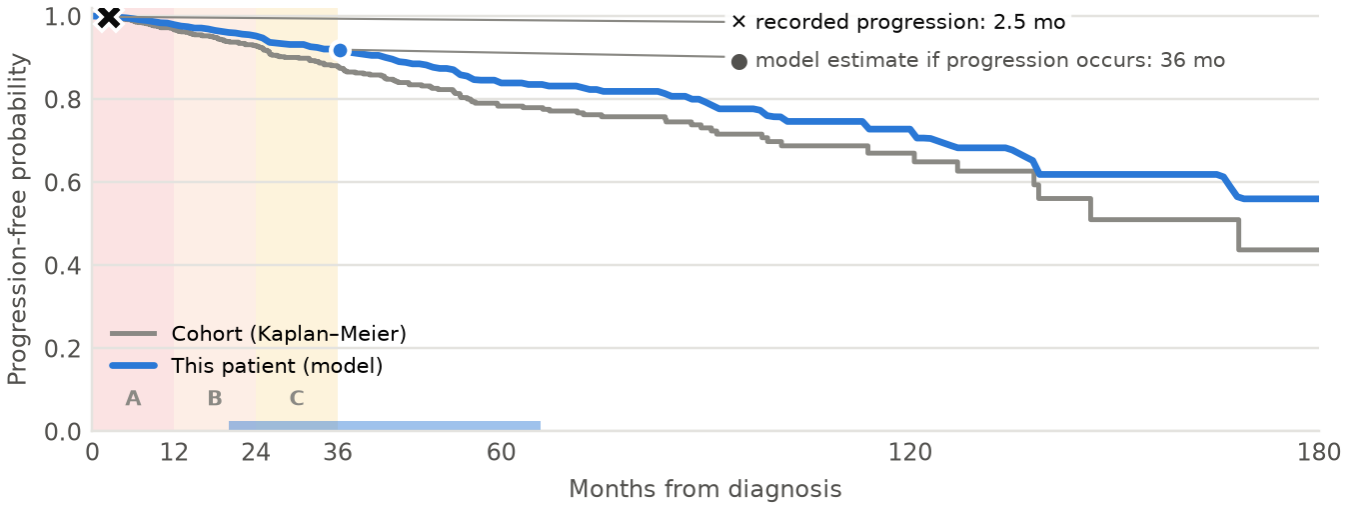
Window > 36 months · IQR 20–66 mo if progression occurs

Recorded progression

2.5 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 34 mo



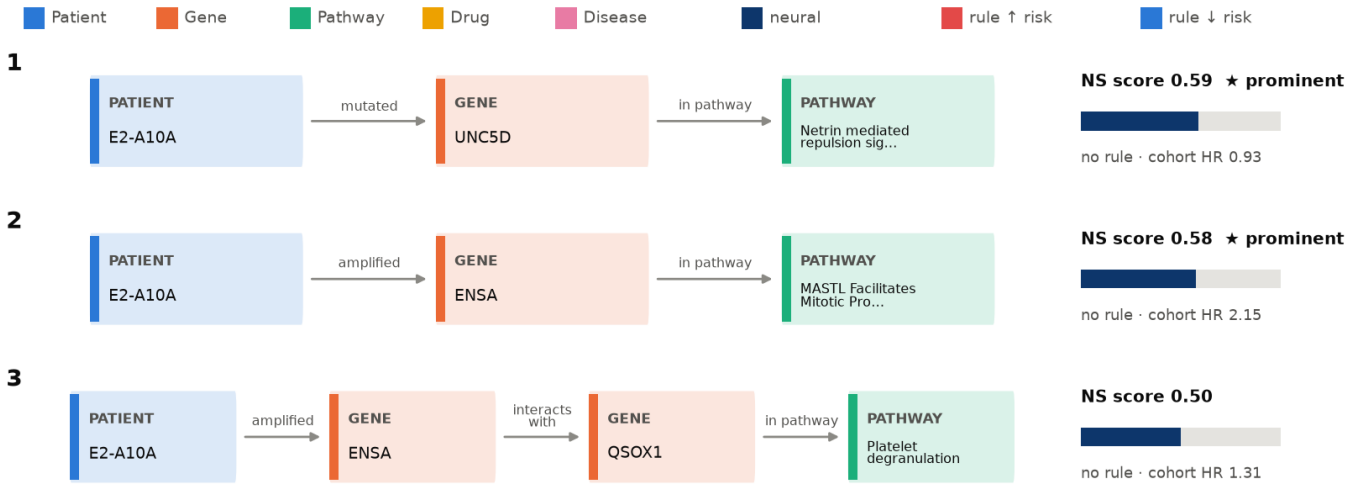
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 41-year-old patient has a LumA tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **32nd risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **UNC5D** to **Netrin mediated repulsion signals**. Progression was recorded at 2.5 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

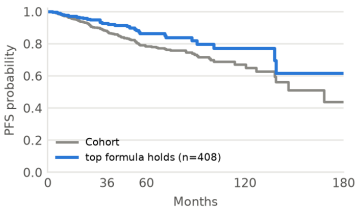


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A10A** mutated → **UNC5D** in pathway → **Netrin mediated repulsion signals**	98	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 91 carriers	**0.59**
2 ★	**TCGA-E2-A10A** amplified → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	96	No learned formula fires on this path; its score is neural only. cohort: HR 2.15, n = 91 carriers	**0.58**
3	**TCGA-E2-A10A** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Platelet degranulation**	84	No learned formula fires on this path; its score is neural only. cohort: HR 1.31, n = 312 carriers	**0.50**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

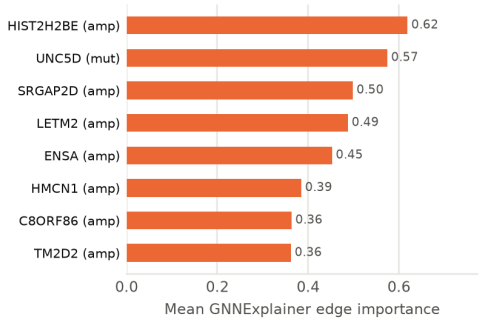
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
+0.058 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Defective B3GALTL causes Peters-plus syndrome (PpS)') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	RL-proposed	44 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
+0.007 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Estrogen-dependent gene expression') ∧ Altered(p, g) ∧ TargetedBy(g, 'Zinc') ⇒ Progress(p) [KG]	RL-proposed	65 pts
-0.004 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	252 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P425Afs*82	13	0.32
ESR1 · driver	Mutation	p.P29Sfs*79	141	0.31
MAP3K1 · driver	Mutation	p.G172Rfs*7	13	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.62
UNC5D	Mutation	p.C354F	1	0.57
SRGAP2D	High-level amplification	GISTIC ±2	0	0.50
LETM2	High-level amplification	GISTIC ±2	0	0.49
ENSA	High-level amplification	GISTIC ±2	1	0.45
HMCN1	High-level amplification	GISTIC ±2	0	0.39
C8ORF86	High-level amplification	GISTIC ±2	0	0.36
TM2D2	High-level amplification	GISTIC ±2	0	0.36



49 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 137 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SD	59	II	LumA	0.71	no progression, followed 14.4 mo
TCGA-OL-A5D8	40	II	LumA	0.71	no progression, followed 32.0 mo
TCGA-B6-A0IM	75	II	LumA	0.69	no progression, followed 127.3 mo
TCGA-A8-A090	74	II	LumA	0.68	no progression, followed 0.0 mo
TCGA-AN-A0AM	56	II	LumB	0.66	no progression, followed 0.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E9-A1NF · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

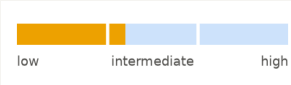
Sample: **TCGA-E9-A1NF-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **0.77** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

40th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

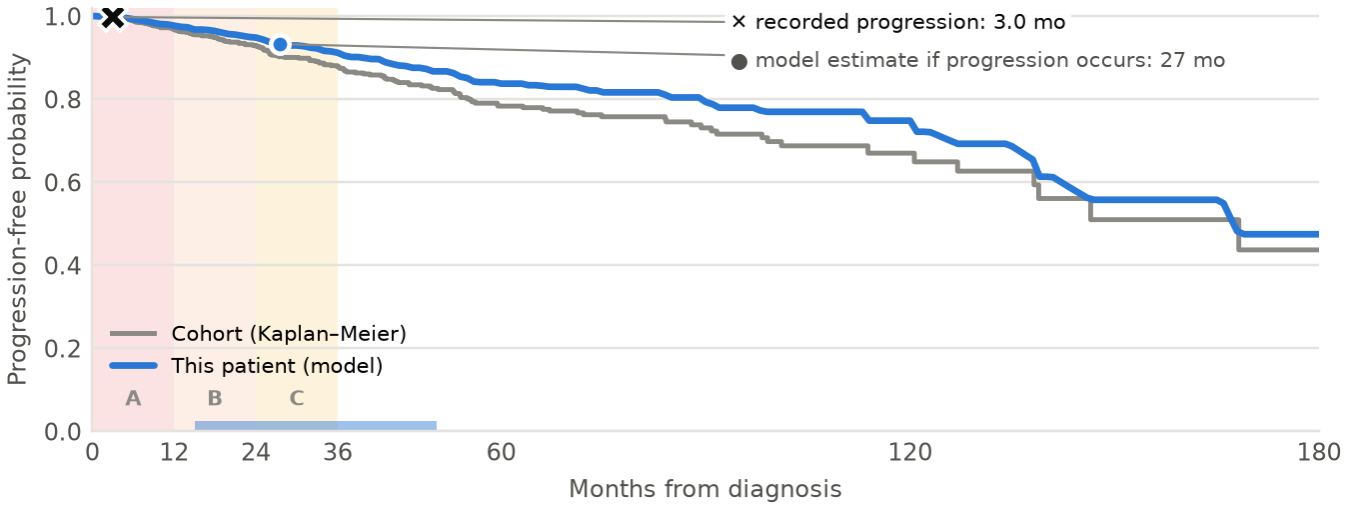
≈ 27 months

Window 24–36 months · IQR 15–50 mo
if progression occurs

Recorded progression

3.0 months

Group A · Early progression
✗ recorded window differs · |error| 25 mo



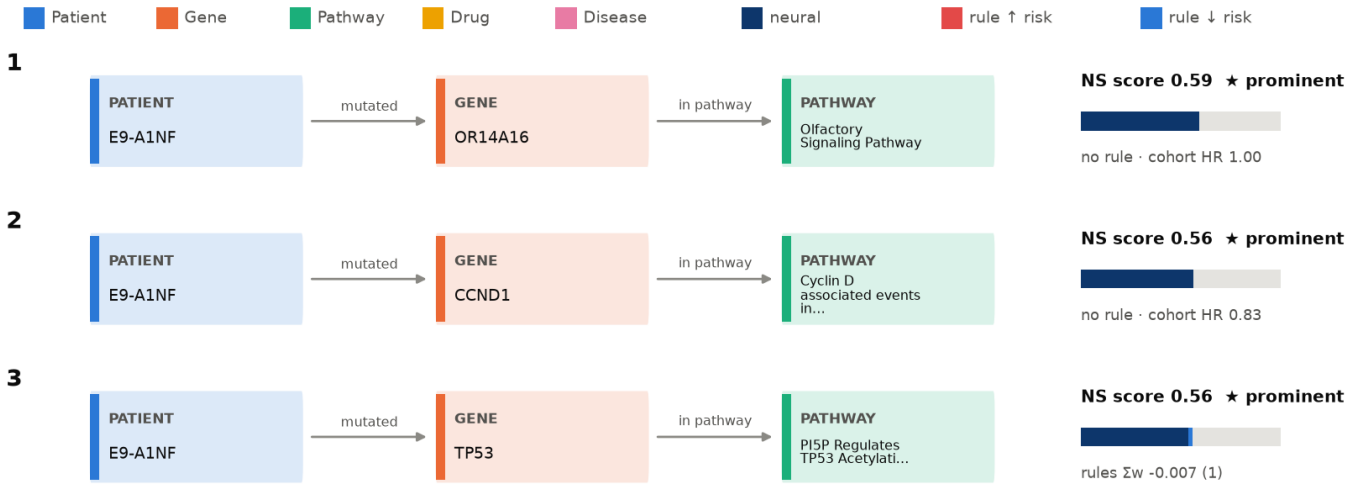
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **40th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **OR14A16** to **Olfactory Signaling Pathway**. Progression was recorded at 3.0 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

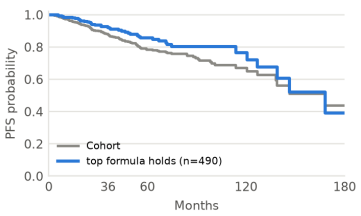


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E9-A1NF** mutated → **OR14A16** in pathway → **Olfactory Signaling Pathway**	99	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 161 carriers	**0.59**
2 ★	**TCGA-E9-A1NF** mutated → **CCND1** in pathway → **Cyclin D associated events in G1**	94	No learned formula fires on this path; its score is neural only. cohort: HR 0.83, n = 187 carriers	**0.56**
3 ★	**TCGA-E9-A1NF** mutated → **TP53** in pathway → **PI5P Regulates TP53 Acetylation**	90	↓ **slower progression** · Σw -0.007 over 1 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.95, n = 283 carriers	**0.56**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

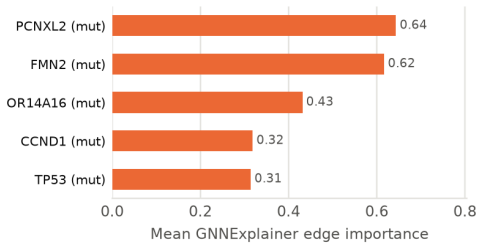
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.117 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	388 pts
+0.019 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	81 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	Mutation	p.L32=	26	0.32
TP53 · driver	Mutation	p.F113C	57	0.31
PCNXL2	Mutation	p.X897_splice	0	0.64
FMN2	Mutation	p.L959F, p.P814S, p.S747F	0	0.62
OR14A16	Mutation	p.A196T	1	0.43



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 2 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A109	64	II	LumB	0.78	no progression, followed 46.6 mo
TCGA-AO-A1KS	69	II	LumB	0.77	no progression, followed 11.5 mo
TCGA-A2-A4S1	66	II	Basal	0.77	no progression, followed 27.0 mo
TCGA-AR-A0TV	66	II	LumB	0.75	no progression, followed 75.2 mo
TCGA-EW-A2FW	52	II	LumB	0.73	no progression, followed 22.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**36**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3C**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A04P-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **4.27** · MSIsensor score: **1.12** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

85th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 3% · 24 mo 9% · 60 mo 29%

Model-estimated progression time

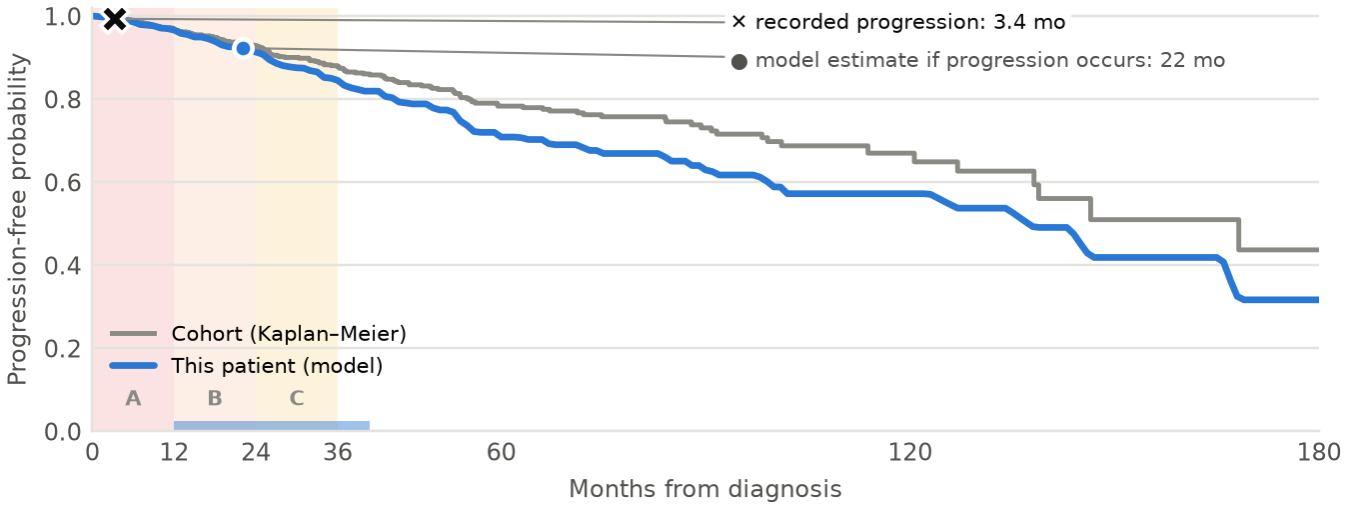
≈ 22 months

Window 12–24 months · IQR 12–41 mo
if progression occurs

Recorded progression

3.4 months

Group A · Early progression
✗ recorded window differs · |error| 19 mo



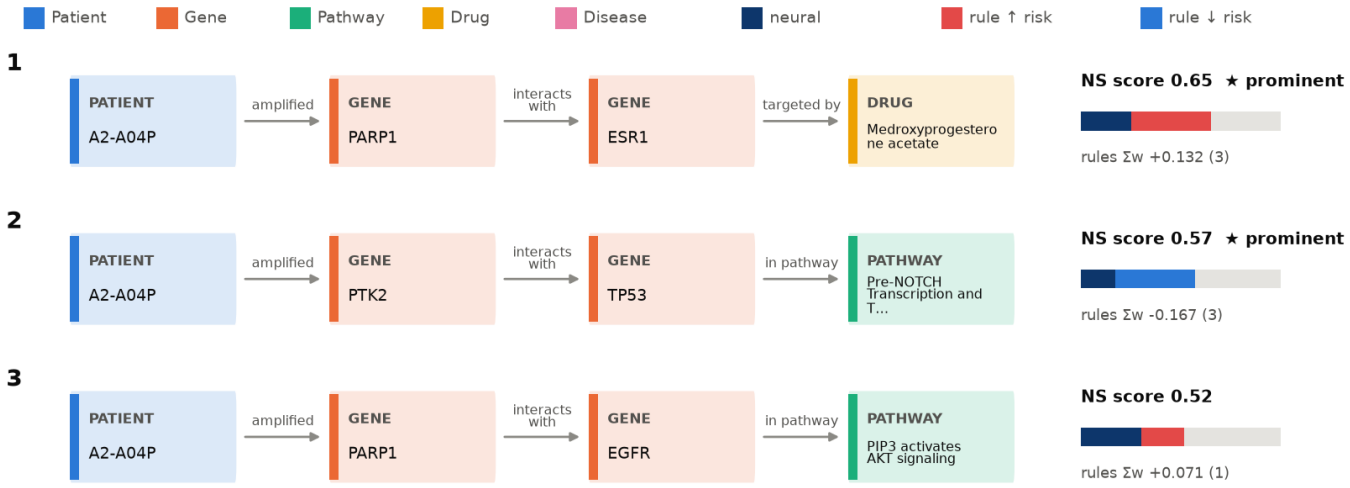
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 36-year-old patient has a Basal tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **85th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'VEGFA-VEGFR2 Pathway')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ion homeostasis')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.141 per SD). The highest-scoring knowledge-graph path runs from **PARP1** to **Medroxyprogesterone acetate**; the learned formulas it instantiates point to faster progression (Σw +0.132). Progression was recorded at 3.4 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

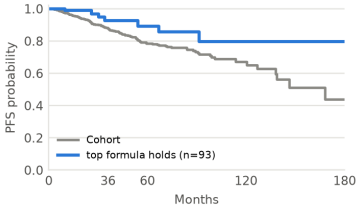


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A2-A04P** amplified → **PARP1** interacts with → **ESR1** targeted by → **Medroxyprogesterone acetate**	42	↑ **faster progression** · Σw +0.132 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 2.45, n = 30 carriers	**0.65**
2 ★	**TCGA-A2-A04P** amplified → **PTK2** interacts with → **TP53** in pathway → **Pre-NOTCH Transcription and Translation**	29	↓ **slower progression** · Σw -0.167 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.86, n = 489 carriers	**0.57**
3	**TCGA-A2-A04P** amplified → **PARP1** interacts with → **EGFR** in pathway → **PIP3 activates AKT signaling**	51	↑ **faster progression** · Σw +0.071 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 0.99, n = 578 carriers	**0.52**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

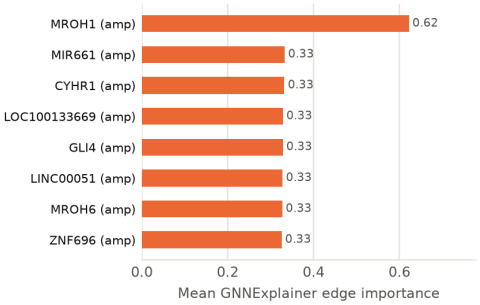
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
-0.097 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Stimuli-sensing channels') ∧ Altered(p, g) ∧ InPathway(g, 'Pyroptosis') ∧ Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	RL-proposed	70 pts
+0.084 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	51 pts
+0.077 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	61 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) [KG]	RL-proposed	71 pts
+0.054 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX2 expression and activity') ⇒ Progress(p) [KG]	single atom	57 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	High-level amplification	GISTIC ±2	141	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.32
PIK3CA · driver	Mutation	p.R88Q	70	0.32
BRCA2 · driver	Mutation	p.Q3206E	14	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
TP53 · driver	Mutation	p.G108Vfs*15	57	0.31
MROH1	High-level amplification	GISTIC ±2	0	0.62
MIR661	High-level amplification	GISTIC ±2	0	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
GLI4	High-level amplification	GISTIC ±2	0	0.33
LINC00051	High-level amplification	GISTIC ±2	0	0.33



149 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 225 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1L7	40	III	LumB	0.73	progressed at 53.2 mo (group D)
TCGA-GM-A3XL	49	II	Basal	0.73	no progression, followed 69.3 mo
TCGA-AR-A0U4	54	II	Basal	0.72	no progression, followed 107.2 mo
TCGA-AR-A0TS	46	II	Basal	0.72	no progression, followed 84.1 mo
TCGA-AR-A251	51	III	Basal	0.72	no progression, followed 99.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-UU-A93S · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T4D**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

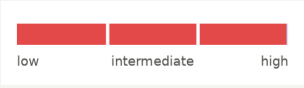
Sample: **TCGA-UU-A93S-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **4.63** · MSIsensor score: **0.48** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

63% by 36 mo

12 mo 23% · 24 mo 45% · 60 mo 88%

Model-estimated progression time

≈ 10 months

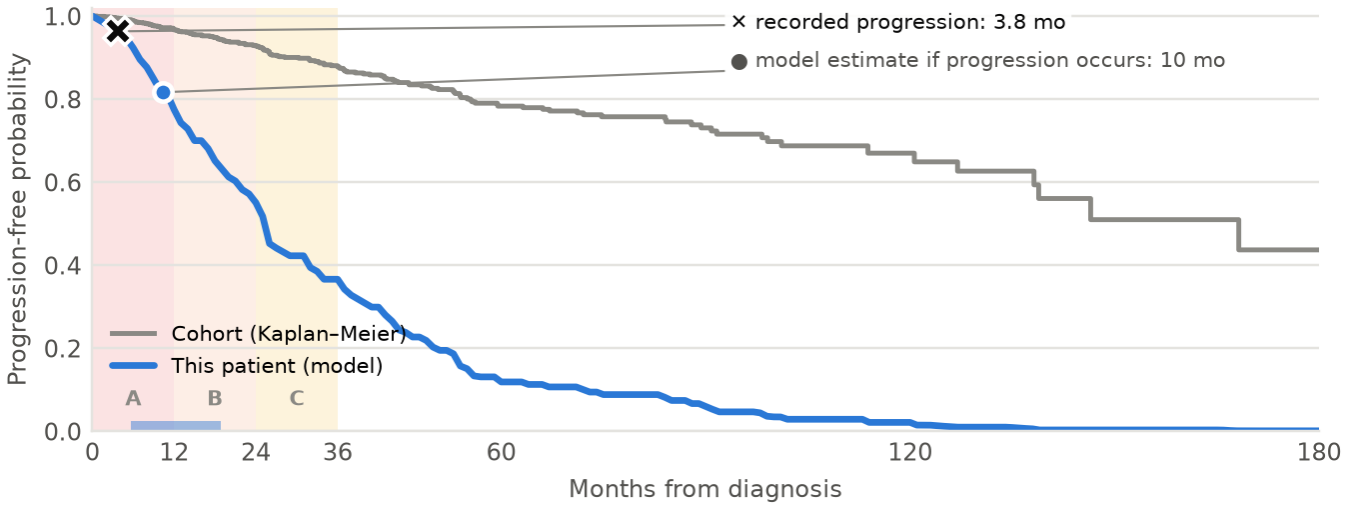
Window ≤ 12 months · IQR 6–19 mo if progression occurs

Recorded progression

3.8 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 7 mo



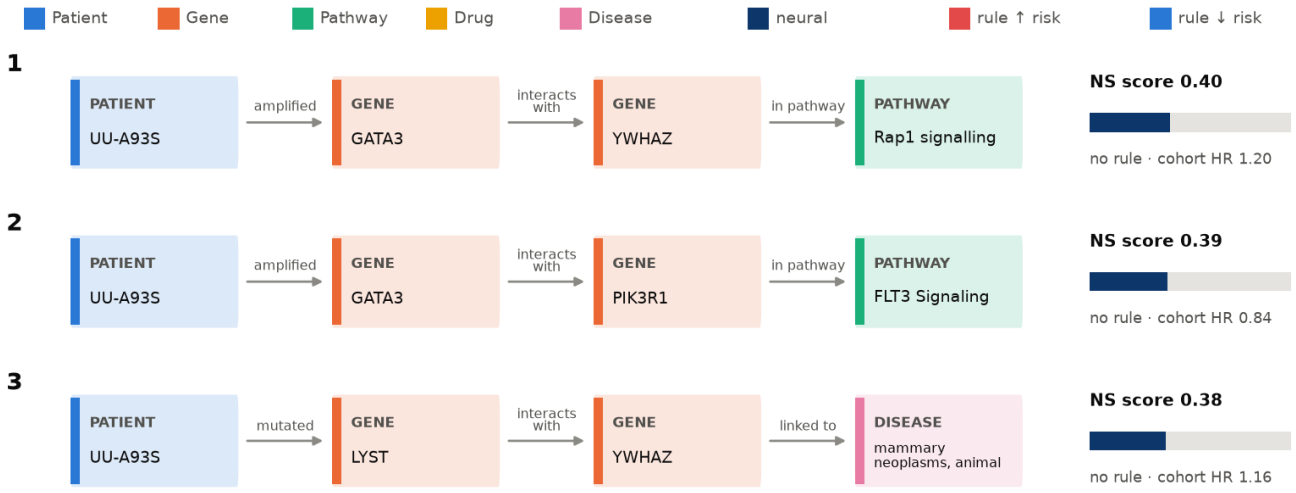
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a Her2 tumour, AJCC stage IV (T4, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.096 per SD). The highest-scoring knowledge-graph path runs from **GATA3** to **Rap1 signalling**. Progression was recorded at 3.8 months; the model's estimate given progression was 10 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

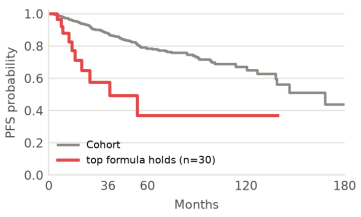


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **YWHAZ** in pathway → **Rap1 signalling**	67	No learned formula fires on this path; its score is neural only. cohort: HR 1.20, n = 100 carriers	**0.40**
2	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **PIK3R1** in pathway → **FLT3 Signaling**	64	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 413 carriers	**0.39**
3	**TCGA-UU-A93S** mutated → **LYST** interacts with → **YWHAZ** linked to → **mammary neoplasms, animal**	63	No learned formula fires on this path; its score is neural only. cohort: HR 1.16, n = 534 carriers	**0.38**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

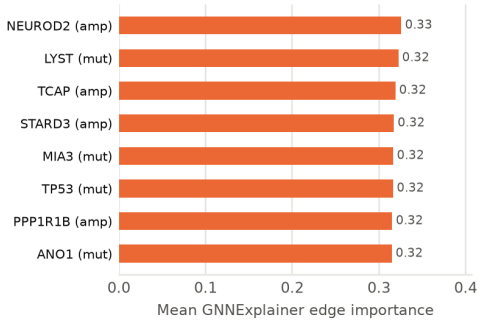
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.096 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	28 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.Q136P	57	0.32
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NEUROD2	High-level amplification	GISTIC ±2	0	0.33
LYST	Mutation	3'UTR	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
MIA3	Mutation	p.S1706*	3	0.32
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
ANO1	Mutation	p.*37*	2	0.32
ZP4	Mutation	p.Y359N	1	0.31
IKZF3	High-level amplification	GISTIC ±2	0	0.31



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 39 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A08J	52	IV	Her2	0.70	progressed at 37.1 mo (group D)
TCGA-E2-A14P	79	III	Her2	0.65	no progression, followed 41.0 mo
TCGA-A8-A08X	43	III	Her2	0.61	progressed at 12.0 mo (group B)
TCGA-B6-A0I9	62	IV	Her2	0.61	progressed at 11.9 mo (group A)
TCGA-BH-A0B7	42	II	Her2	0.60	no progression, followed 84.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**—**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-C8-A137-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **0.24** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

64th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

≈ 41 months

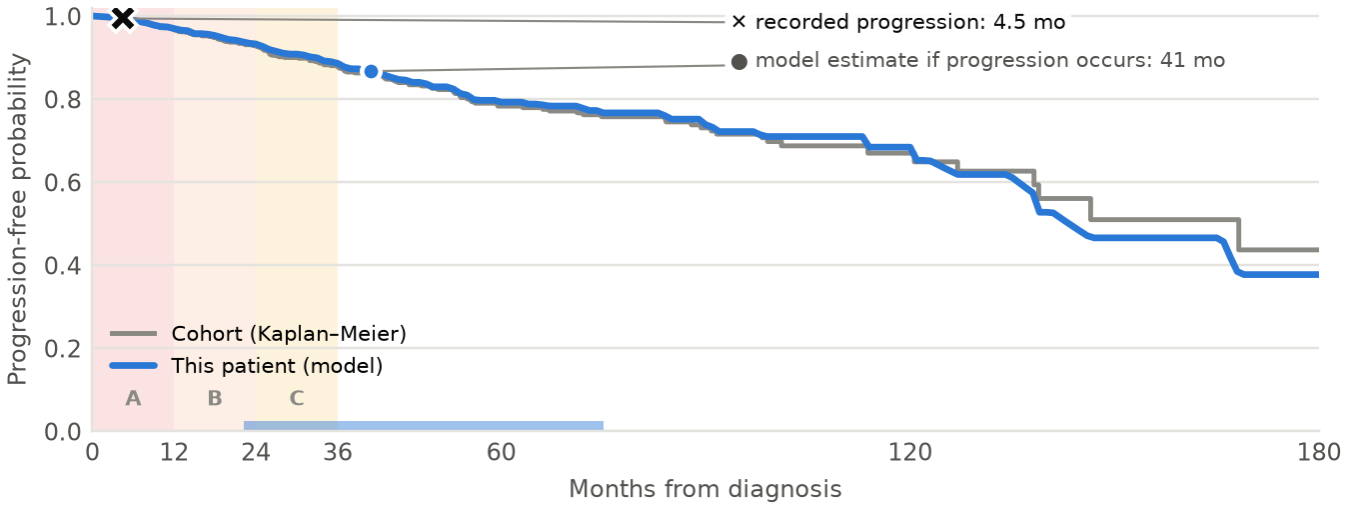
Window > 36 months · IQR 22–75 mo if progression occurs

Recorded progression

4.5 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 36 mo



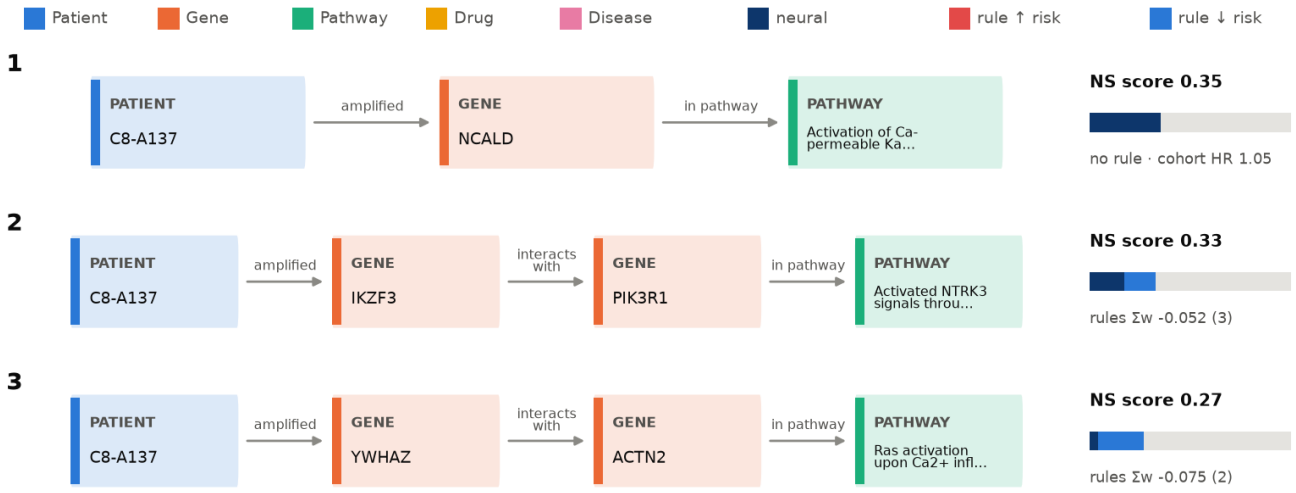
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a Her2 tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **64th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*Stage(p, II)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **NCALD** to **Activation of Ca-permeable Kainate Receptor**. Progression was recorded at 4.5 months; the model's estimate given progression was 41 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

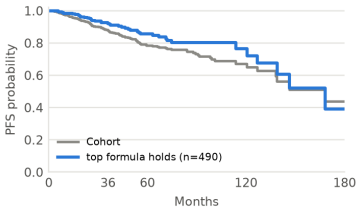


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-C8-A137** amplified → **NCALD** in pathway → **Activation of Ca-permeable Kainate Receptor**	59	No learned formula fires on this path; its score is neural only. cohort: HR 1.05, n = 100 carriers	**0.35**
2	**TCGA-C8-A137** amplified → **IKZF3** interacts with → **PIK3R1** in pathway → **Activated NTRK3 signals through PI3K**	28	↓ **slower progression** · Σw -0.052 over 3 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.86, n = 309 carriers	**0.33**
3	**TCGA-C8-A137** amplified → **YWHAZ** interacts with → **ACTN2** in pathway → **Ras activation upon Ca2+ influx through NMDA receptor**	7	↓ **slower progression** · Σw -0.075 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 1.04, n = 103 carriers	**0.27**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

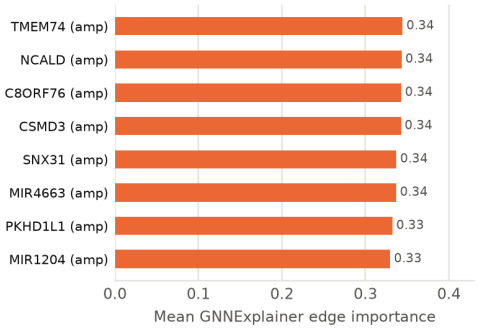
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts
+0.035 ↑ risk	Age(p) < 40 ⇒ Progress(p)	single atom	65 pts
-0.034 ↓ risk	LowAneuploidy(p) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Deactivation of the beta-catenin transactivating complex') ⇒ Progress(p) [KG]	RL-proposed	41 pts
+0.023 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ⇒ Progress(p) [KG]	mined pair	55 pts
-0.007 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	306 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
TMEM74	High-level amplification	GISTIC ±2	0	0.34
NCALD	High-level amplification	GISTIC ±2	1	0.34
C8ORF76	High-level amplification	GISTIC ±2	0	0.34
CSMD3	High-level amplification	GISTIC ±2	0	0.34
SNX31	High-level amplification	GISTIC ±2	0	0.34
MIR4663	High-level amplification	GISTIC ±2	0	0.34
PKHD1L1	High-level amplification	GISTIC ±2	0	0.33
MIR1204	High-level amplification	GISTIC ±2	0	0.33
AARD	High-level amplification	GISTIC ±2	0	0.33



148 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916) … and 38 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A15S	34	II	LumB	0.81	no progression, followed 14.1 mo
TCGA-C8-A135	64	II	Her2	0.80	no progression, followed 12.9 mo
TCGA-AN-A04C	51	II	Her2	0.79	no progression, followed 1.8 mo
TCGA-C8-A12Z	45	II	Her2	0.78	no progression, followed 12.6 mo
TCGA-AR-A24U	47	II	Her2	0.77	no progression, followed 102.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**36**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AO-A0JA-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.11** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

69th

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

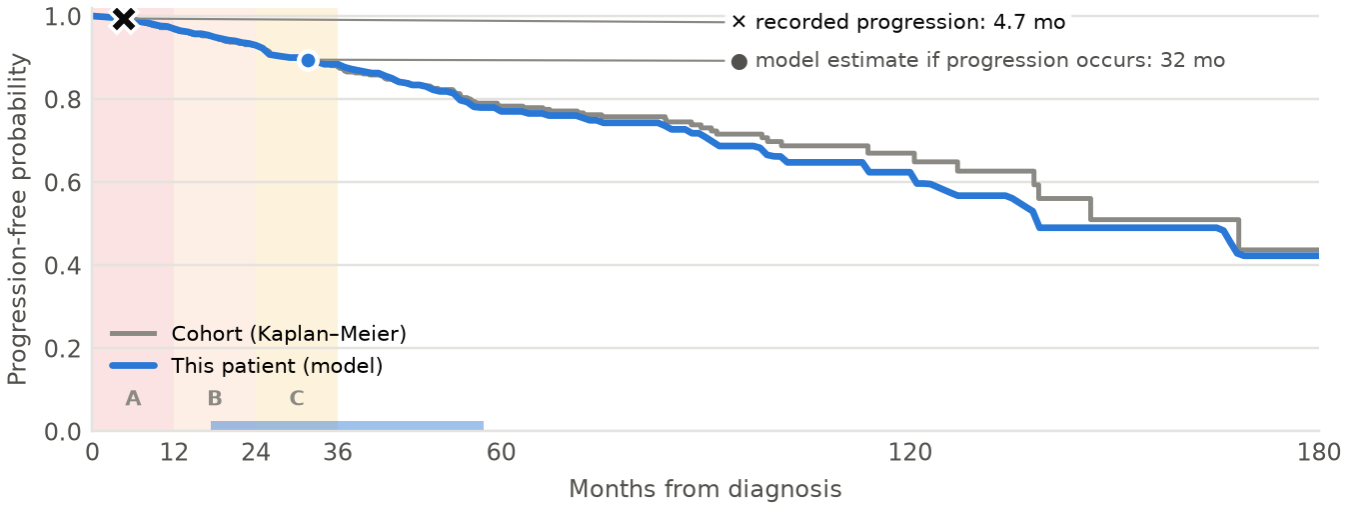
≈ 32 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

4.7 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 27 mo



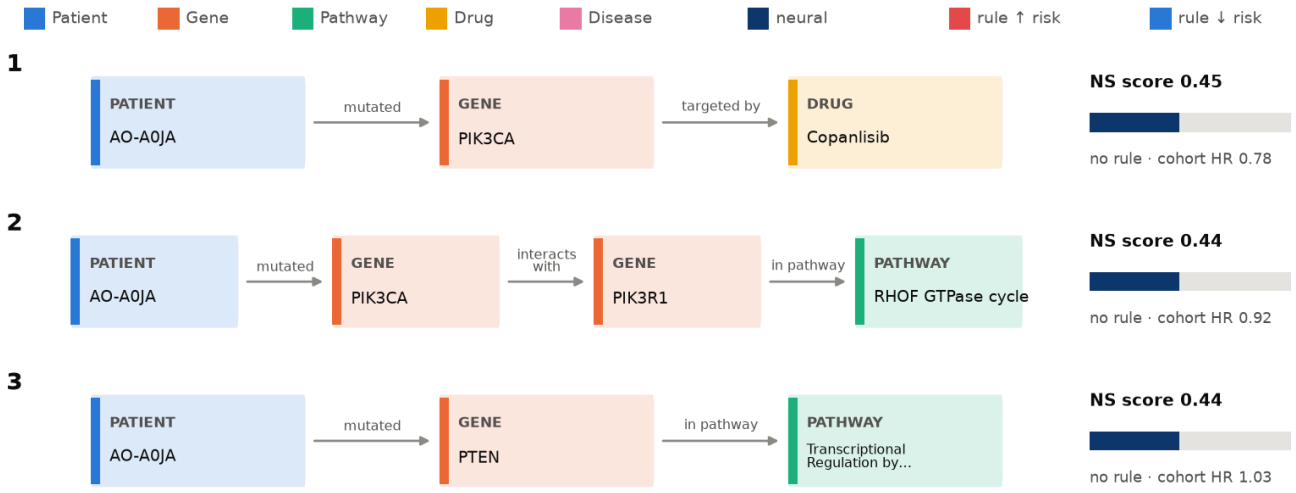
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 36-year-old patient has a LumA tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **69th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Copanlisib**. Progression was recorded at 4.7 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

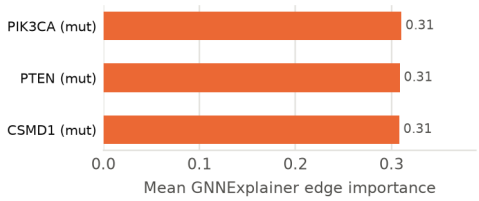
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AO-A0JA** mutated → **PIK3CA** targeted by → **Copanlisib**	75	No learned formula fires on this path; its score is neural only. cohort: HR 0.78, n = 308 carriers	**0.45**
2	**TCGA-AO-A0JA** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **RHOF GTPase cycle**	74	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 135 carriers	**0.44**
3	**TCGA-AO-A0JA** mutated → **PTEN** in pathway → **Transcriptional Regulation by MECP2**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.03, n = 181 carriers	**0.44**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.E542K	70	0.31
PTEN · driver	Mutation	p.Y88*	22	0.31
CSMD1	Mutation	p.D875H, p.S2505*	0	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1R3	70	III	LumA	0.79	no progression, followed 2.6 mo
TCGA-D8-A27L	49	III	LumA	0.73	no progression, followed 16.4 mo
TCGA-S3-A6ZH	29	III	LumA	0.70	no progression, followed 21.1 mo
TCGA-A7-A0D9	37	II	LumA	0.69	no progression, followed 37.4 mo
TCGA-BH-A0BC	60	III	LumA	0.69	no progression, followed 32.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A1LK · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**84**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**—**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

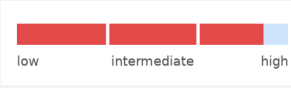
Sample: **TCGA-E2-A1LK-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **2.89** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 32%

Model-estimated progression time

≈ 7 months

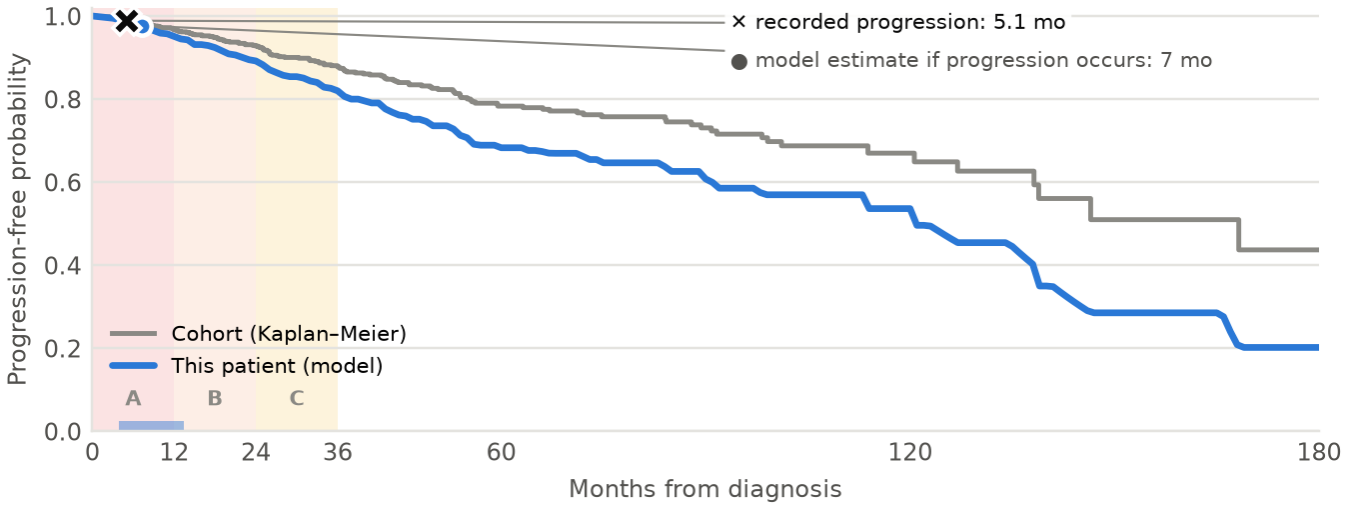
Window ≤ 12 months · IQR 4–13 mo if progression occurs

Recorded progression

5.1 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 2 mo



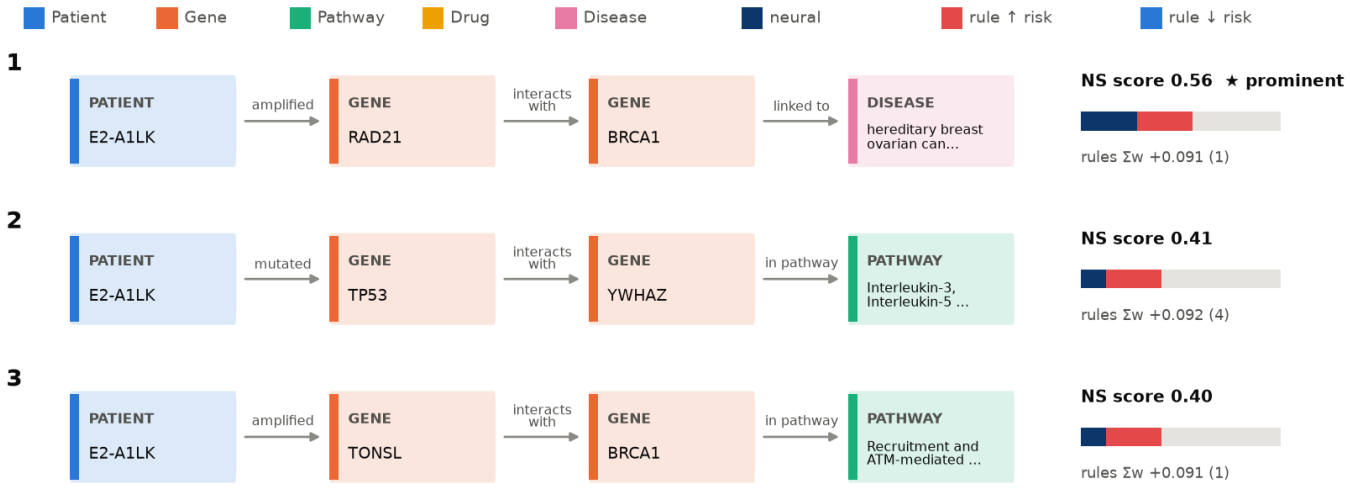
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 84-year-old patient has a Basal tumour, AJCC stage III (T4, N3, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T4)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.113 per SD). The highest-scoring knowledge-graph path runs from **RAD21** to **hereditary breast ovarian cancer syndrome**; the learned formulas it instantiates point to faster progression (Σw +0.091). Progression was recorded at 5.1 months; the model's estimate given progression was 7 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

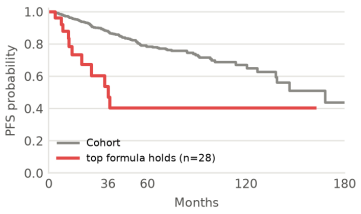


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A1LK** amplified → **RAD21** interacts with → **BRCA1** linked to → **hereditary breast ovarian cancer syndrome**	47	↑ **faster progression** · Σw +0.091 over 1 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 0.99, n = 798 carriers	**0.56**
2	**TCGA-E2-A1LK** mutated → **TP53** interacts with → **YWHAZ** in pathway → **Interleukin-3, Interleukin-5 and GM-CSF signaling**	21	↑ **faster progression** · Σw +0.092 over 4 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 0.92, n = 373 carriers	**0.41**
3	**TCGA-E2-A1LK** amplified → **TONSL** interacts with → **BRCA1** in pathway → **Recruitment and ATM-mediated phosphorylation of repair and signaling proteins at DNA double strand breaks**	21	↑ **faster progression** · Σw +0.091 over 1 formula(s) Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) cohort: HR 0.92, n = 329 carriers	**0.40**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

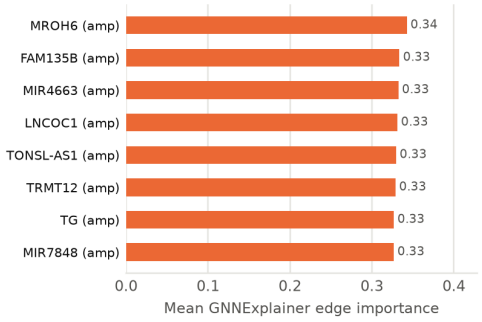
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.113 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	26 pts
+0.097 ↑ risk	NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	50 pts
+0.091 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Platelet degranulation ') ∧ Altered(p, g) ∧ InPathway(g, 'TP53 Regulates Transcription of DNA Repair Genes') ⇒ Progress(p) [KG]	RL-proposed	84 pts
+0.079 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Formation of the beta-catenin:TCF transactivating complex') ∧ Altered(p, g) ∧ InPathway(g, 'HATs acetylate histones') ∧ Altered(p, g) ∧ InPathway(g, 'Purinergic signaling in leishmaniasis infection') ⇒ Progress(p) [KG]	RL-proposed	49 pts
-0.067 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	RL-proposed	39 pts
-0.034 ↓ risk	Amplified(p, ZFHX4) ⇒ Progress(p)	single atom	55 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
TP53 · driver	Mutation	p.A86Cfs*63	57	0.31
MROH6	High-level amplification	GISTIC ±2	0	0.34
FAM135B	High-level amplification	GISTIC ±2	0	0.33
MIR4663	High-level amplification	GISTIC ±2	0	0.33
LNCOC1	High-level amplification	GISTIC ±2	0	0.33
TONSL-AS1	High-level amplification	GISTIC ±2	0	0.33
TRMT12	High-level amplification	GISTIC ±2	1	0.33
TG	High-level amplification	GISTIC ±2	1	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33



338 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840) … and 146 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YH	53	III	LumB	0.87	no progression, followed 21.7 mo
TCGA-C8-A131	82	III	Basal	0.85	no progression, followed 13.5 mo
TCGA-A8-A06P	63	III	LumA	0.85	no progression, followed 13.0 mo
TCGA-A2-A0CU	73	II	LumA	0.84	no progression, followed 5.2 mo
TCGA-A8-A09Q	83	III	LumB	0.84	no progression, followed 25.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A7-A3RF · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

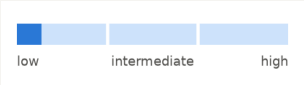
Sample: **TCGA-A7-A3RF-01** · Aneuploidy score: **0** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

9th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

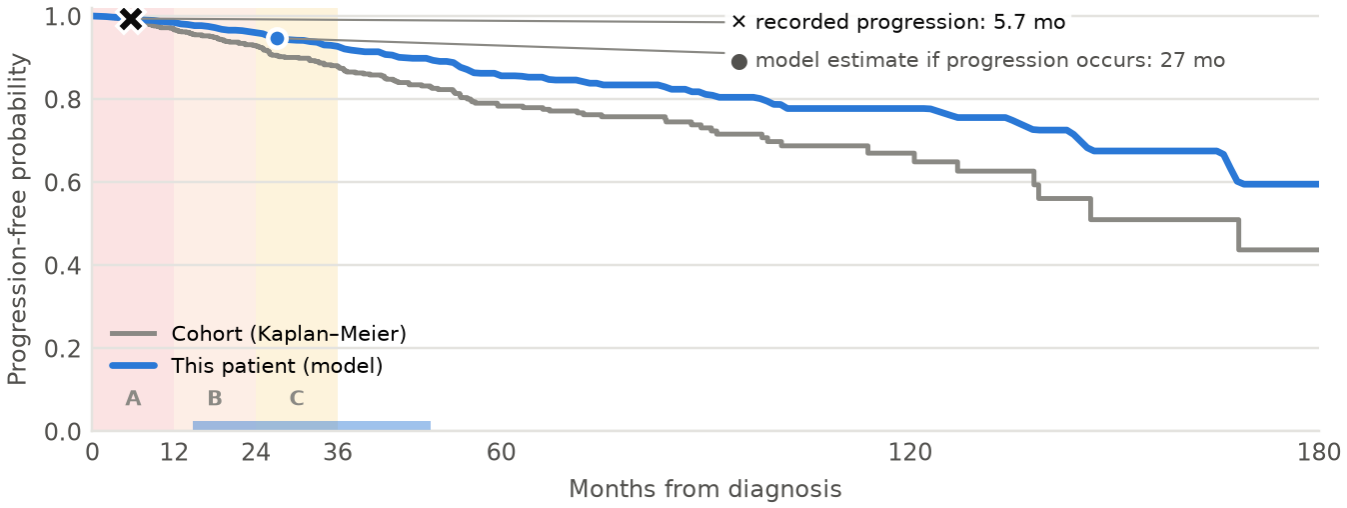
≈ 27 months

Window 24–36 months · IQR 15–50 mo
if progression occurs

Recorded progression

5.7 months

Group A · Early progression
✗ recorded window differs · |error| 21 mo



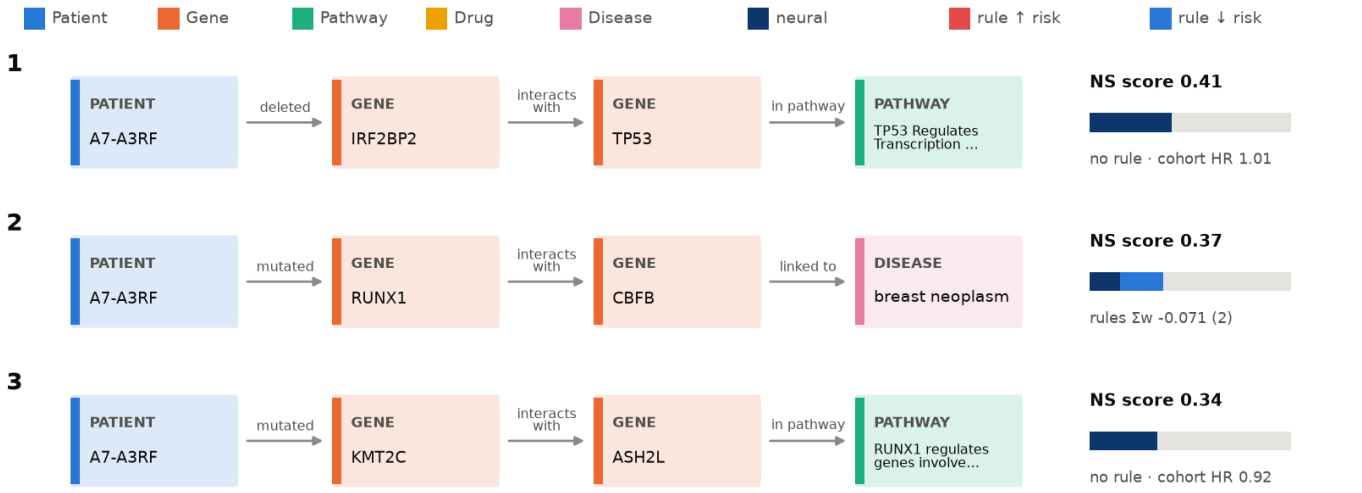
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **9th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **IRF2BP2** to **TP53 Regulates Transcription of Genes Involved in Cytochrome C Release**. Progression was recorded at 5.7 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

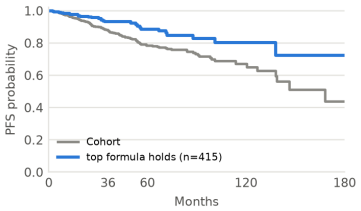


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A7-A3RF** deleted → **IRF2BP2** interacts with → **TP53** in pathway → **TP53 Regulates Transcription of Genes Involved in Cytochrome C Release**	68	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 364 carriers	**0.41**
2	**TCGA-A7-A3RF** mutated → **RUNX1** interacts with → **CBFB** linked to → **breast neoplasm**	25	↓ **slower progression** · Σw -0.071 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known') ⇒ Progress(p) cohort: HR 0.94, n = 792 carriers	**0.37**
3	**TCGA-A7-A3RF** mutated → **KMT2C** interacts with → **ASH2L** in pathway → **RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function**	56	No learned formula fires on this path; its score is neural only. cohort: HR 0.92, n = 254 carriers	**0.34**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

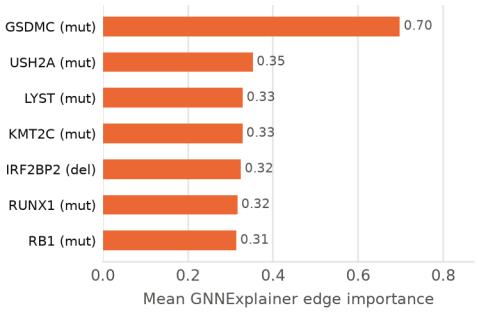
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
+0.078 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	79 pts
-0.048 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 interacts with co-factors whose precise effect on RUNX1 targets is not known') ⇒ Progress(p) [KG]	mined pair	68 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
-0.023 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of RUNX1 Expression and Activity') ⇒ Progress(p) [KG]	mined pair	162 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q1937*	4	0.33
RUNX1 · driver	Mutation	3'UTR	18	0.32
RB1 · driver	Mutation	p.S866Afs*7	25	0.31
GSDMC	Mutation	p.R34C	0	0.70
USH2A	Mutation	p.S1051=	0	0.35
LYST	Mutation	p.P1758Q	0	0.33
IRF2BP2	Homozygous deletion	GISTIC ±2	0	0.32



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XV	84	II	LumA	0.79	no progression, followed 15.2 mo
TCGA-AO-A0J8	61	II	LumA	0.73	no progression, followed 22.4 mo
TCGA-AO-A125	72	II	LumA	0.73	no progression, followed 113.6 mo
TCGA-A2-A0EN	70	II	LumA	0.73	no progression, followed 134.4 mo
TCGA-A8-A0AD	83	I	LumA	0.71	no progression, followed 38.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**90**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AC-A23H-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **142.67** · MSIsensor score: **0.54** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

82nd

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 28%

Model-estimated progression time

≈ 9 months

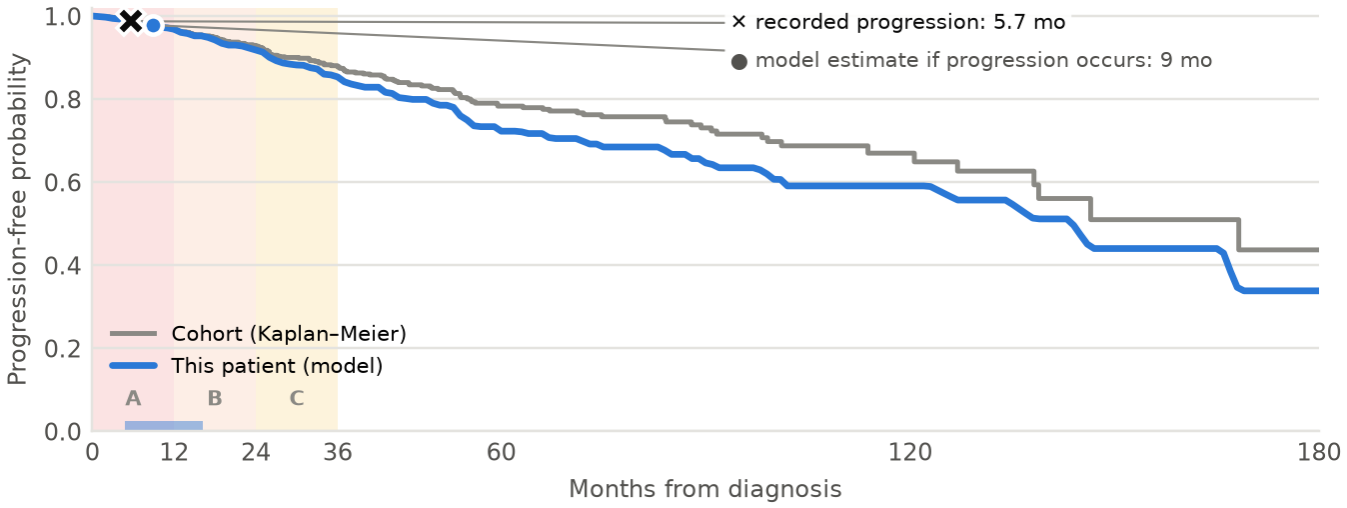
Window ≤ 12 months · IQR 5–16 mo if progression occurs

Recorded progression

5.7 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 3 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 90-year-old patient has a Her2 tumour, AJCC stage II (T2, missing, M0). The fused neuro-symbolic model places them in the **82nd risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'VEGFA-VEGFR2 Pathway')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ion homeostasis')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.141 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Stimuli-sensing channels**; the learned formulas it instantiates point to slower progression (Σw -0.188). Progression was recorded at 5.7 months; the model's estimate given progression was 9 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

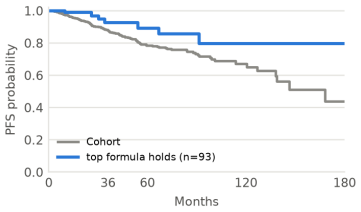


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AC-A23H** mutated → **RYR2** in pathway → **Stimuli-sensing channels**	70	↓ **slower progression** · Σw -0.188 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.78, n = 261 carriers	**0.82**
2 ★	**TCGA-AC-A23H** mutated → **ESR1** targeted by → **Estriol**	59	↑ **faster progression** · Σw +0.132 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 2.45, n = 30 carriers	**0.75**
3 ★	**TCGA-AC-A23H** mutated → **NID1** in pathway → **Laminin interactions**	76	↓ **slower progression** · Σw -0.067 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Laminin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'NCAM signaling for neurite out-growth') ∧ Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ⇒ Progress(p) cohort: HR 1.09, n = 101 carriers	**0.66**
4 ★	**TCGA-AC-A23H** amplified → **TPCN2** interacts with → **FGFR1** in pathway → **Signaling by plasma membrane FGFR1 fusions**	59	↑ **faster progression** · Σw +0.090 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'Signaling by activated point mutants of FGFR1') ∧ Altered(p, g) ∧ InPathway(g, 'Striated Muscle Contraction') ∧ Altered(p, g) ∧ InPathway(g, 'RHOA GTPase cycle') ⇒ Progress(p) cohort: HR 1.02, n = 139 carriers	**0.63**
5 ★	**TCGA-AC-A23H** amplified → **CCND1** interacts with → **TCHH** in pathway → **Formation of the cornified envelope**	77	↑ **faster progression** · Σw +0.048 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 0.95, n = 147 carriers	**0.61**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

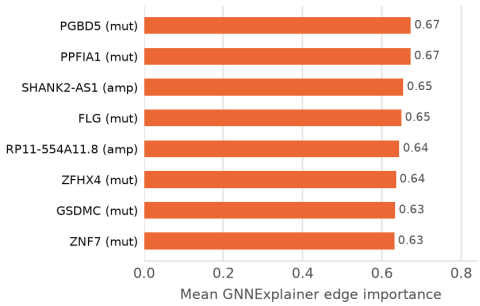
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
-0.089 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Sensing of DNA Double Strand Breaks') ⇒ Progress(p) [KG]	single atom	39 pts
+0.085 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Signaling by activated point mutants of FGFR1') ∧ Altered(p, g) ∧ InPathway(g, 'Striated Muscle Contraction') ∧ Altered(p, g) ∧ InPathway(g, 'RHOA GTPase cycle') ⇒ Progress(p) [KG]	RL-proposed	37 pts
+0.079 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) [KG]	mined pair	40 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) [KG]	RL-proposed	71 pts
-0.067 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Laminin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'NCAM signaling for neurite out-growth') ∧ Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ⇒ Progress(p) [KG]	RL-proposed	37 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.D226N	9	0.32
NF1 · driver	Mutation	p.S1896L	6	0.32
MAP3K1 · driver	Mutation	p.R283T	13	0.32
SF3B1 · driver	Mutation	p.L353=, p.L464F	4	0.32
ARID1A · driver	Mutation	p.M1273I	10	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
KMT2C · driver	Mutation	p.D3264N, p.D4344H, p.E3724K	4	0.32
BRCA2 · driver	Mutation	p.D1033H, p.D687H	14	0.31
ATM · driver	Mutation	p.I105=	29	0.31
PIK3CA · driver	Mutation	p.D603H, p.L456=, p.L989V	70	0.31
RB1 · driver	Mutation	p.G802=	25	0.31
ESR1 · driver	Mutation	p.L448=	141	0.31



177 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010); SF3B1 → H3B-8800 (DB14017); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021) … and 279 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A12L	67	II	Her2	0.77	no progression, followed 11.9 mo
TCGA-AN-A046	68	II	LumA	0.75	no progression, followed 0.3 mo
TCGA-BH-A0BV	78	II	LumA	0.72	no progression, followed 49.9 mo
TCGA-A2-A0D1	76	II	Her2	0.71	no progression, followed 34.6 mo
TCGA-E2-A105	79	II	LumA	0.70	no progression, followed 43.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-Z7-A8R5 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**Yes**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-Z7-A8R5-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

67th

High risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 22%

Model-estimated progression time

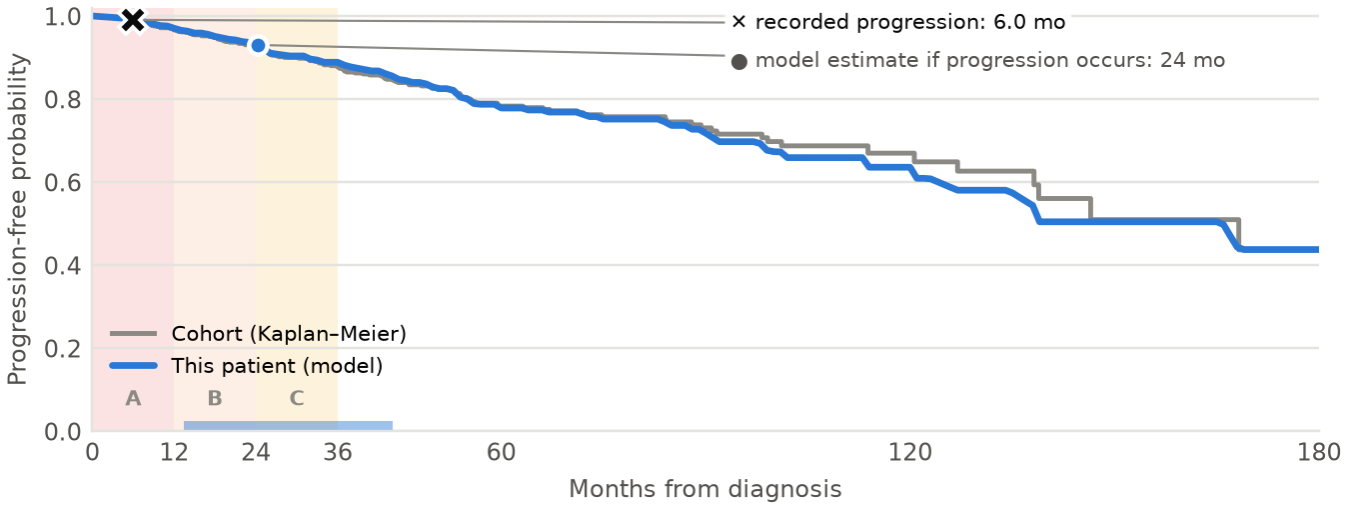
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

6.0 months

Group A · Early progression
✗ recorded window differs · |error| 18 mo



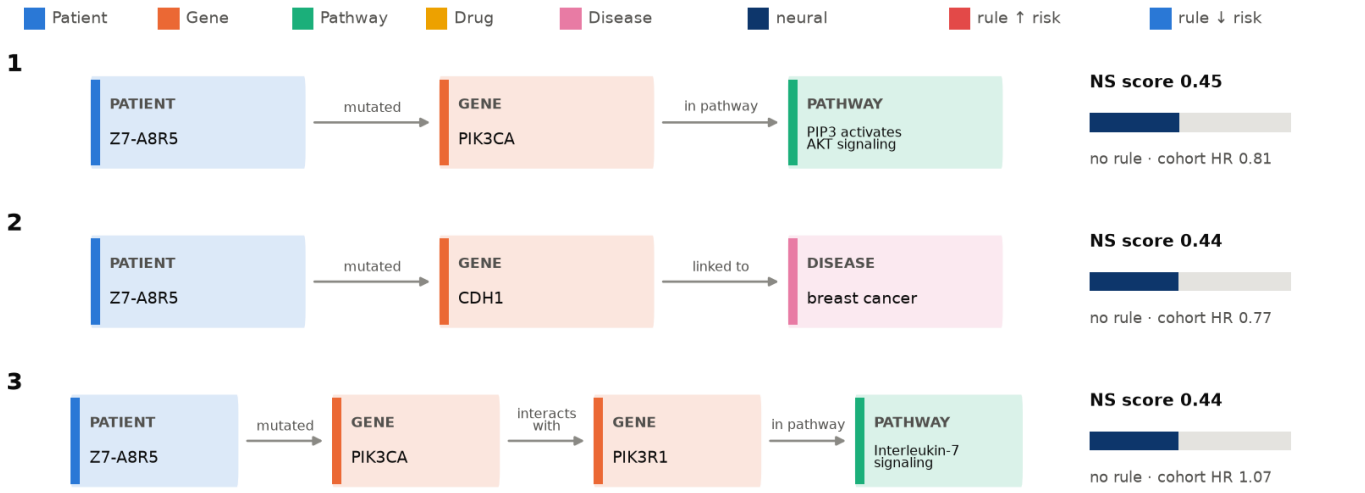
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **67th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PIK3CA** to **PIP3 activates AKT signaling**. Progression was recorded at 6.0 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

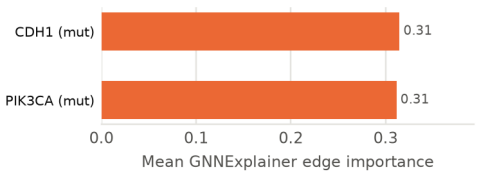
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-Z7-A8R5** mutated → **PIK3CA** in pathway → **PIP3 activates AKT signaling**	74	No learned formula fires on this path; its score is neural only. cohort: HR 0.81, n = 581 carriers	**0.45**
2	**TCGA-Z7-A8R5** mutated → **CDH1** linked to → **breast cancer**	74	No learned formula fires on this path; its score is neural only. cohort: HR 0.77, n = 794 carriers	**0.44**
3	**TCGA-Z7-A8R5** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Interleukin-7 signaling**	74	No learned formula fires on this path; its score is neural only. cohort: HR 1.07, n = 39 carriers	**0.44**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.L71Sfs*12	17	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A1AL	60	III	LumA	0.89	no progression, followed 97.7 mo
TCGA-A7-A425	70	III	LumA	0.79	progressed at 13.8 mo (group B)
TCGA-LD-A66U	44	II	LumA	0.78	no progression, followed 21.2 mo
TCGA-A2-A4S2	62	III	LumA	0.78	no progression, followed 21.1 mo
TCGA-A2-A0YL	48	III	LumA	0.77	no progression, followed 48.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A14Z · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

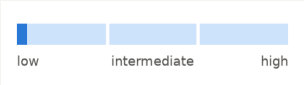
Sample: **TCGA-E2-A14Z-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **0.13** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 12%

Model-estimated progression time

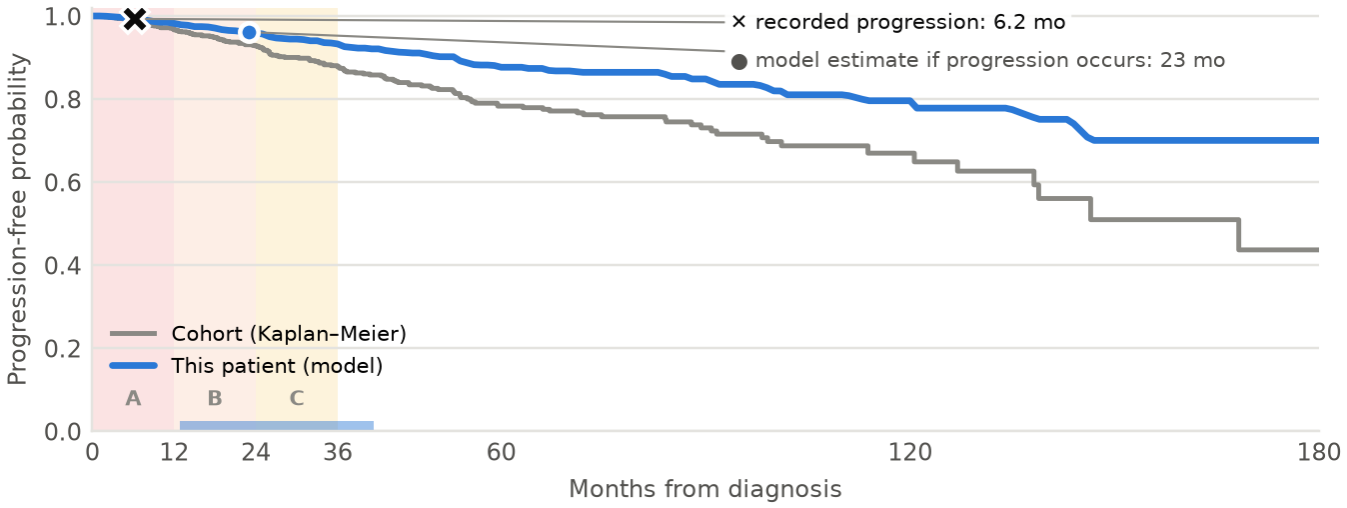
≈ 23 months

Window 12–24 months · IQR 13–41 mo
if progression occurs

Recorded progression

6.2 months

Group A · Early progression
✗ recorded window differs · |error| 17 mo



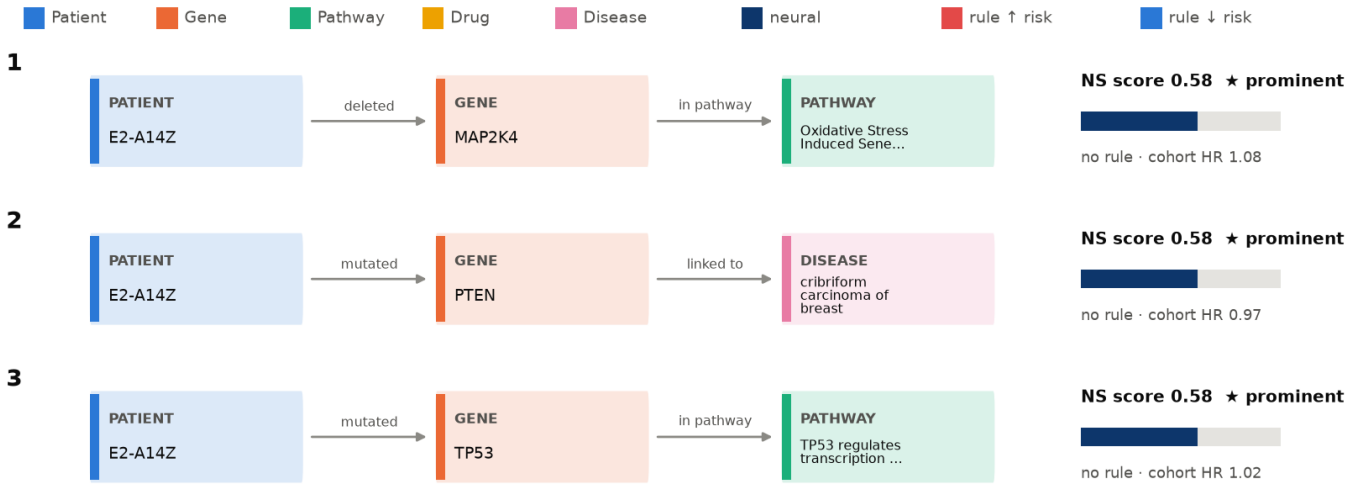
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **MAP2K4** to **Oxidative Stress Induced Senescence**. Progression was recorded at 6.2 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

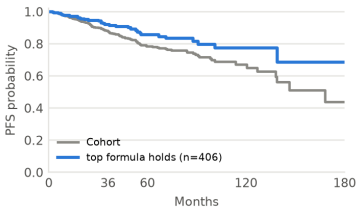


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A14Z** deleted → **MAP2K4** in pathway → **Oxidative Stress Induced Senescence**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.08, n = 318 carriers	**0.58**
2 ★	**TCGA-E2-A14Z** mutated → **PTEN** linked to → **cribriform carcinoma of breast**	97	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 674 carriers	**0.58**
3 ★	**TCGA-E2-A14Z** mutated → **TP53** in pathway → **TP53 regulates transcription of several additional cell death genes whose specific roles in p53-dependent apoptosis remain uncertain**	97	No learned formula fires on this path; its score is neural only. cohort: HR 1.02, n = 382 carriers	**0.58**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

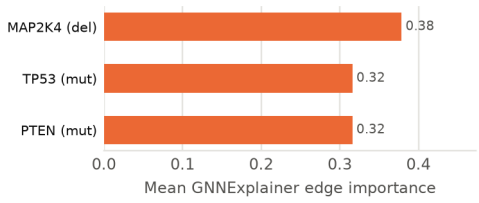
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
-0.055 ↓ risk	Stage(p, I) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	81 pts
-0.051 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	118 pts
-0.010 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	390 pts
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.38
TP53 · driver	Mutation	p.R282W	57	0.32
PTEN · driver	Mutation	p.D252Rfs*42	22	0.32



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XA	64	I	LumA	0.94	no progression, followed 27.6 mo
TCGA-E2-A1IN	60	I	LumA	0.88	no progression, followed 22.2 mo
TCGA-AO-A03V	41	I	LumA	0.87	no progression, followed 44.4 mo
TCGA-A8-A06O	60	I	LumB	0.86	no progression, followed 13.0 mo
TCGA-GM-A2DK	58	I	LumA	0.84	no progression, followed 87.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

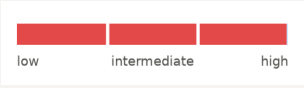
Sample: **TCGA-LL-A73Z-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

70% by 36 mo

12 mo 27% · 24 mo 47% · 60 mo 88%

Model-estimated progression time

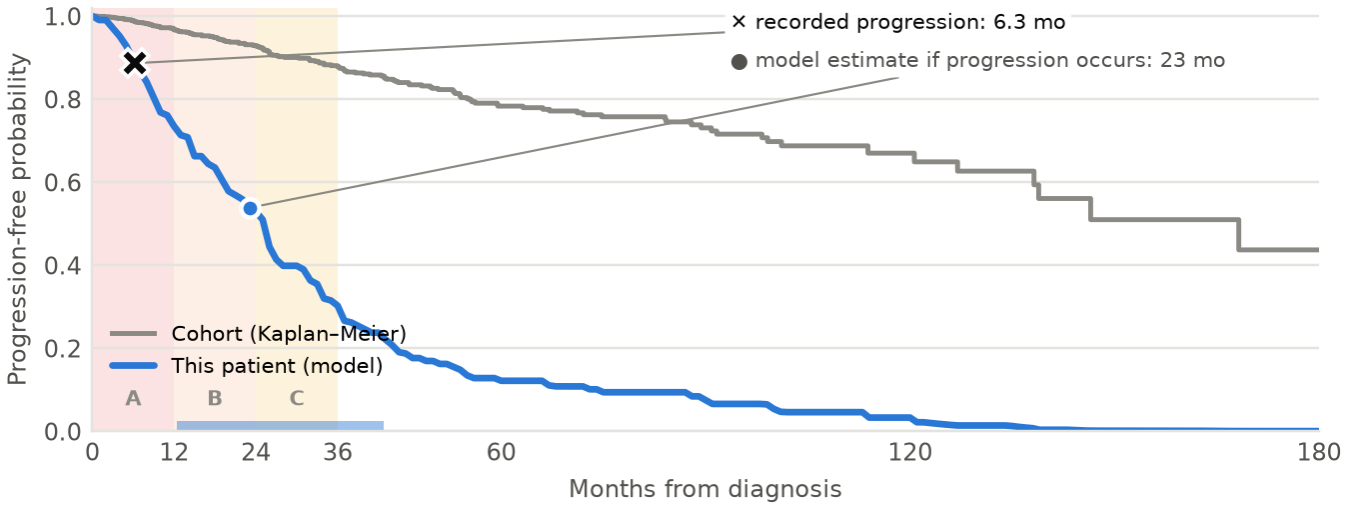
≈ 23 months

Window 12–24 months · IQR 12–43 mo
if progression occurs

Recorded progression

6.3 months

Group A · Early progression
✗ recorded window differs · |error| 17 mo



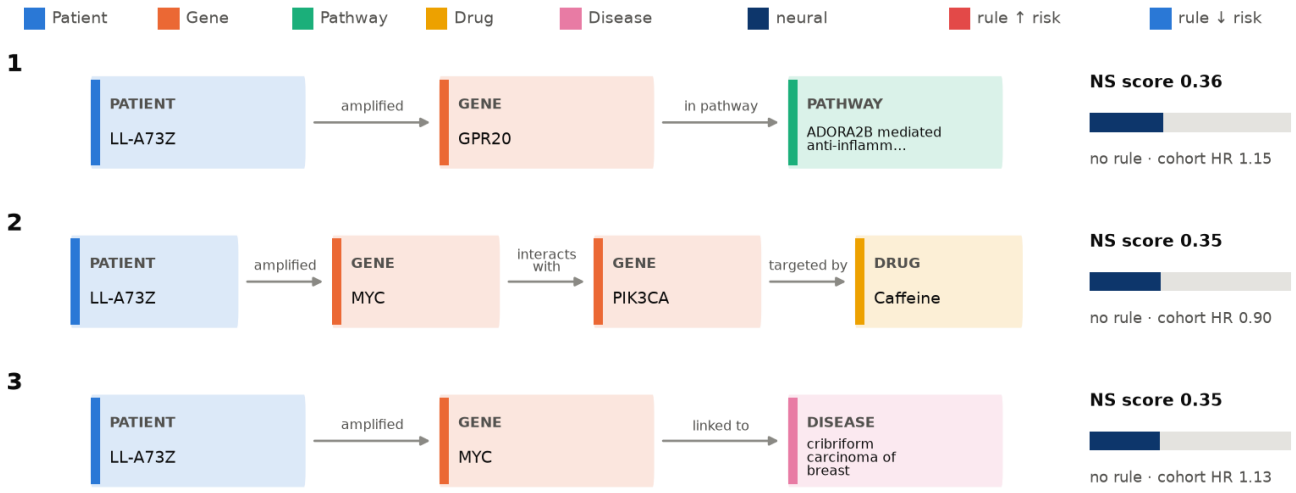
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a Normal tumour, AJCC stage IV (T1, missing, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The strongest learned formula that holds for them (*TStage(p, T1)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.042 per SD). The highest-scoring knowledge-graph path runs from **GPR20** to **ADORA2B mediated anti-inflammatory cytokines production**. Progression was recorded at 6.3 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

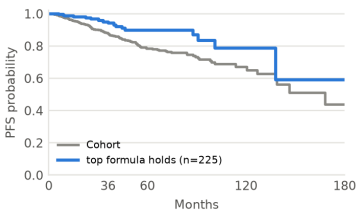


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-LL-A73Z** amplified → **GPR20** in pathway → **ADORA2B mediated anti-inflammatory cytokines production**	61	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 206 carriers	**0.36**
2	**TCGA-LL-A73Z** amplified → **MYC** interacts with → **PIK3CA** targeted by → **Caffeine**	59	No learned formula fires on this path; its score is neural only. cohort: HR 0.90, n = 319 carriers	**0.35**
3	**TCGA-LL-A73Z** amplified → **MYC** linked to → **cribriform carcinoma of breast**	58	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 664 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

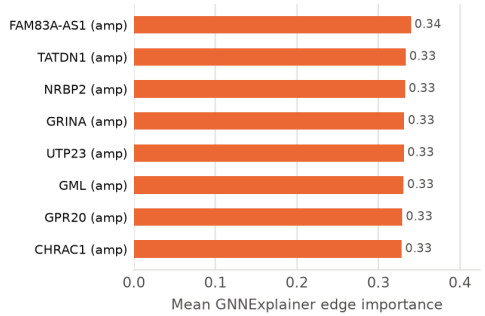
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.042 ↓ risk	TStage(p, T1) ⇒ Progress(p)	single atom	225 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
CDH1 · driver	Mutation	p.D254Y	17	0.31
PTEN · driver	Mutation	p.V290Sfs*8	22	0.31
FAM83A-AS1	High-level amplification	GISTIC ±2	0	0.34
TATDN1	High-level amplification	GISTIC ±2	0	0.33
NRBP2	High-level amplification	GISTIC ±2	0	0.33
GRINA	High-level amplification	GISTIC ±2	0	0.33
UTP23	High-level amplification	GISTIC ±2	0	0.33
GML	High-level amplification	GISTIC ±2	0	0.33
GPR20	High-level amplification	GISTIC ±2	2	0.33
CHRAC1	High-level amplification	GISTIC ±2	0	0.33
MIR6893	High-level amplification	GISTIC ±2	0	0.33



235 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); PTEN → Phosphatidylethanolamine (DB04327); OPLAH → Glutamic acid (DB00142); GPT → Pyridoxal phosphate (DB00114); GPT → Glutamic acid (DB00142); GPT → Alanine (DB00160); GPT → Phenelzine (DB00780); TOP1MT → Irinotecan (DB00762); TOP1MT → Topotecan (DB01030) … and 73 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PF	50	I	missing	0.73	no progression, followed 14.4 mo
TCGA-B6-A0X1	48	II	Basal	0.72	progressed at 16.1 mo (group B)
TCGA-BH-A0C0	62	II	LumB	0.72	no progression, followed 41.8 mo
TCGA-EW-A1PD	61	II	missing	0.71	no progression, followed 13.9 mo
TCGA-EW-A1P1	68	III	Normal	0.70	progressed at 33.3 mo (group C)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

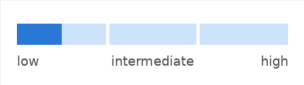
Sample: **TCGA-E2-A10B-01** · Aneuploidy score: **35** · TMB (nonsyn., /Mb): **1.53** · MSIsensor score: **0.98** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

17th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

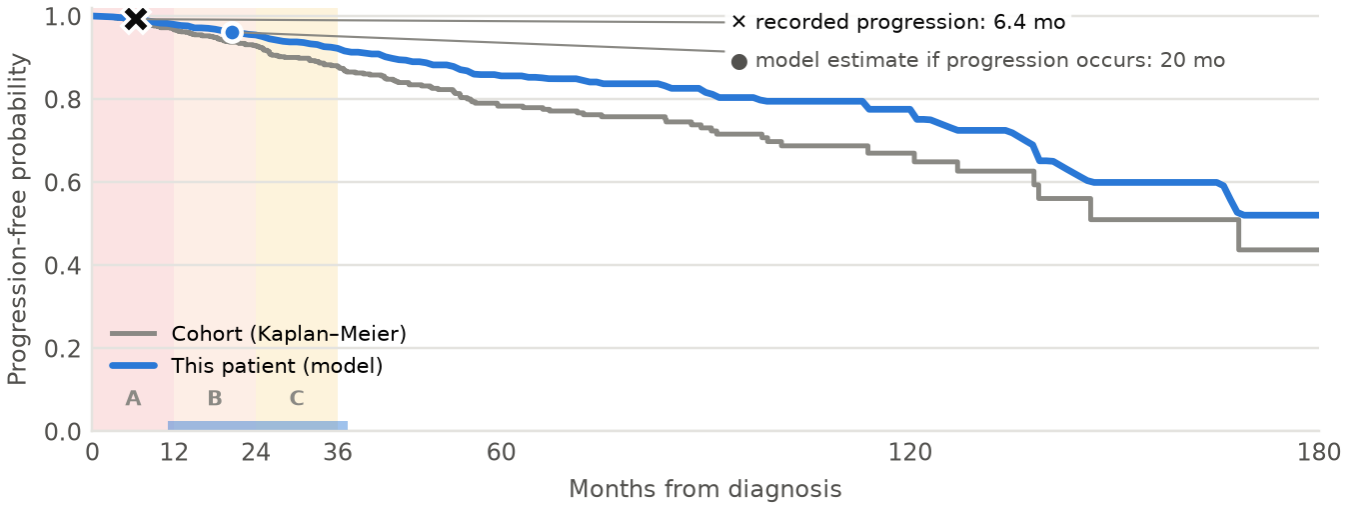
≈ 20 months

Window 12–24 months · IQR 11–38 mo
if progression occurs

Recorded progression

6.4 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 14 mo



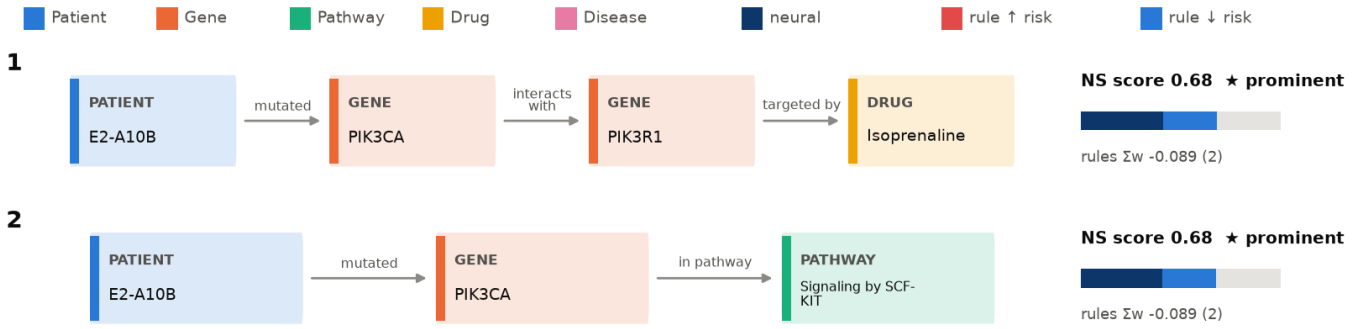
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **17th risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Isoprenaline**; the learned formulas it instantiates point to slower progression (Σw -0.089). Progression was recorded at 6.4 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

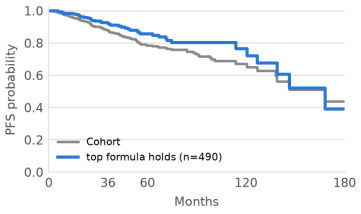


2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A10B** mutated → **PIK3CA** interacts with → **PIK3R1** targeted by → **Isoprenaline**	69	↓ **slower progression** · Σw -0.089 over 2 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.74, n = 140 carriers	**0.68**
2 ★	**TCGA-E2-A10B** mutated → **PIK3CA** in pathway → **Signaling by SCF-KIT**	68	↓ **slower progression** · Σw -0.089 over 2 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.98, n = 373 carriers	**0.68**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

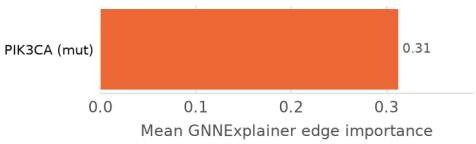
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.31



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YC	59	II	LumA	0.99	progressed at 24.9 mo (group C)
TCGA-AR-A0TR	68	II	LumA	0.98	no progression, followed 5.3 mo
TCGA-EW-A1P5	77	II	LumA	0.96	no progression, followed 23.1 mo
TCGA-E9-A227	42	II	LumA	0.93	no progression, followed 32.1 mo
TCGA-BH-A0BJ	41	II	LumA	0.93	no progression, followed 21.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-EW-A1P8-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **1.23** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

84th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 28%

Model-estimated progression time

≈ 17 months

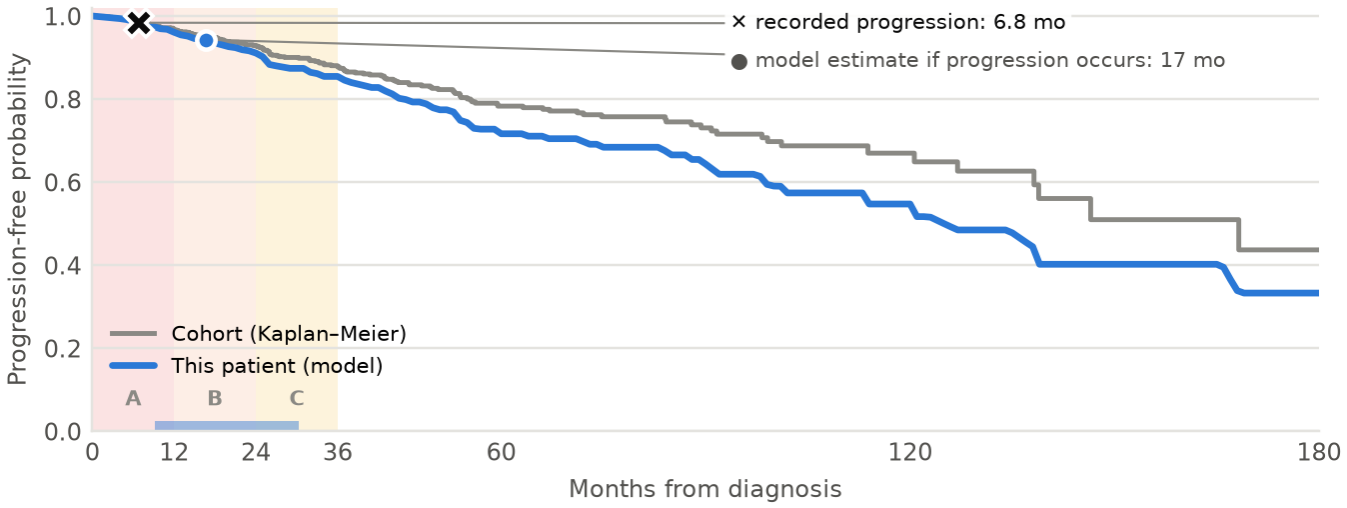
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

6.8 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 10 mo



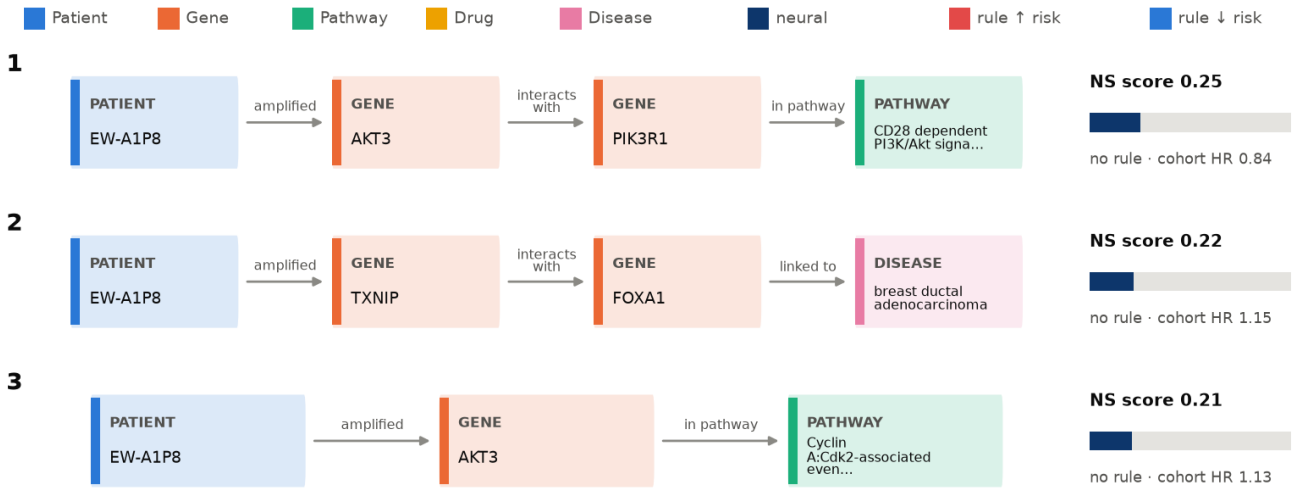
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a Basal tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **84th risk percentile** (high tier). The strongest learned formula that holds for them (*Stage(p, III)* ∧ *Amplified(p, FAM72C)* ⇒ *Progress(p)*) raises the progression hazard (weight +0.040 per SD). The highest-scoring knowledge-graph path runs from **AKT3** to **CD28 dependent PI3K/Akt signaling**. Progression was recorded at 6.8 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

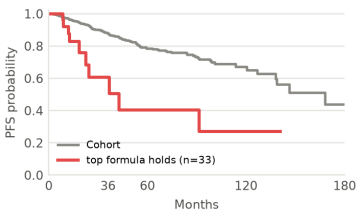


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-EW-A1P8** amplified → **AKT3** interacts with → **PIK3R1** in pathway → **CD28 dependent PI3K/Akt signaling**	42	No learned formula fires on this path; its score is neural only. cohort: HR 0.84, n = 413 carriers	**0.25**
2	**TCGA-EW-A1P8** amplified → **TXNIP** interacts with → **FOXA1** linked to → **breast ductal adenocarcinoma**	36	No learned formula fires on this path; its score is neural only. cohort: HR 1.15, n = 519 carriers	**0.22**
3	**TCGA-EW-A1P8** amplified → **AKT3** in pathway → **Cyclin A:Cdk2-associated events at S phase entry**	35	No learned formula fires on this path; its score is neural only. cohort: HR 1.13, n = 270 carriers	**0.21**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

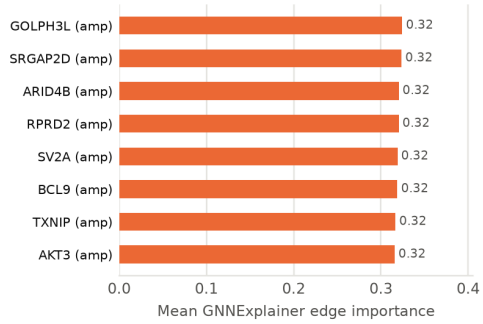
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.040 ↑ risk	Stage(p, III) ∧ Amplified(p, FAM72C) ⇒ Progress(p)	mined pair	33 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R282W	57	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
GOLPH3L	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
ARID4B	High-level amplification	GISTIC ±2	3	0.32
RPRD2	High-level amplification	GISTIC ±2	1	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32
BCL9	High-level amplification	GISTIC ±2	3	0.32
TXNIP	High-level amplification	GISTIC ±2	4	0.32
AKT3	High-level amplification	GISTIC ±2	24	0.32
PCNXL2	Mutation	p.R298P	0	0.32
ITGA10	High-level amplification	GISTIC ±2	3	0.31



27 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); SV2A → Levetiracetam (DB01202) … and 8 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1ND	75	II	Basal	0.68	no progression, followed 41.6 mo
TCGA-BH-A0E0	38	III	Basal	0.68	no progression, followed 4.4 mo
TCGA-A8-A07U	66	III	Basal	0.67	no progression, followed 25.0 mo
TCGA-BH-A0B3	53	II	Basal	0.65	no progression, followed 39.6 mo
TCGA-A2-A0T2	66	IV	Basal	0.64	progressed at 8.4 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**70**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

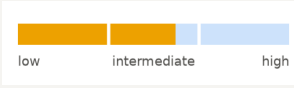
Sample: **TCGA-BH-A18T-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.43** · MSIsensor score: **1.63** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

58th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

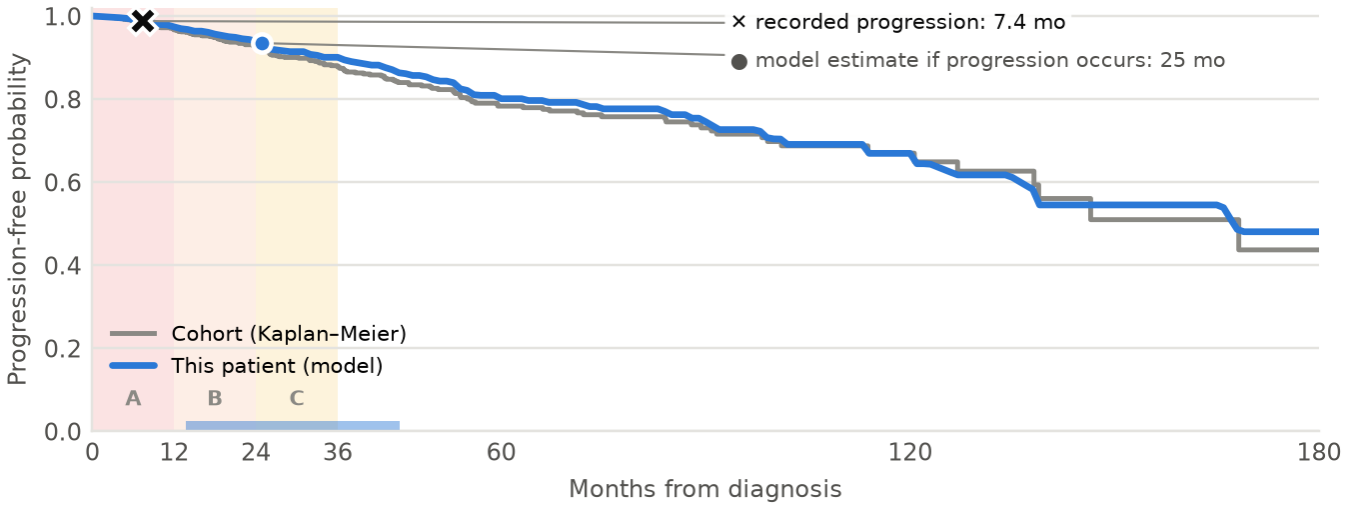
≈ 25 months

Window 24–36 months · IQR 14–45 mo
if progression occurs

Recorded progression

7.4 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 17 mo



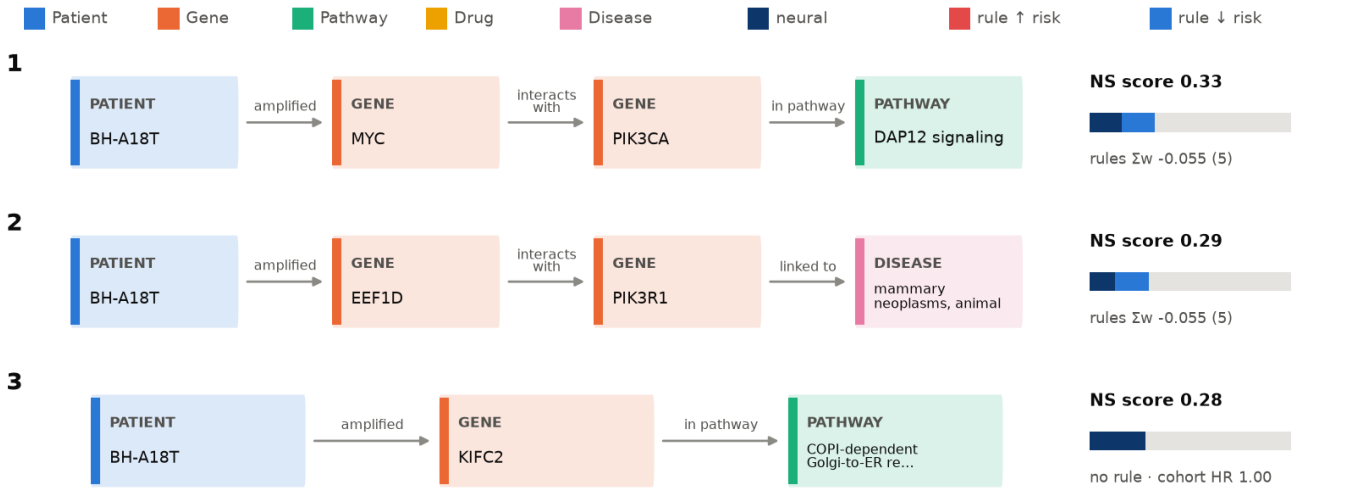
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 70-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **58th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **MYC** to **DAP12 signaling**; the learned formulas it instantiates point to slower progression (Σw -0.055). Progression was recorded at 7.4 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

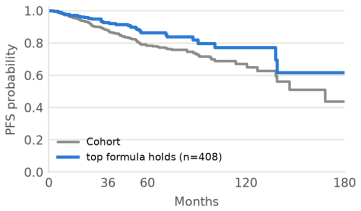


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-BH-A18T** amplified → **MYC** interacts with → **PIK3CA** in pathway → **DAP12 signaling**	26	↓ **slower progression** · Σw -0.055 over 5 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.76, n = 334 carriers	**0.33**
2	**TCGA-BH-A18T** amplified → **EEF1D** interacts with → **PIK3R1** linked to → **mammary neoplasms, animal**	21	↓ **slower progression** · Σw -0.055 over 5 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 1.16, n = 534 carriers	**0.29**
3	**TCGA-BH-A18T** amplified → **KIFC2** in pathway → **COPI-dependent Golgi-to-ER retrograde traffic**	46	No learned formula fires on this path; its score is neural only. cohort: HR 1.00, n = 187 carriers	**0.28**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

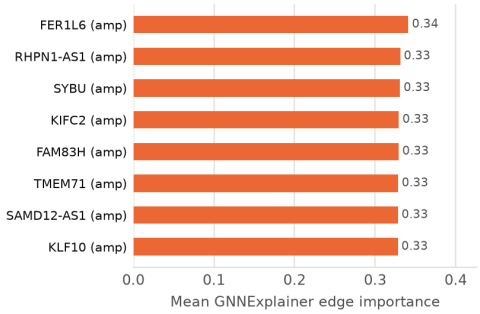
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.004 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	252 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
TP53 · driver	Mutation	p.Y107D	57	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
FOXA1 · driver	Mutation	p.A153V	9	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
FER1L6	High-level amplification	GISTIC ±2	0	0.34
RHPN1-AS1	High-level amplification	GISTIC ±2	0	0.33
SYBU	High-level amplification	GISTIC ±2	0	0.33
KIFC2	High-level amplification	GISTIC ±2	2	0.33
FAM83H	High-level amplification	GISTIC ±2	0	0.33
TMEM71	High-level amplification	GISTIC ±2	0	0.33
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.33



316 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 99 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0FX	52	II	Basal	0.89	no progression, followed 0.3 mo
TCGA-BH-A0DL	64	II	Basal	0.89	no progression, followed 78.3 mo
TCGA-E2-A150	48	II	Basal	0.89	no progression, followed 63.6 mo
TCGA-E2-A14R	62	II	Basal	0.87	progressed at 31.9 mo (group C)
TCGA-EW-A1P4	43	II	Basal	0.86	no progression, followed 29.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Metaplastic Breast Cancer**
Sex	**Female**	OncoTree code	**MBC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**AMR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8575/3**

Sample: **TCGA-AC-A2QJ-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

95th

High risk tier · fused neuro-symbolic model



Probability of progression

20% by 36 mo

12 mo 5% · 24 mo 12% · 60 mo 37%

Model-estimated progression time

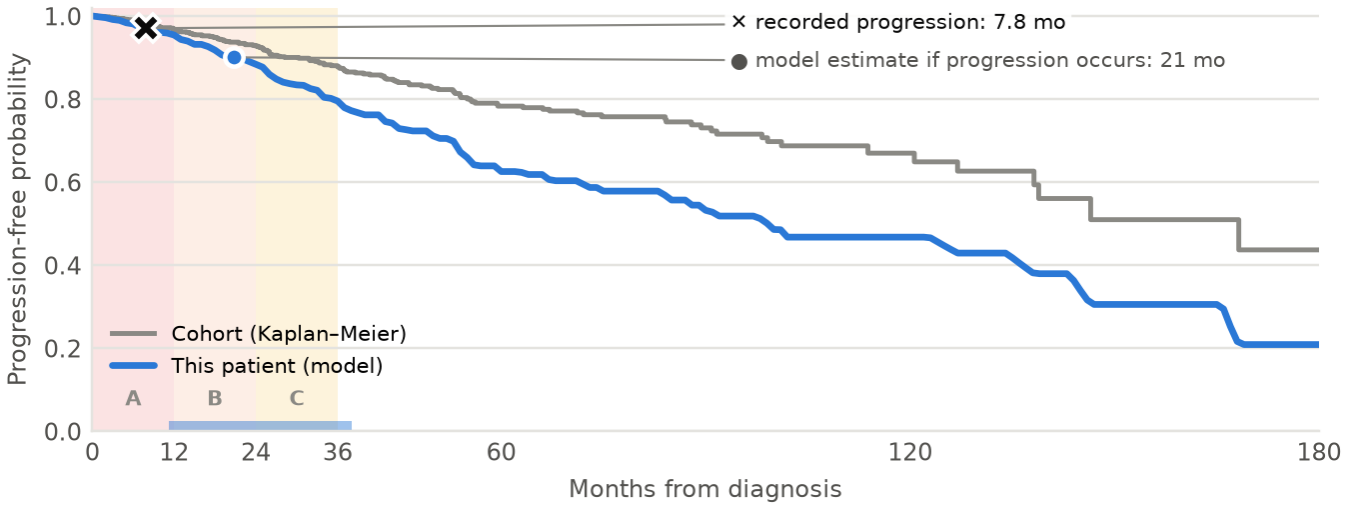
≈ 21 months

Window 12–24 months · IQR 11–38 mo
if progression occurs

Recorded progression

7.8 months

Group A · Early progression
✗ recorded window differs · |error| 13 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a subtype-unassigned tumour, AJCC stage III (T4, N0, M0). The fused neuro-symbolic model places them in the **95th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N0) ⇒ Progress(p)*) lowers the progression hazard (weight -0.304 per SD). Progression was recorded at 7.8 months; the model's estimate given progression was 21 months.

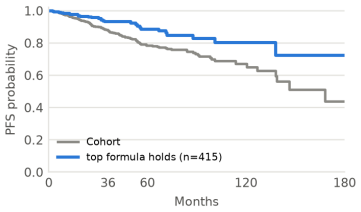
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
+0.134 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	27 pts
+0.078 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	79 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A3J0	62	II	missing	0.65	no progression, followed 10.3 mo
TCGA-E9-A5UP	63	II	missing	0.65	no progression, followed 26.4 mo
TCGA-A7-A3IY	71	I	missing	0.62	no progression, followed 11.3 mo
TCGA-BH-A1FR	73	III	LumA	0.61	progressed at 54.0 mo (group D)
TCGA-A8-A07S	73	II	missing	0.61	no progression, followed 8.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A8-A091 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

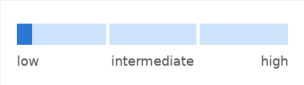
Sample: **TCGA-A8-A091-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.28** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

6th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 1% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

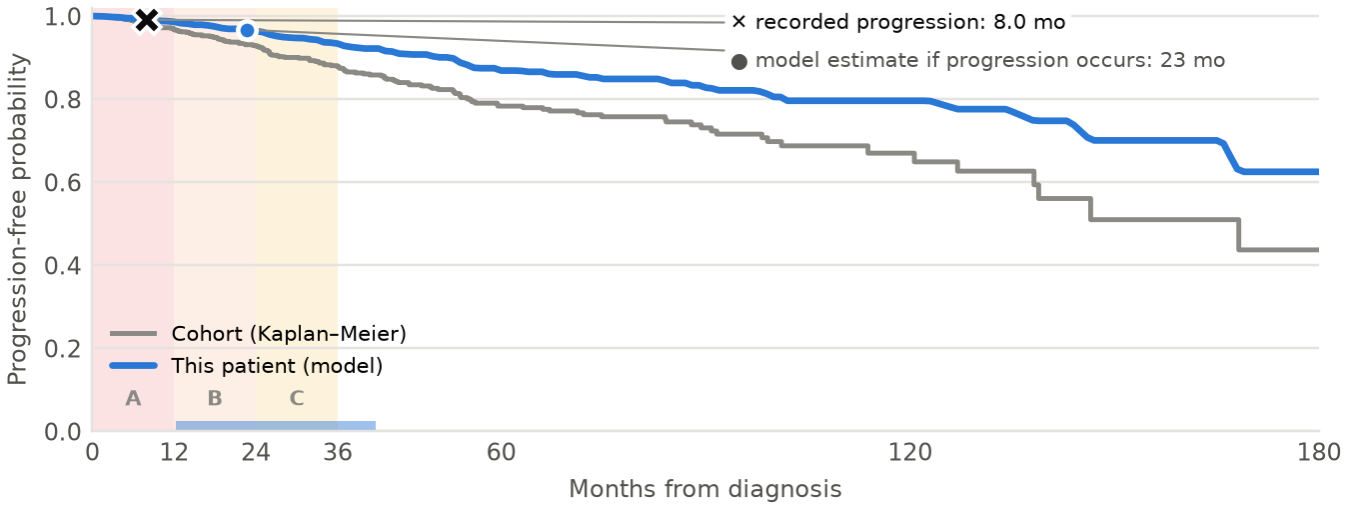
≈ 23 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

8.0 months

Group A · Early progression
✗ recorded window differs · |error| 15 mo



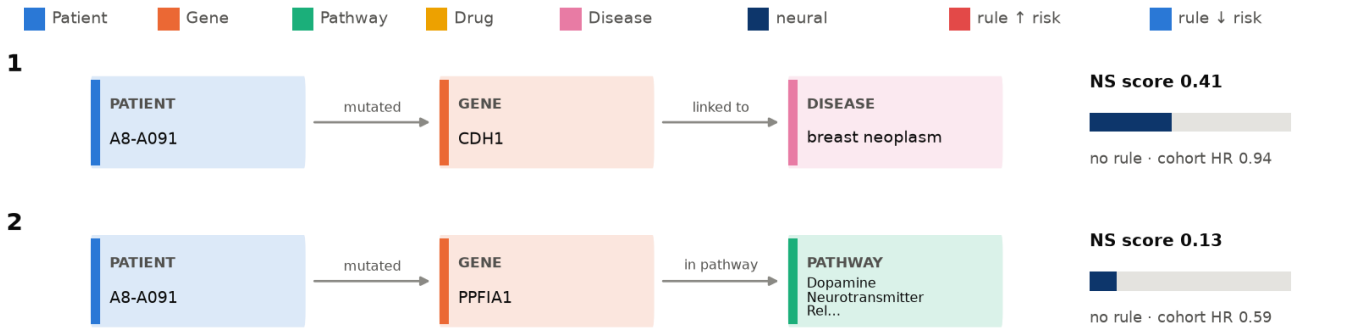
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **6th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **CDH1** to **breast neoplasm**. Progression was recorded at 8.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

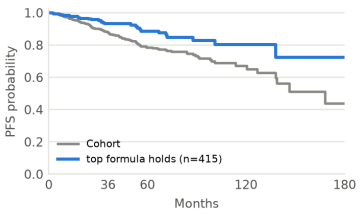


None of the 23 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A8-A091** mutated → **CDH1** linked to → **breast neoplasm**	68	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 792 carriers	**0.41**
2	**TCGA-A8-A091** mutated → **PPFIA1** in pathway → **Dopamine Neurotransmitter Release Cycle**	22	No learned formula fires on this path; its score is neural only. cohort: HR 0.59, n = 129 carriers	**0.13**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

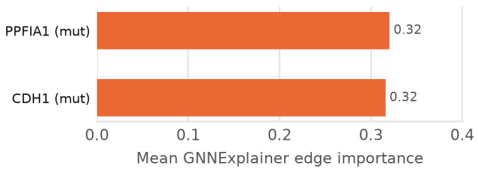
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
-0.032 ↓ risk	NStage(p, N0) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	194 pts
-0.016 ↓ risk	Subtype(p, LumA) ⇒ Progress(p)	single atom	408 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.E243K	17	0.32
PPFIA1	Mutation	p.A638T	6	0.32



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A12B	63	II	LumA	0.87	no progression, followed 98.3 mo
TCGA-B6-A2IU	62	II	LumA	0.86	progressed at 138.9 mo (group D)
TCGA-E9-A1R6	63	II	LumA	0.83	no progression, followed 11.1 mo
TCGA-GM-A2DM	57	II	LumA	0.82	no progression, followed 106.1 mo
TCGA-AR-A2LN	65	II	LumA	0.80	no progression, followed 38.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A0T2 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**66**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A0T2-01** · Aneuploidy score: **34** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **1.48** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

72% by 36 mo

12 mo 23% · 24 mo 50% · 60 mo 93%

Model-estimated progression time

≈ 9 months

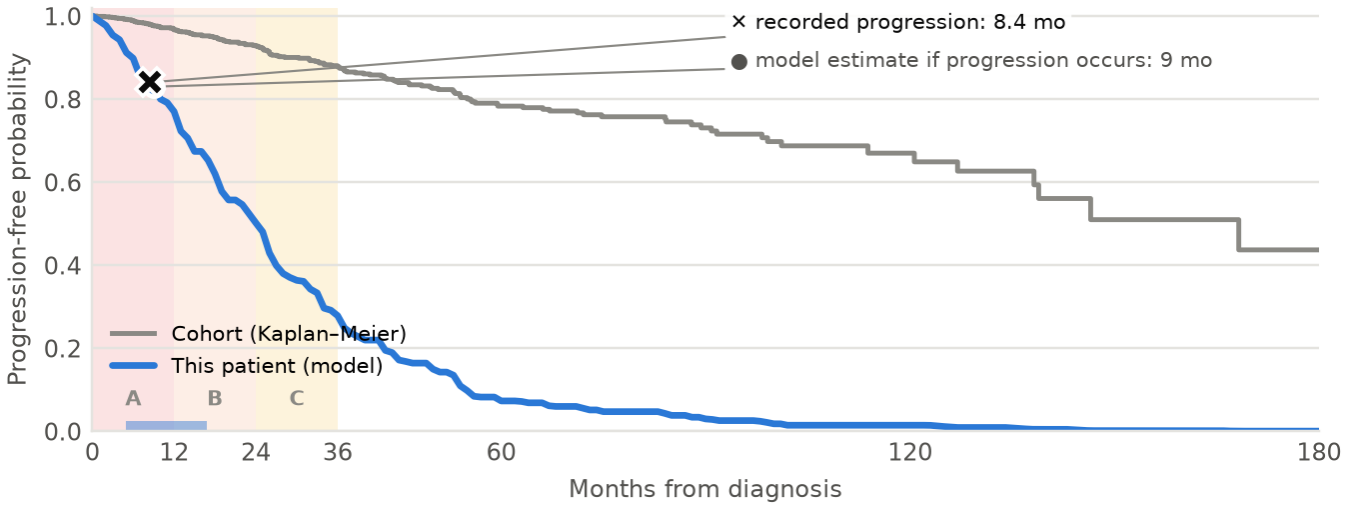
Window ≤ 12 months · IQR 5–17 mo if progression occurs

Recorded progression

8.4 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 1 mo



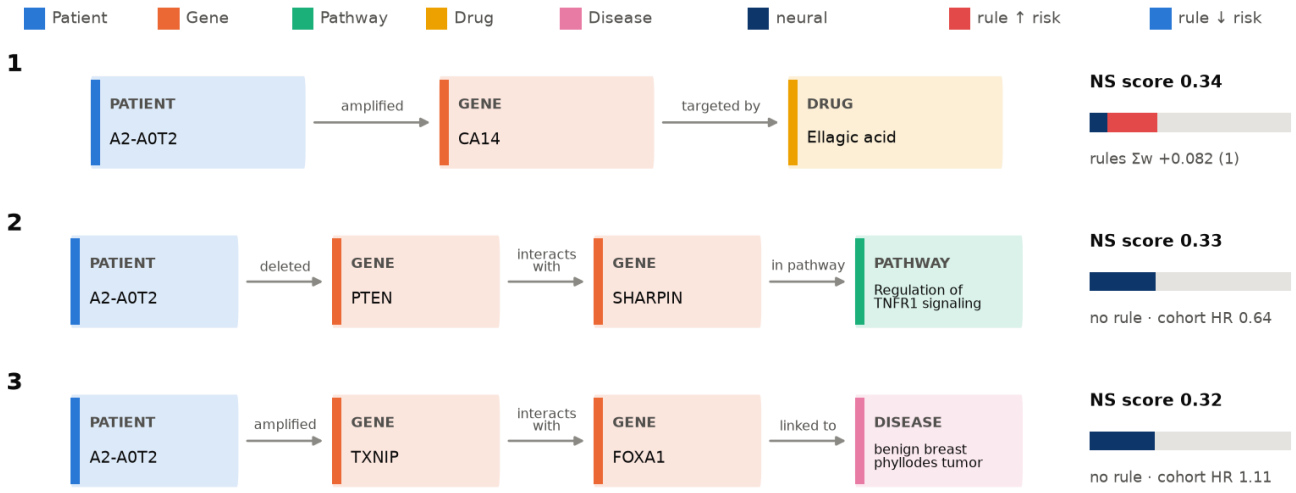
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 66-year-old patient has a Basal tumour, AJCC stage IV (T3, N3, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *TargetedBy(g, 'Ellagic acid')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.082 per SD). The highest-scoring knowledge-graph path runs from **CA14** to **Ellagic acid**; the learned formulas it instantiates point to faster progression (Σw +0.082). Progression was recorded at 8.4 months; the model's estimate given progression was 9 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

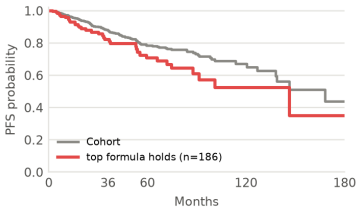


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-A2-A0T2** amplified → **CA14** targeted by → **Ellagic acid**	15	↑ **faster progression** · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 1.75, n = 186 carriers	**0.34**
2	**TCGA-A2-A0T2** deleted → **PTEN** interacts with → **SHARPIN** in pathway → **Regulation of TNFR1 signaling**	55	No learned formula fires on this path; its score is neural only. cohort: HR 0.64, n = 208 carriers	**0.33**
3	**TCGA-A2-A0T2** amplified → **TXNIP** interacts with → **FOXA1** linked to → **benign breast phyllodes tumor**	54	No learned formula fires on this path; its score is neural only. cohort: HR 1.11, n = 737 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

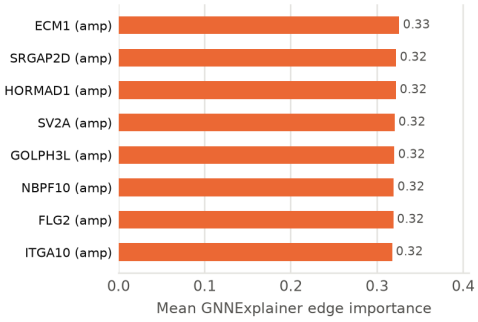
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.077 ↑ risk	NStage(p, N3) ⇒ Progress(p)	single atom	61 pts
+0.068 ↑ risk	Radiation(p, no) ∧ Altered(p, g) ∧ InPathway(g, 'Mitochondrial tRNA aminoacylation') ⇒ Progress(p) [KG]	RL-proposed	40 pts
+0.023 ↑ risk	Radiation(p, no) ⇒ Progress(p)	single atom	348 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.32
TP53 · driver	Mutation	p.Y220C	57	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
ECM1	High-level amplification	GISTIC ±2	1	0.33
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
HORMAD1	High-level amplification	GISTIC ±2	0	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32
GOLPH3L	High-level amplification	GISTIC ±2	0	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
FLG2	High-level amplification	GISTIC ±2	1	0.32
ITGA10	High-level amplification	GISTIC ±2	3	0.32
ARNT	High-level amplification	GISTIC ±2	6	0.32



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); SV2A → Levetiracetam (DB01202) … and 20 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0E0	38	III	Basal	0.70	no progression, followed 4.4 mo
TCGA-EW-A1P8	58	III	Basal	0.64	progressed at 6.8 mo (group A)
TCGA-BH-A0B3	53	II	Basal	0.62	no progression, followed 39.6 mo
TCGA-BH-A0AV	52	I	Basal	0.62	no progression, followed 59.8 mo
TCGA-AC-A2BK	78	III	Basal	0.59	no progression, followed 73.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-LL-A5YM · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**88**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**—**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8541/3**

Sample: **TCGA-LL-A5YM-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0.2** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

96th

High risk tier · fused neuro-symbolic model



Probability of progression

23% by 36 mo

12 mo 5% · 24 mo 13% · 60 mo 41%

Model-estimated progression time

≈ 16 months

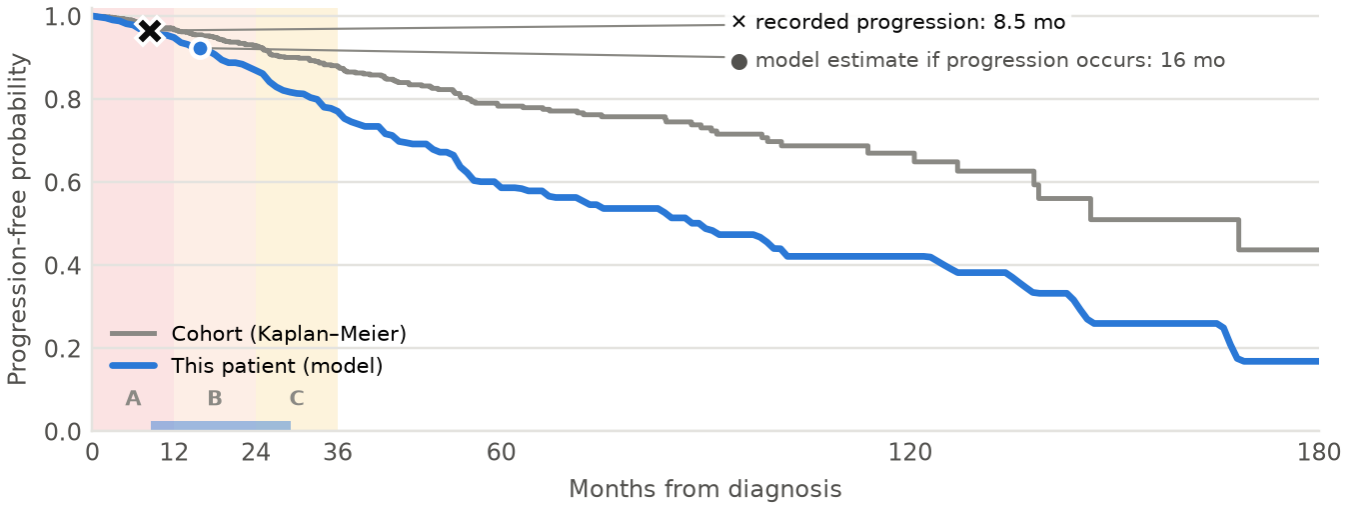
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

8.5 months

Group A · Early progression

✗ recorded window differs · |error| 7 mo



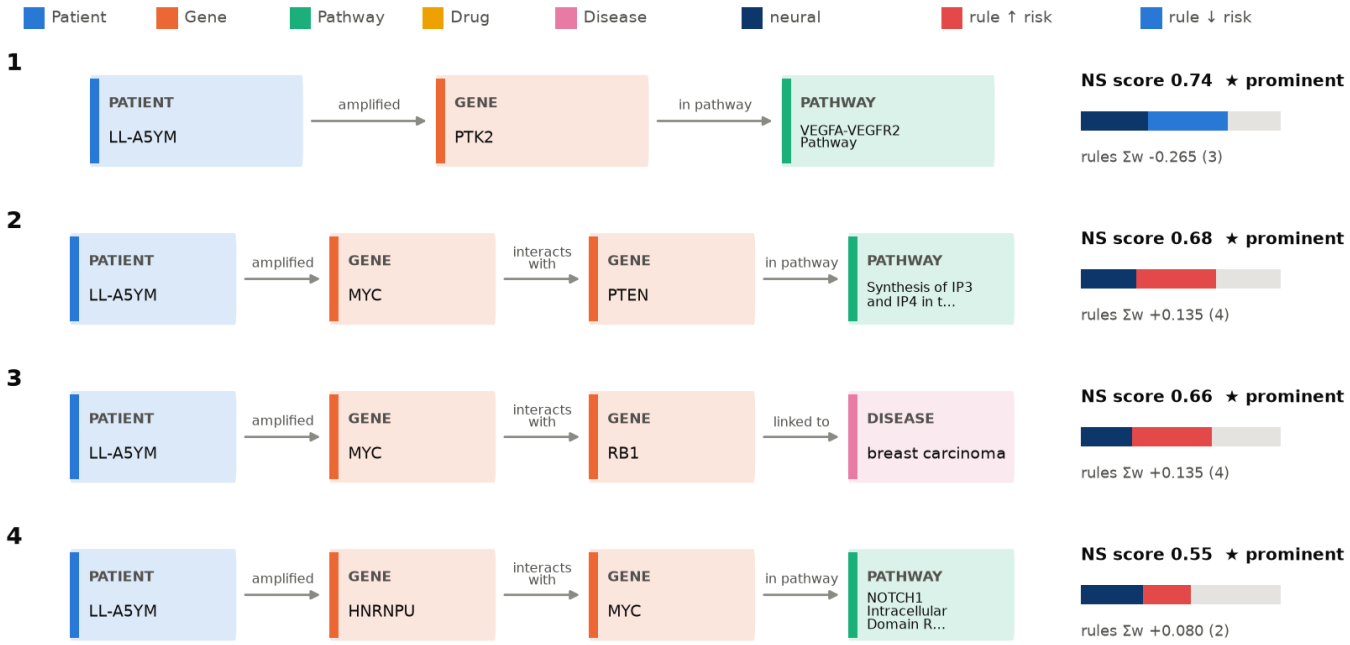
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 88-year-old patient has a LumB tumour, AJCC stage III (T4, missing, M0). The fused neuro-symbolic model places them in the **96th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *InPathway(g, 'VEGFA-VEGFR2 Pathway')* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Ion homeostasis')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.141 per SD). The highest-scoring knowledge-graph path runs from **PTK2** to **VEGFA-VEGFR2 Pathway**; the learned formulas it instantiates point to slower progression (Σw -0.265). Progression was recorded at 8.5 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

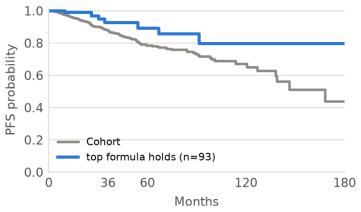


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-LL-A5YM** amplified → **PTK2** in pathway → **VEGFA-VEGFR2 Pathway**	56	↓ **slower progression** · Σw -0.265 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) cohort: HR 0.85, n = 375 carriers	**0.74**
2 ★	**TCGA-LL-A5YM** amplified → **MYC** interacts with → **PTEN** in pathway → **Synthesis of IP3 and IP4 in the cytosol**	46	↑ **faster progression** · Σw +0.135 over 4 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 1.31, n = 99 carriers	**0.68**
3 ★	**TCGA-LL-A5YM** amplified → **MYC** interacts with → **RB1** linked to → **breast carcinoma**	43	↑ **faster progression** · Σw +0.135 over 4 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 1.11, n = 782 carriers	**0.66**
4 ★	**TCGA-LL-A5YM** amplified → **HNRNPU** interacts with → **MYC** in pathway → **NOTCH1 Intracellular Domain Regulates Transcription**	52	↑ **faster progression** · Σw +0.080 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) cohort: HR 1.38, n = 129 carriers	**0.55**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

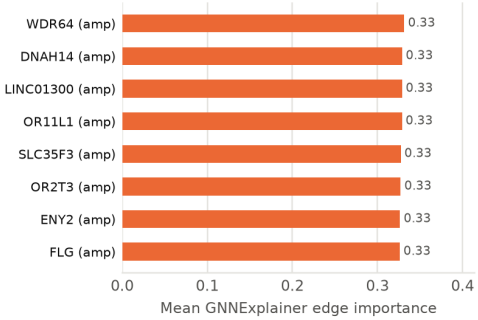
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.141 ↓ risk	Altered(p, g) ∧ InPathway(g, 'VEGFA-VEGFR2 Pathway') ∧ Altered(p, g) ∧ InPathway(g, 'Ion homeostasis') ⇒ Progress(p) [KG]	RL-proposed	93 pts
+0.134 ↑ risk	TStage(p, T4) ⇒ Progress(p)	single atom	27 pts
+0.101 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Mitochondrial translation initiation') ⇒ Progress(p) [KG]	single atom	104 pts
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PIP3 activates AKT signaling') ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Insulin-like Growth Factor (IGF) transport and uptake by Insulin-like Growth Factor Binding Proteins (IGFBPs)') ∧ Altered(p, g) ∧ InPathway(g, 'Interleukin-4 and Interleukin-13 signaling') ⇒ Progress(p) [KG]	RL-proposed	71 pts
+0.071 ↑ risk	Altered(p, g) ∧ InPathway(g, 'HDACs deacetylate histones') ⇒ Progress(p) [KG]	single atom	97 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P432Tfs*41	13	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
WDR64	High-level amplification	GISTIC ±2	0	0.33
DNAH14	High-level amplification	GISTIC ±2	0	0.33
LINC01300	High-level amplification	GISTIC ±2	0	0.33
OR11L1	High-level amplification	GISTIC ±2	1	0.33
SLC35F3	High-level amplification	GISTIC ±2	0	0.33
OR2T3	High-level amplification	GISTIC ±2	1	0.33
ENY2	High-level amplification	GISTIC ±2	1	0.33
FLG	High-level amplification	GISTIC ±2	1	0.33
FAM72C	High-level amplification	GISTIC ±2	0	0.33
PLD5	High-level amplification	GISTIC ±2	0	0.33



184 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); NID1 → Urokinase (DB00013); KCNK9 → Doxapram (DB00561); KCNK9 → Halothane (DB01159); CA14 → Benzthiazide (DB00562); CA14 → Cyclothiazide (DB00606); CA14 → Acetazolamide (DB00819); CA14 → Zonisamide (DB00909); CA14 → Ellagic acid (DB08846) … and 60 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XG	86	III	LumA	0.81	no progression, followed 14.7 mo
TCGA-AN-A0FK	88	III	LumA	0.80	no progression, followed 7.0 mo
TCGA-A8-A09B	58	III	LumA	0.76	no progression, followed 12.0 mo
TCGA-A8-A06Y	66	II	LumA	0.74	progressed at 26.0 mo (group C)
TCGA-AN-A0XP	69	III	LumA	0.73	no progression, followed 0.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EAS**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

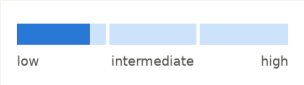
Sample: **TCGA-AO-A126-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.16** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

27th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

≈ 43 months

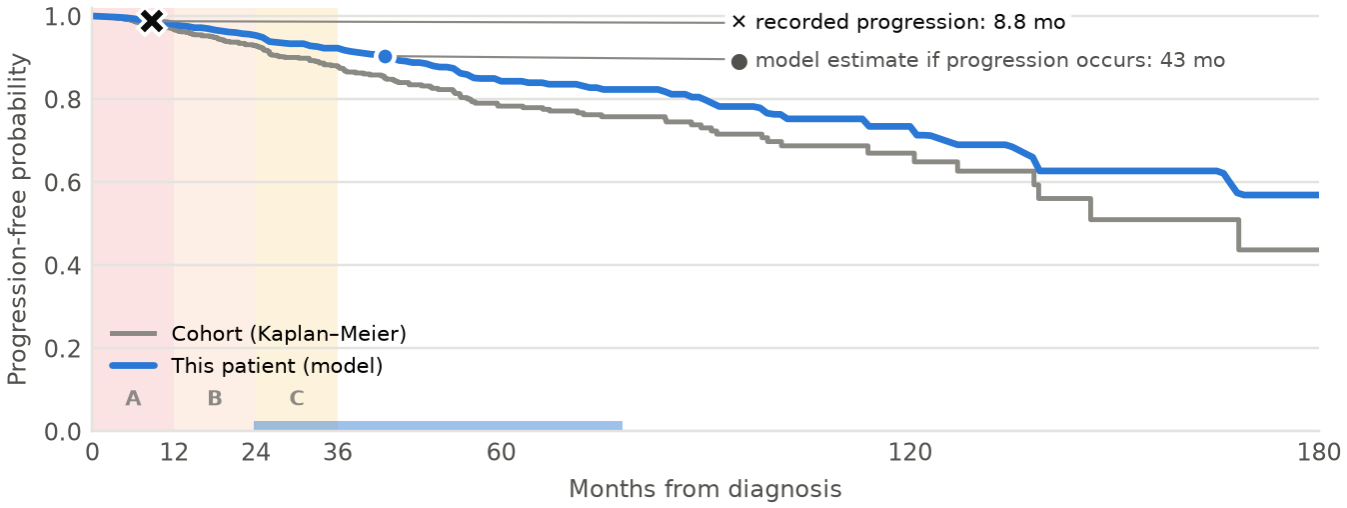
Window > 36 months · IQR 24–78 mo if progression occurs

Recorded progression

8.8 months

■ **Group A** · Early progression

✗ recorded window differs · |error| 34 mo



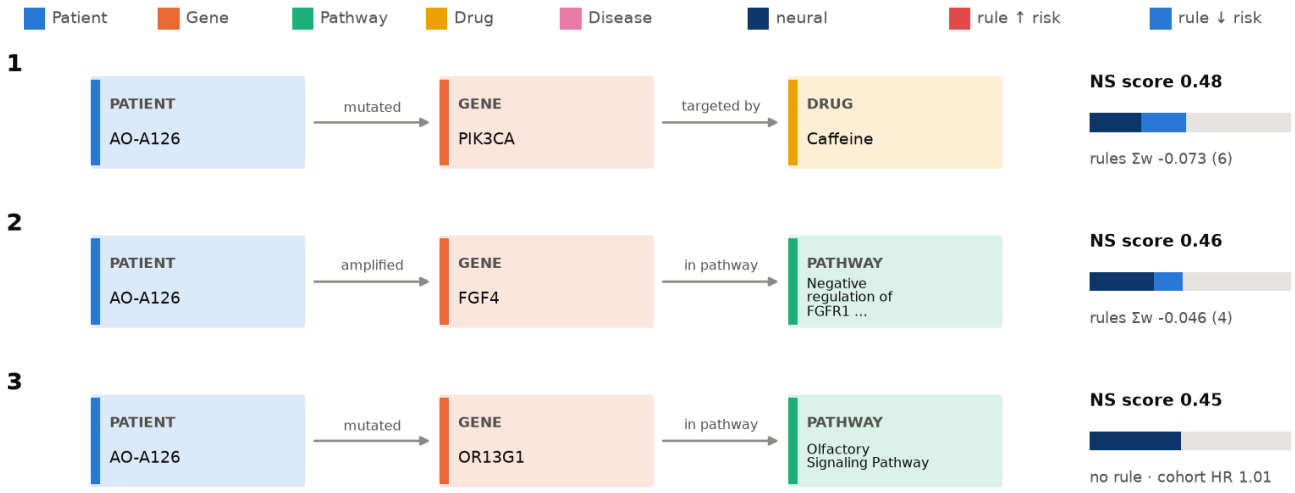
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **27th risk percentile** (low tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.099 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Caffeine**; the learned formulas it instantiates point to slower progression (Σw -0.073). Progression was recorded at 8.8 months; the model's estimate given progression was 43 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

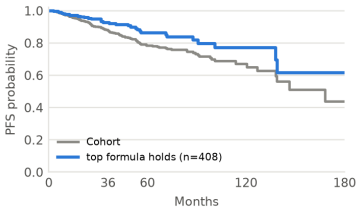


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AO-A126** mutated → **PIK3CA** targeted by → **Caffeine**	43	↓ **slower progression** · Σw -0.073 over 6 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.75, n = 324 carriers	**0.48**
2	**TCGA-AO-A126** amplified → **FGF4** in pathway → **Negative regulation of FGFR1 signaling**	54	↓ **slower progression** · Σw -0.046 over 4 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.91, n = 207 carriers	**0.46**
3	**TCGA-AO-A126** mutated → **OR13G1** in pathway → **Olfactory Signaling Pathway**	75	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 158 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

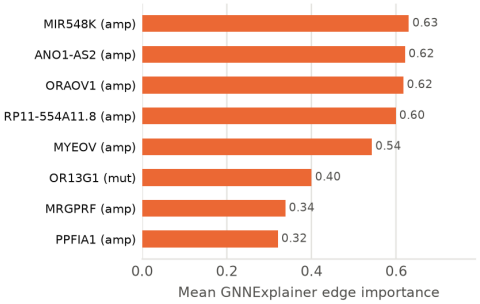
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.099 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	398 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.018 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ TargetedBy(g, 'ATP') ⇒ Progress(p) [KG]	mined pair	162 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.009 ↓ risk	NStage(p, N0) ∧ Altered(p, g) ∧ InPathway(g, 'Synthesis of PIPs at the plasma membrane') ⇒ Progress(p) [KG]	mined pair	212 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.63
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.62
ORAOV1	High-level amplification	GISTIC ±2	0	0.62
RP11-554A11.8	High-level amplification	GISTIC ±2	0	0.60
MYEOV	High-level amplification	GISTIC ±2	0	0.54
OR13G1	Mutation	p.I42S	1	0.40
MRGPRF	High-level amplification	GISTIC ±2	0	0.34
PPFIA1	High-level amplification	GISTIC ±2	6	0.32
KCNK9	Mutation	p.H188=	4	0.32
FGF19	High-level amplification	GISTIC ±2	13	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 8 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A26J	49	II	LumA	0.92	no progression, followed 20.6 mo
TCGA-E9-A1QZ	61	II	LumA	0.89	no progression, followed 24.8 mo
TCGA-C8-A12Y	44	II	LumA	0.88	no progression, followed 48.5 mo
TCGA-A8-A093	61	II	LumA	0.85	no progression, followed 18.0 mo
TCGA-BH-A18I	53	II	LumA	0.83	no progression, followed 35.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**Yes**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A1AR-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **1.57** · MSIsensor score: **0.03** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 34%

Model-estimated progression time

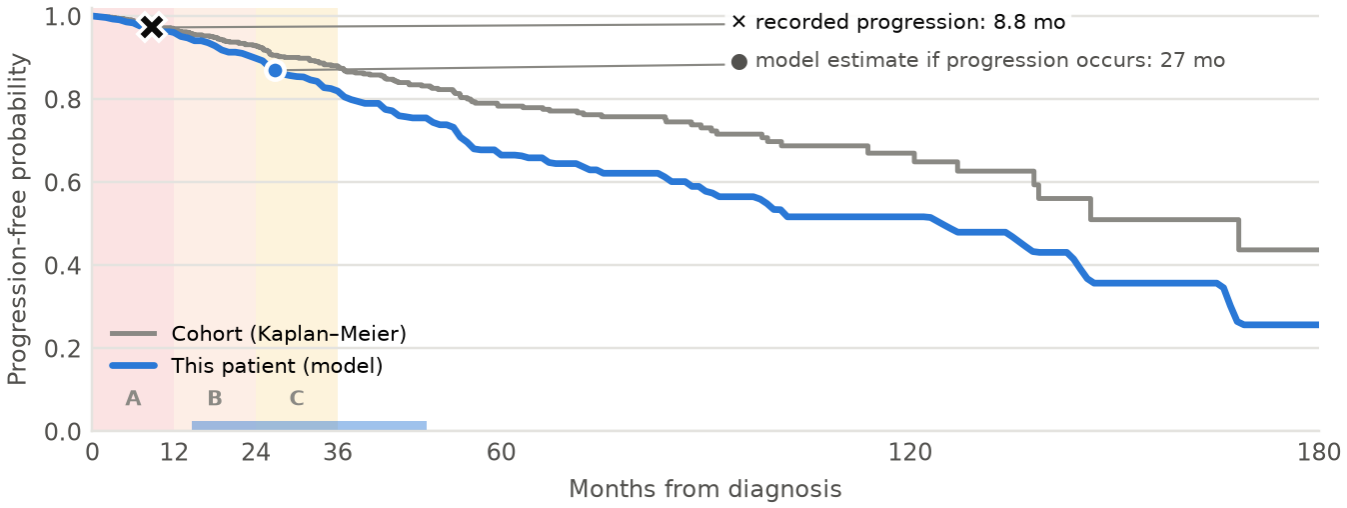
≈ 27 months

Window 24–36 months · IQR 15–49 mo
if progression occurs

Recorded progression

8.8 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 18 mo



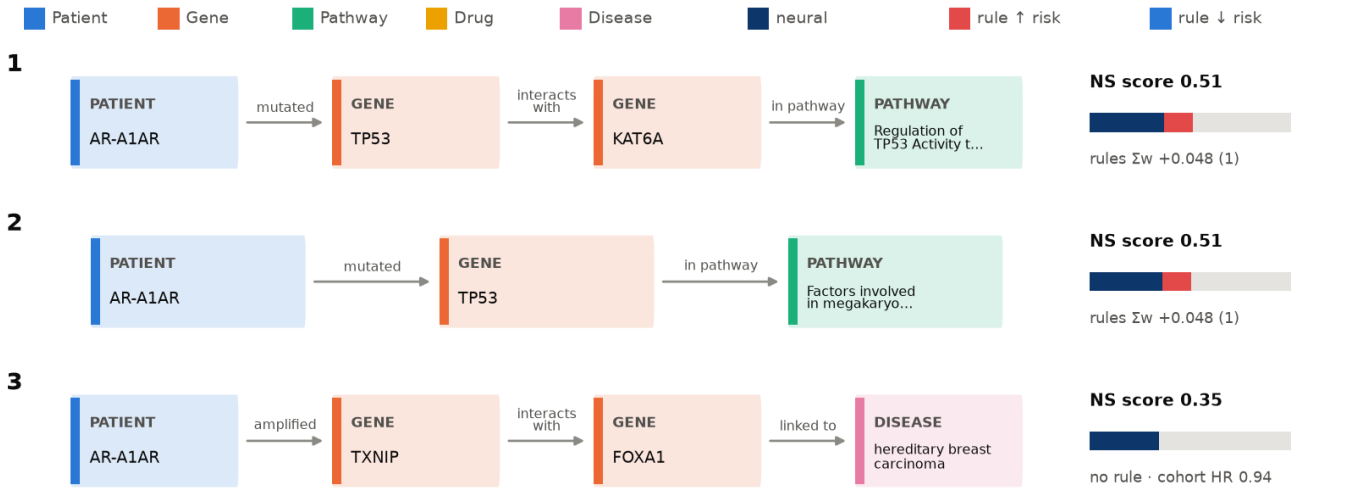
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a Basal tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p)*) raises the progression hazard (weight +0.082 per SD). The highest-scoring knowledge-graph path runs from **TP53** to **Regulation of TP53 Activity through Acetylation**; the learned formulas it instantiates point to faster progression (Σw +0.048). Progression was recorded at 8.8 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

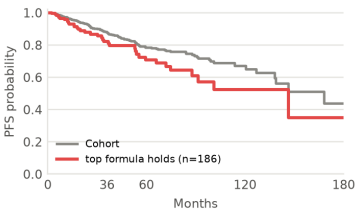


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A1AR** mutated → **TP53** interacts with → **KAT6A** in pathway → **Regulation of TP53 Activity through Acetylation**	61	↑ **faster progression** · Σw +0.048 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Transcriptional Regulation by VENTX') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) cohort: HR 0.95, n = 410 carriers	**0.51**
2	**TCGA-AR-A1AR** mutated → **TP53** in pathway → **Factors involved in megakaryocyte development and platelet production**	60	↑ **faster progression** · Σw +0.048 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Transcriptional Regulation by VENTX') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) cohort: HR 0.88, n = 424 carriers	**0.51**
3	**TCGA-AR-A1AR** amplified → **TXNIP** interacts with → **FOXA1** linked to → **hereditary breast carcinoma**	58	No learned formula fires on this path; its score is neural only. cohort: HR 0.94, n = 792 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

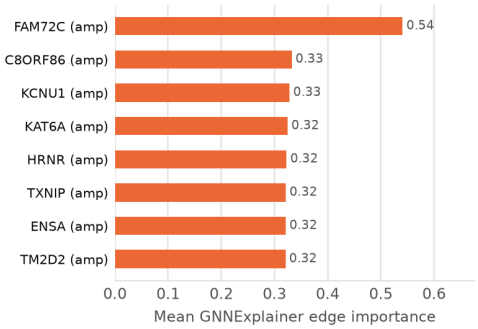
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.064 ↑ risk	Stage(p, III) ∧ Altered(p, g) ∧ InPathway(g, 'Integrin cell surface interactions') ⇒ Progress(p) [KG]	RL-proposed	59 pts
+0.048 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ∧ Altered(p, g) ∧ InPathway(g, 'Transcriptional Regulation by VENTX') ∧ Altered(p, g) ∧ InPathway(g, 'Factors involved in megakaryocyte development and platelet production') ⇒ Progress(p) [KG]	RL-proposed	64 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Homozygous deletion	GISTIC ±2	4	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
TP53 · driver	Mutation	p.S90Pfs*33	57	0.31
FAM72C	High-level amplification	GISTIC ±2	0	0.54
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
KCNU1	High-level amplification	GISTIC ±2	1	0.33
KAT6A	High-level amplification	GISTIC ±2	2	0.32
HRNR	High-level amplification	GISTIC ±2	3	0.32
TXNIP	High-level amplification	GISTIC ±2	4	0.32
ENSA	High-level amplification	GISTIC ±2	1	0.32
TM2D2	High-level amplification	GISTIC ±2	0	0.32



62 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 66 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0E6	69	I	Basal	0.79	no progression, followed 9.6 mo
TCGA-GM-A2DF	53	II	Basal	0.77	no progression, followed 70.8 mo
TCGA-AN-A0XN	68	III	LumA	0.76	no progression, followed 0.3 mo
TCGA-AR-A1AJ	83	I	Basal	0.75	no progression, followed 101.0 mo
TCGA-D8-A1XD	36	III	missing	0.69	no progression, followed 17.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**9020/3**

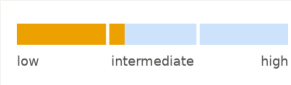
Sample: **TCGA-OL-A97C-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **2** · MSIsensor score: **0.05** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

40th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 17%

Model-estimated progression time

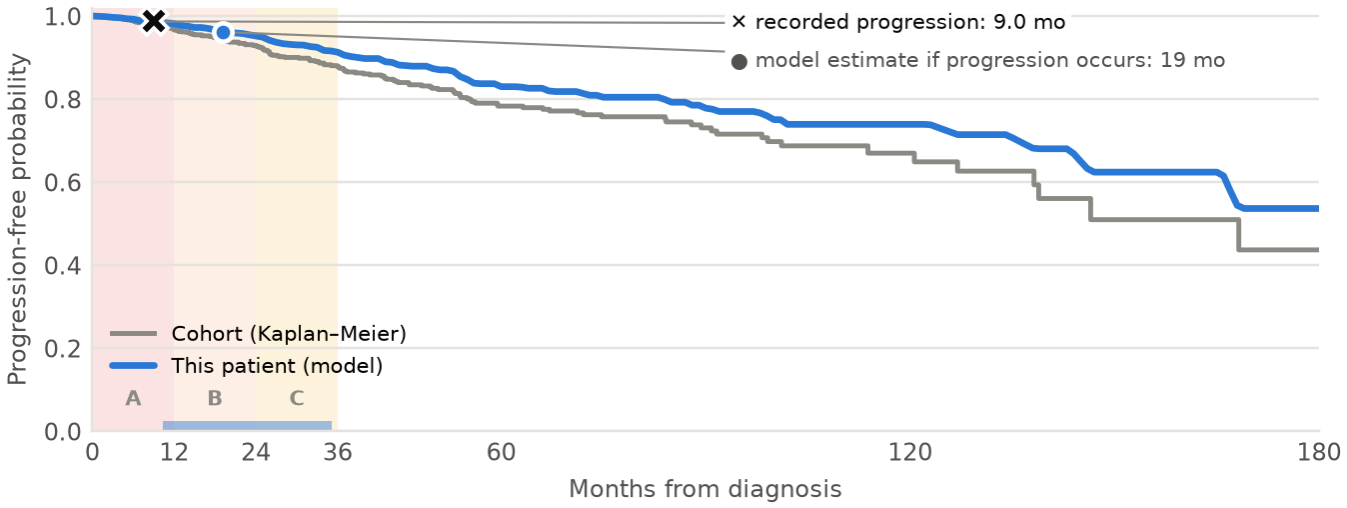
≈ 19 months

Window 12–24 months · IQR 10–35 mo
if progression occurs

Recorded progression

9.0 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 10 mo



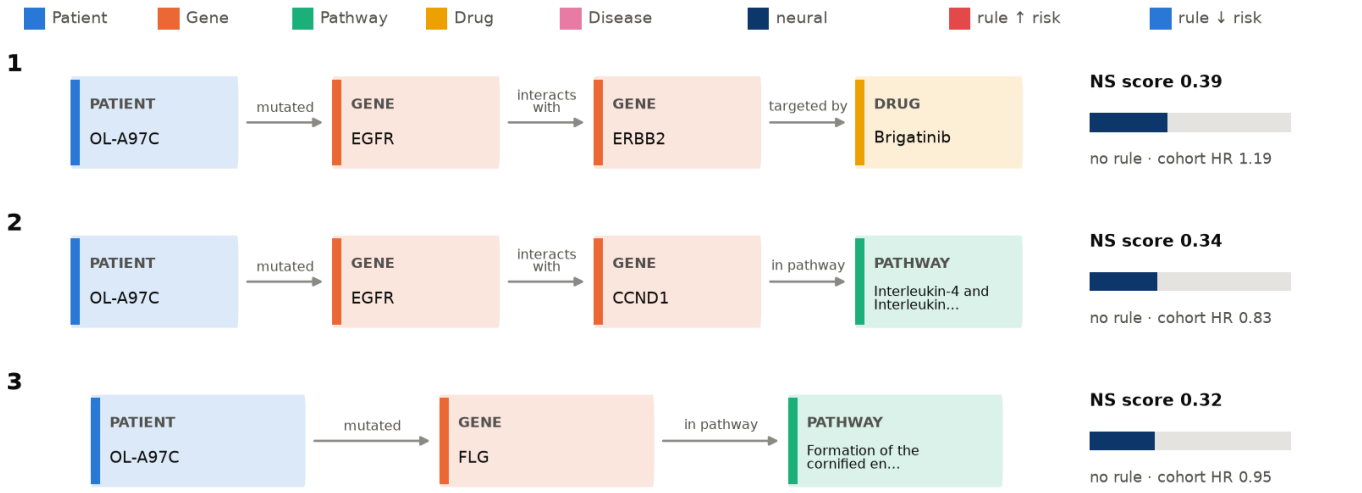
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a Normal tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **40th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.304 per SD). The highest-scoring knowledge-graph path runs from **EGFR** to **Brigatinib**. Progression was recorded at 9.0 months; the model's estimate given progression was 19 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

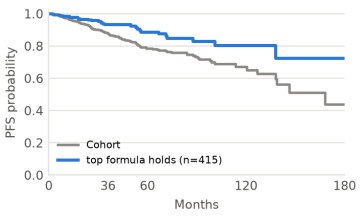


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-OL-A97C** mutated → **EGFR** interacts with → **ERBB2** targeted by → **Brigatinib**	65	No learned formula fires on this path; its score is neural only. cohort: HR 1.19, n = 148 carriers	**0.39**
2	**TCGA-OL-A97C** mutated → **EGFR** interacts with → **CCND1** in pathway → **Interleukin-4 and Interleukin-13 signaling**	56	No learned formula fires on this path; its score is neural only. cohort: HR 0.83, n = 516 carriers	**0.34**
3	**TCGA-OL-A97C** mutated → **FLG** in pathway → **Formation of the cornified envelope**	54	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 147 carriers	**0.32**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

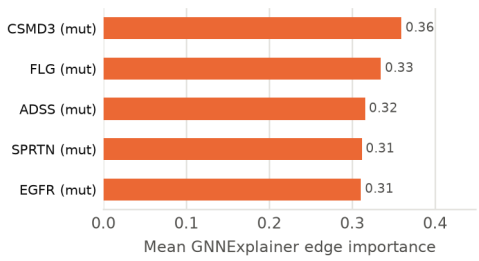
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.304 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	408 pts
+0.078 ↑ risk	Histology(p, other) ⇒ Progress(p)	single atom	79 pts
-0.034 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	481 pts
+0.023 ↑ risk	Radiation(p, no) ⇒ Progress(p)	single atom	348 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
EGFR · driver	Mutation	p.A289V	73	0.31
CSMD3	Mutation	p.L49V	0	0.36
FLG	Mutation	p.G2143E	1	0.33
ADSS	Mutation	3'UTR	0	0.32
SPRTN	Mutation	p.R140H	1	0.31



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

EGFR → Cetuximab (DB00002); EGFR → Lidocaine (DB00281); EGFR → Gefitinib (DB00317); EGFR → Erlotinib (DB00530); EGFR → Lapatinib (DB01259); EGFR → Panitumumab (DB01269); EGFR → Alvocidib (DB03496); EGFR → IGN311 (DB04988); EGFR → Matuzumab (DB05101); EGFR → Vandetanib (DB05294) … and 21 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A2LQ	59	II	Normal	0.73	no progression, followed 40.5 mo
TCGA-BH-A6R9	61	II	Normal	0.68	no progression, followed 5.3 mo
TCGA-AO-A1KO	46	II	Normal	0.67	no progression, followed 20.4 mo
TCGA-PL-A8LY	30	II	Normal	0.60	no progression, followed 0.3 mo
TCGA-AC-A7VC	56	II	Basal	0.58	no progression, followed 0.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**72**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**Yes**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

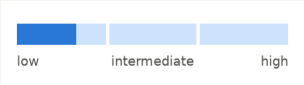
Sample: **TCGA-A7-A26H-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **3.67** · MSIsensor score: **0.18** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

22nd

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

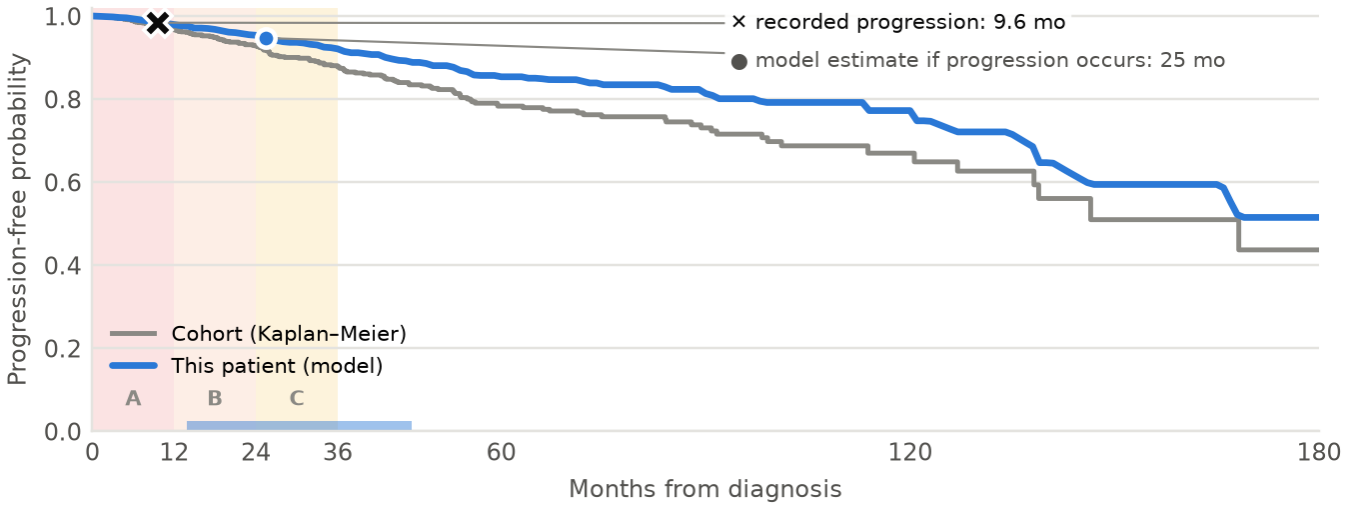
≈ 25 months

Window 24–36 months · IQR 14–47 mo
if progression occurs

Recorded progression

9.6 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 16 mo



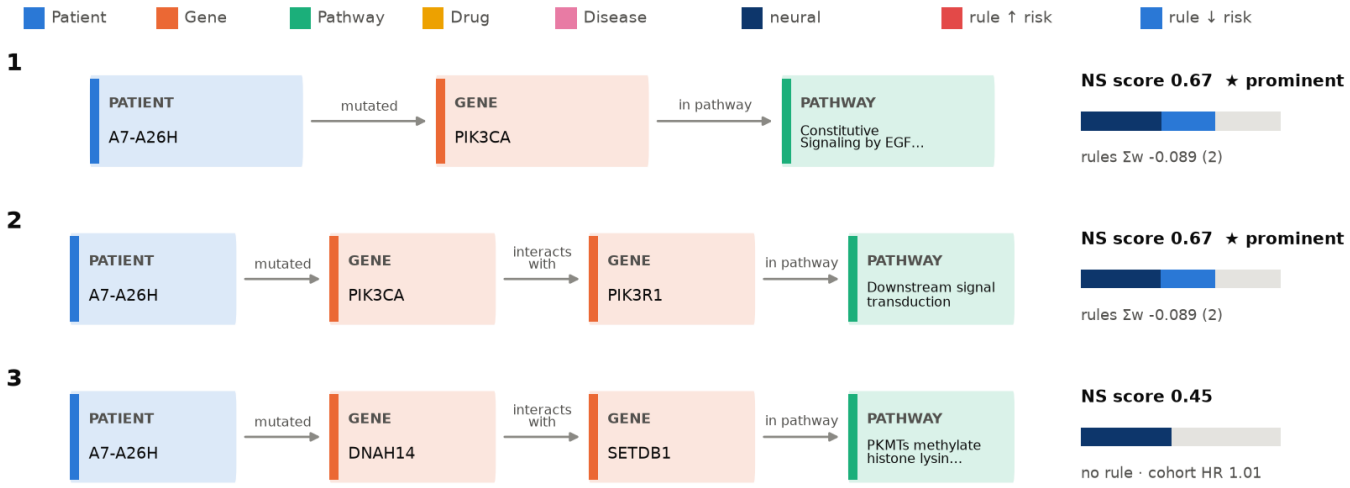
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 72-year-old patient has a LumA tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **22nd risk percentile** (low tier). The strongest learned formula that holds for them (*Stage(p, II) ⇒ Progress(p)*) lowers the progression hazard (weight -0.151 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **Constitutive Signaling by EGFRvIII**; the learned formulas it instantiates point to slower progression (Σw -0.089). Progression was recorded at 9.6 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

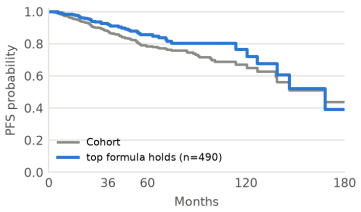


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-A7-A26H** mutated → **PIK3CA** in pathway → **Constitutive Signaling by EGFRvIII**	67	↓ **slower progression** · Σw -0.089 over 2 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.80, n = 329 carriers	**0.67**
2 ★	**TCGA-A7-A26H** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **Downstream signal transduction**	67	↓ **slower progression** · Σw -0.089 over 2 formula(s) Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) cohort: HR 0.98, n = 373 carriers	**0.67**
3	**TCGA-A7-A26H** mutated → **DNAH14** interacts with → **SETDB1** in pathway → **PKMTs methylate histone lysines**	75	No learned formula fires on this path; its score is neural only. cohort: HR 1.01, n = 324 carriers	**0.45**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

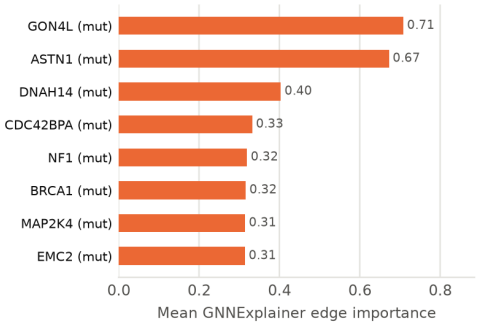
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.151 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	482 pts
-0.125 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	120 pts
-0.048 ↓ risk	Subtype(p, LumA) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	292 pts
-0.041 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'RAF/MAP kinase cascade') ⇒ Progress(p) [KG]	mined pair	337 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Mutation	p.L852=	6	0.32
BRCA1 · driver	Mutation	p.T688Vfs*12	29	0.32
MAP2K4 · driver	Mutation	p.L385P	12	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
GON4L	Mutation	p.E1520Q	0	0.71
ASTN1	Mutation	p.C699*	0	0.67
DNAH14	Mutation	p.*742*	0	0.40
CDC42BPA	Mutation	p.E889K	4	0.33
EMC2	Mutation	p.E57K	0	0.31



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A145	80	II	LumA	0.85	no progression, followed 13.5 mo
TCGA-BH-A0W5	77	II	LumA	0.84	no progression, followed 42.3 mo
TCGA-AO-A0JF	68	II	LumA	0.81	no progression, followed 65.1 mo
TCGA-E9-A1R0	58	II	LumA	0.81	no progression, followed 28.3 mo
TCGA-E2-A1L6	44	II	LumA	0.79	no progression, followed 54.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AR-A5QQ · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**68**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8575/3**

Sample: **TCGA-AR-A5QQ-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **2.23** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 32%

Model-estimated progression time

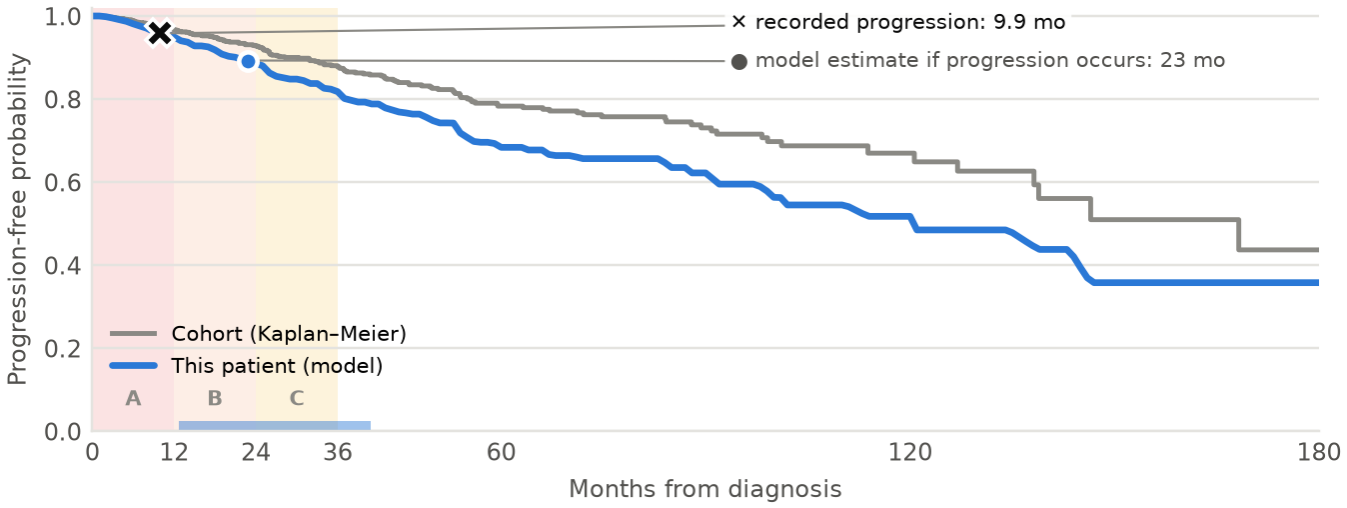
≈ 23 months

Window 12–24 months · IQR 13–41 mo
if progression occurs

Recorded progression

9.9 months

Group A · Early progression
✗ recorded window differs · |error| 13 mo



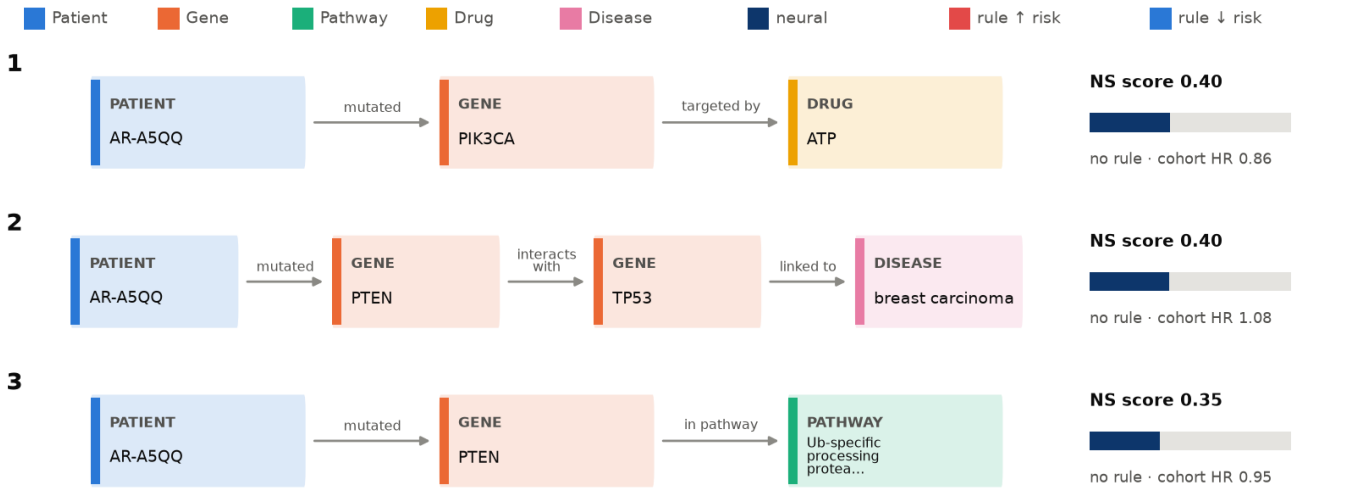
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 68-year-old patient has a Basal tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PIK3CA** to **ATP**. Progression was recorded at 9.9 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

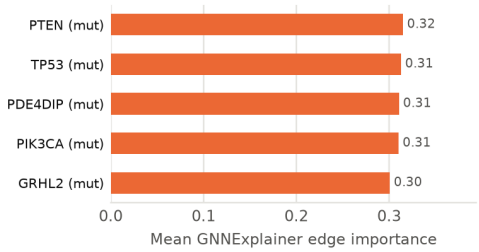
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-AR-A5QQ** mutated → **PIK3CA** targeted by → **ATP**	67	No learned formula fires on this path; its score is neural only. cohort: HR 0.86, n = 321 carriers	**0.40**
2	**TCGA-AR-A5QQ** mutated → **PTEN** interacts with → **TP53** linked to → **breast carcinoma**	66	No learned formula fires on this path; its score is neural only. cohort: HR 1.08, n = 782 carriers	**0.40**
3	**TCGA-AR-A5QQ** mutated → **PTEN** in pathway → **Ub-specific processing proteases**	58	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 460 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Mutation	p.R130Q	22	0.32
TP53 · driver	Mutation	p.R209Kfs*6	57	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PDE4DIP	Mutation	p.H445R	0	0.31
GRHL2	Mutation	p.D336Y	0	0.30



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3CA → ATP (DB00171) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IE	38	III	LumA	0.82	progressed at 27.4 mo (group C)
TCGA-C8-A133	65	III	LumA	0.68	no progression, followed 0.0 mo
TCGA-E9-A1NC	61	II	Basal	0.64	no progression, followed 39.6 mo
TCGA-A7-A26F	55	I	Basal	0.64	no progression, followed 24.3 mo
TCGA-BH-A0WA	82	I	Basal	0.62	no progression, followed 23.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-D8-A1Y1 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**80**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-D8-A1Y1-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **6.8** · MSIsensor score: **6.24** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

78th

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 26%

Model-estimated progression time

≈ 13 months

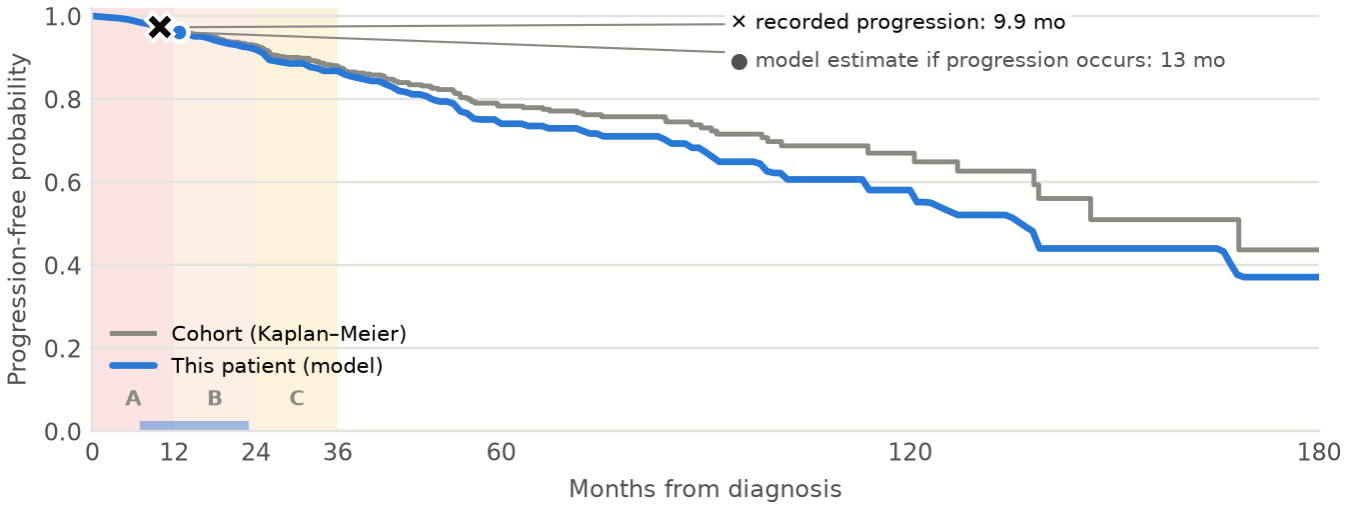
Window 12–24 months · IQR 7–23 mo if progression occurs

Recorded progression

9.9 months

Group A · Early progression

✗ recorded window differs · |error| 3 mo



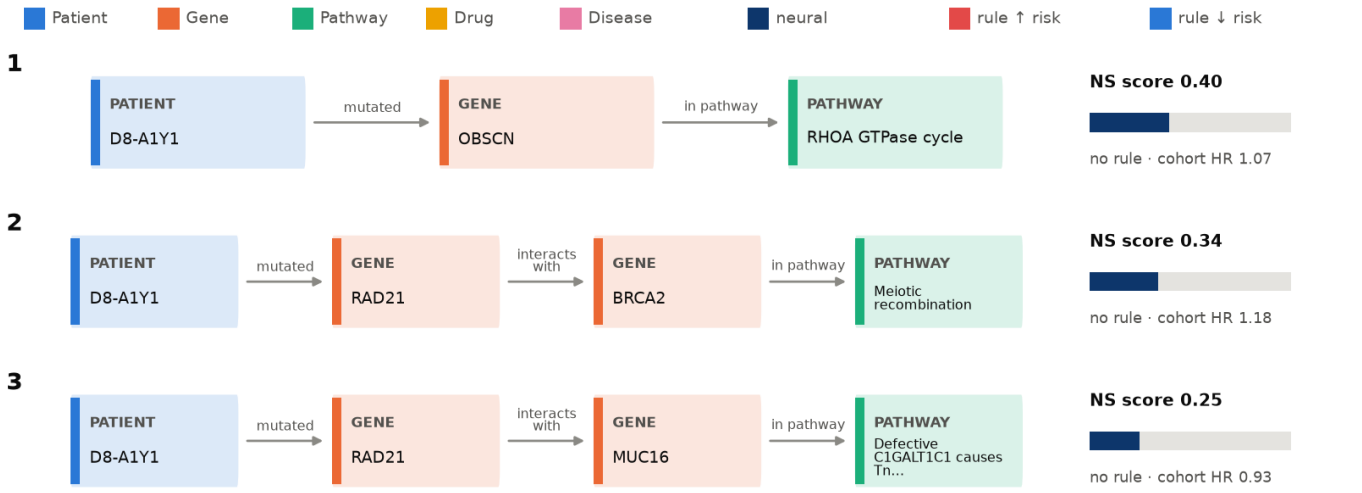
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 80-year-old patient has a LumB tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **78th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **OBSCN** to **RHOA GTPase cycle**. Progression was recorded at 9.9 months; the model's estimate given progression was 13 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

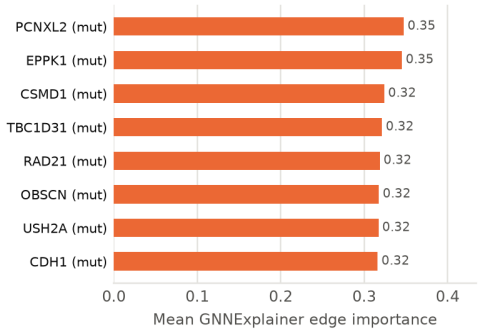
#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-D8-A1Y1** mutated → **OBSCN** in pathway → **RHOA GTPase cycle**	66	No learned formula fires on this path; its score is neural only. cohort: HR 1.07, n = 233 carriers	**0.40**
2	**TCGA-D8-A1Y1** mutated → **RAD21** interacts with → **BRCA2** in pathway → **Meiotic recombination**	57	No learned formula fires on this path; its score is neural only. cohort: HR 1.18, n = 107 carriers	**0.34**
3	**TCGA-D8-A1Y1** mutated → **RAD21** interacts with → **MUC16** in pathway → **Defective C1GALT1C1 causes Tn polyagglutination syndrome (TNPS)**	41	No learned formula fires on this path; its score is neural only. cohort: HR 0.93, n = 119 carriers	**0.25**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

No learned formula with a non-zero weight fires for this patient; the symbolic risk is the baseline of the logic model.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.X646_splice	17	0.32
MAP2K4 · driver	Mutation	p.D173Tfs*24	12	0.32
PIK3CA · driver	Mutation	p.E542K, p.F70=	70	0.31
PCNXL2	Mutation	p.T592M	0	0.35
EPPK1	Mutation	p.Q1841*	0	0.35
CSMD1	Mutation	p.G3126S	0	0.32
TBC1D31	Mutation	p.Q205R	0	0.32
RAD21	Mutation	p.A377G	7	0.32
OBSCN	Mutation	p.V4325M	8	0.32
USH2A	Mutation	p.D4968=	0	0.32
TAF2	Mutation	p.L828=	9	0.31
HMCN1	Mutation	p.P2507=	0	0.31



16 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); ATP1A4 → Rubidium Rb-82 (DB09479). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R5	61	III	LumA	0.68	progressed at 6.0 mo (group A)
TCGA-A2-A1G4	71	III	LumB	0.63	no progression, followed 19.6 mo
TCGA-AR-A1AL	60	III	LumA	0.61	no progression, followed 97.7 mo
TCGA-BH-A1FJ	66	III	LumB	0.57	no progression, followed 63.4 mo
TCGA-EW-A1IZ	53	III	LumA	0.56	no progression, followed 18.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E9-A243 · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

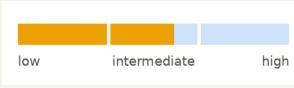
Sample: **TCGA-E9-A243-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **2.3** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

58th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 20%

Model-estimated progression time

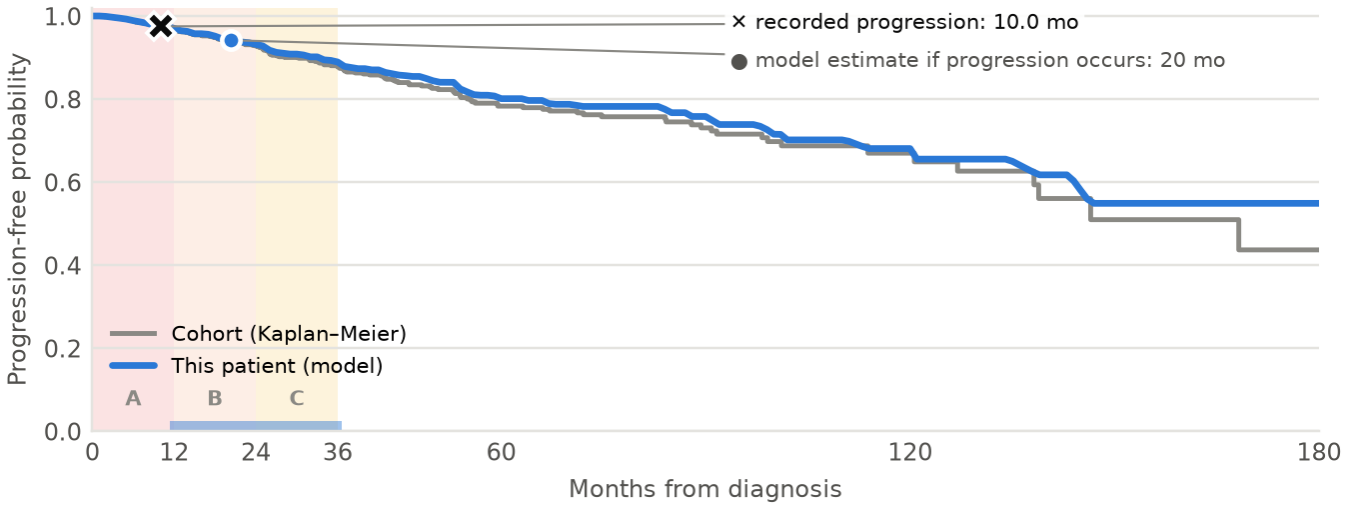
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

10.0 months

Group A · Early progression
✗ recorded window differs · |error| 10 mo



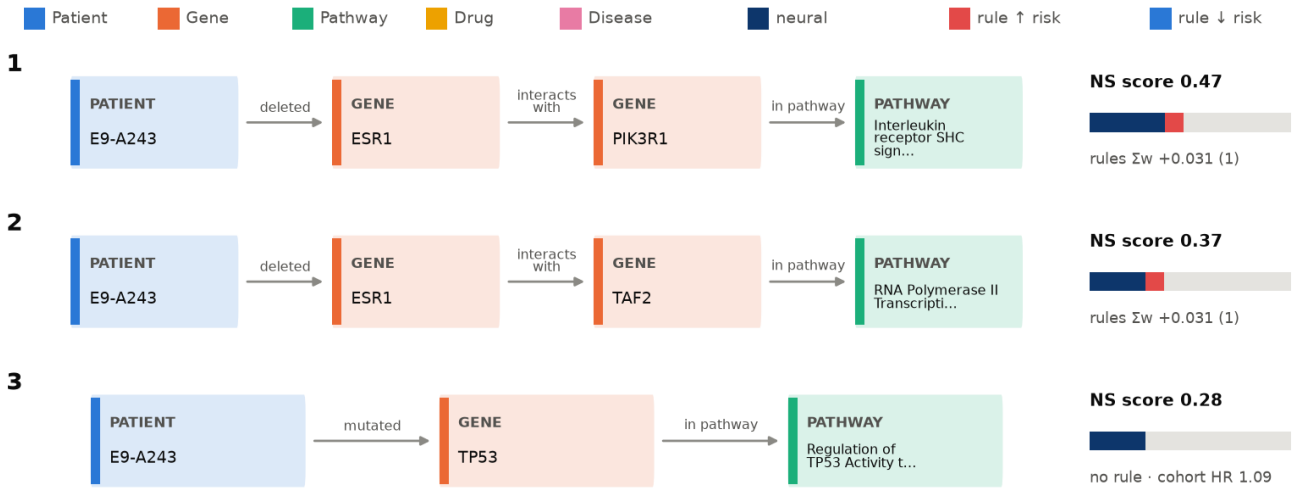
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **58th risk percentile** (intermediate tier). The strongest learned formula that holds for them (*NStage(p, N0)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.095 per SD). The highest-scoring knowledge-graph path runs from **ESR1** to **Interleukin receptor SHC signaling**; the learned formulas it instantiates point to faster progression (Σw +0.031). Progression was recorded at 10.0 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

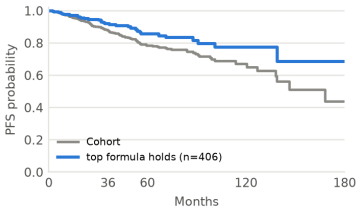


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-E9-A243** deleted → **ESR1** interacts with → **PIK3R1** in pathway → **Interleukin receptor SHC signaling**	62	↑ **faster progression** · Σw +0.031 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Nuclear signaling by ERBB4') ⇒ Progress(p) cohort: HR 0.97, n = 324 carriers	**0.47**
2	**TCGA-E9-A243** deleted → **ESR1** interacts with → **TAF2** in pathway → **RNA Polymerase II Transcription Initiation**	46	↑ **faster progression** · Σw +0.031 over 1 formula(s) Altered(p, g) ∧ InPathway(g, 'Nuclear signaling by ERBB4') ⇒ Progress(p) cohort: HR 1.06, n = 135 carriers	**0.37**
3	**TCGA-E9-A243** mutated → **TP53** in pathway → **Regulation of TP53 Activity through Phosphorylation**	46	No learned formula fires on this path; its score is neural only. cohort: HR 1.09, n = 418 carriers	**0.28**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

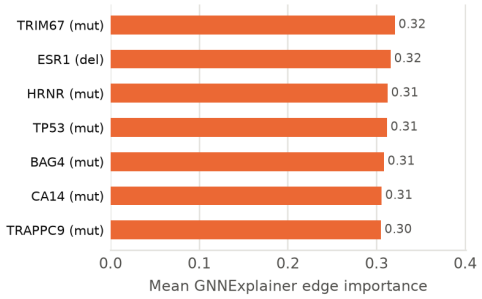
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.095 ↓ risk	NStage(p, N0) ⇒ Progress(p)	single atom	397 pts
+0.031 ↑ risk	Altered(p, g) ∧ InPathway(g, 'Nuclear signaling by ERBB4') ⇒ Progress(p) [KG]	single atom	29 pts
-0.006 ↓ risk	Stage(p, II) ⇒ Progress(p)	single atom	487 pts
-0.005 ↓ risk	HighAneuploidy(p) ⇒ Progress(p)	single atom	214 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	Homozygous deletion	GISTIC ±2	141	0.32
TP53 · driver	Mutation	p.W91*, p.Y220S	57	0.31
TRIM67	Mutation	p.D606N	0	0.32
HRNR	Mutation	p.Q396*	3	0.31
BAG4	Mutation	p.S303=	4	0.31
CA14	Mutation	p.V10E	6	0.31
TRAPPC9	Mutation	p.R553=	2	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 123 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XX	49	II	Basal	0.87	no progression, followed 47.3 mo
TCGA-E2-A1B6	44	II	Basal	0.87	no progression, followed 28.5 mo
TCGA-AR-A1AI	47	II	Basal	0.86	progressed at 19.3 mo (group B)
TCGA-A7-A26I	65	II	Basal	0.85	no progression, followed 21.7 mo
TCGA-C8-A12V	55	II	Basal	0.85	no progression, followed 12.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Male**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AC-A62V-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **2.27** · MSIsensor score: **0.07** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

42% by 36 mo

12 mo 10% · 24 mo 25% · 60 mo 67%

Model-estimated progression time

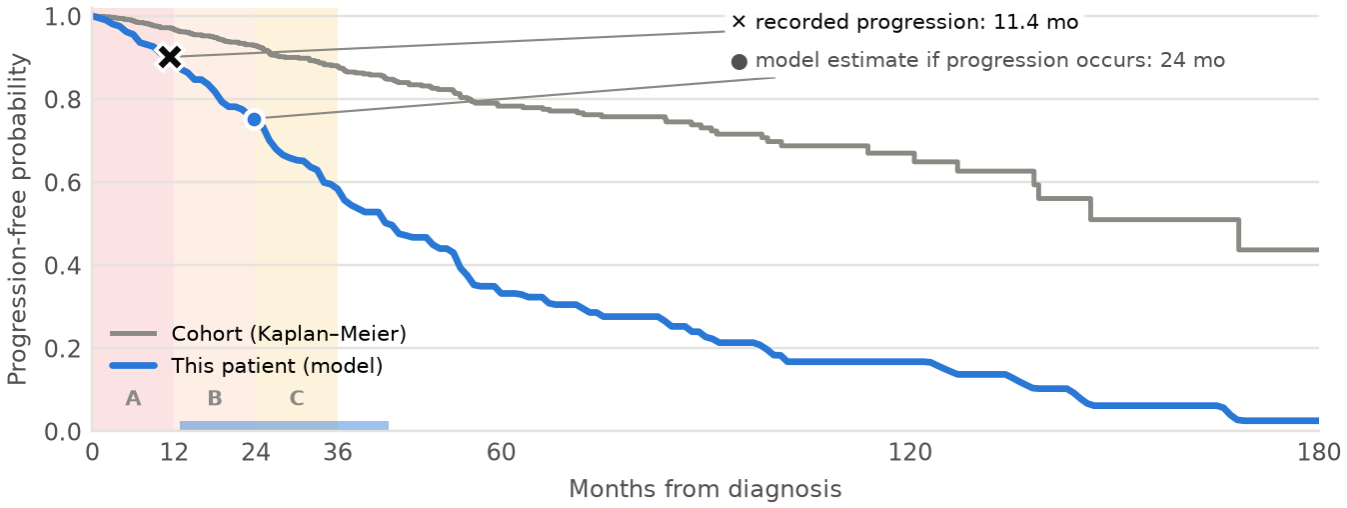
≈ 24 months

Window 12–24 months · IQR 13–43 mo
if progression occurs

Recorded progression

11.4 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 12 mo



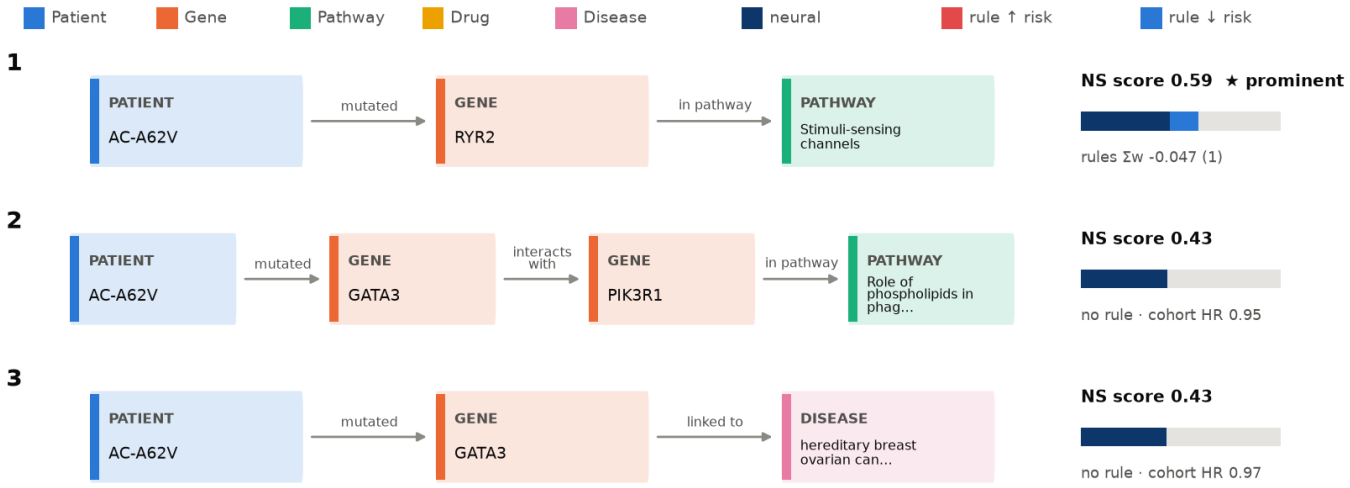
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a subtype-unassigned tumour, AJCC stage IV (T2, N1, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *TargetedBy(g, 'Flecainide')* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.047 per SD). The highest-scoring knowledge-graph path runs from **RYR2** to **Stimuli-sensing channels**; the learned formulas it instantiates point to slower progression (Σw -0.047). Progression was recorded at 11.4 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

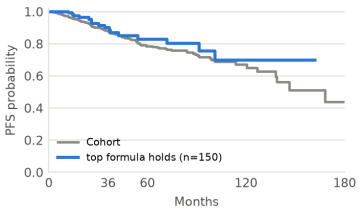


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-AC-A62V** mutated → **RYR2** in pathway → **Stimuli-sensing channels**	74	↓ **slower progression** · Σw -0.047 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) cohort: HR 0.78, n = 261 carriers	**0.59**
2	**TCGA-AC-A62V** mutated → **GATA3** interacts with → **PIK3R1** in pathway → **Role of phospholipids in phagocytosis**	72	No learned formula fires on this path; its score is neural only. cohort: HR 0.95, n = 312 carriers	**0.43**
3	**TCGA-AC-A62V** mutated → **GATA3** linked to → **hereditary breast ovarian cancer syndrome**	71	No learned formula fires on this path; its score is neural only. cohort: HR 0.97, n = 797 carriers	**0.43**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

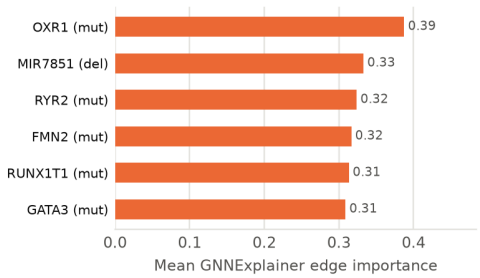
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.047 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	147 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.R329Kfs*21	13	0.31
OXR1	Mutation	p.R662H	0	0.39
MIR7851	Homozygous deletion	GISTIC ±2	0	0.33
RYR2	Mutation	p.S4006=	7	0.32
FMN2	Mutation	p.R1581I	0	0.32
RUNX1T1	Mutation	3'UTR	0	0.31



6 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085); RYR2 → Calcium citrate (DB11093); RYR2 → Calcium Phosphate (DB11348); RYR2 → Calcium phosphate dihydrate (DB14481). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment

recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FH	47	IV	LumA	0.82	progressed at 34.0 mo (group C)
TCGA-BH-A0HL	56	II	missing	0.69	no progression, followed 2.4 mo
TCGA-C8-A273	29	II	LumA	0.56	no progression, followed 16.9 mo
TCGA-AO-A03P	54	II	LumB	0.55	progressed at 63.2 mo (group D)
TCGA-AR-A1AV	68	II	missing	0.55	no progression, followed 61.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**71**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A1LE-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.02** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

95th

High risk tier · fused neuro-symbolic model



Probability of progression

23% by 36 mo

12 mo 7% · 24 mo 13% · 60 mo 37%

Model-estimated progression time

≈ 16 months

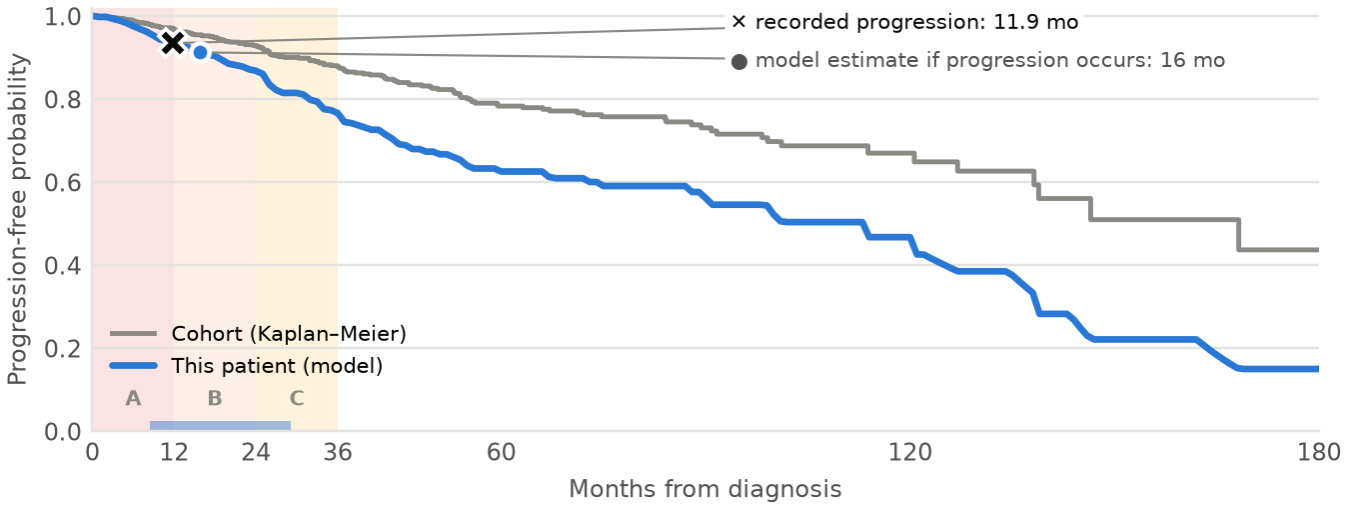
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

11.9 months

Group A · Early progression

✗ recorded window differs · |error| 4 mo



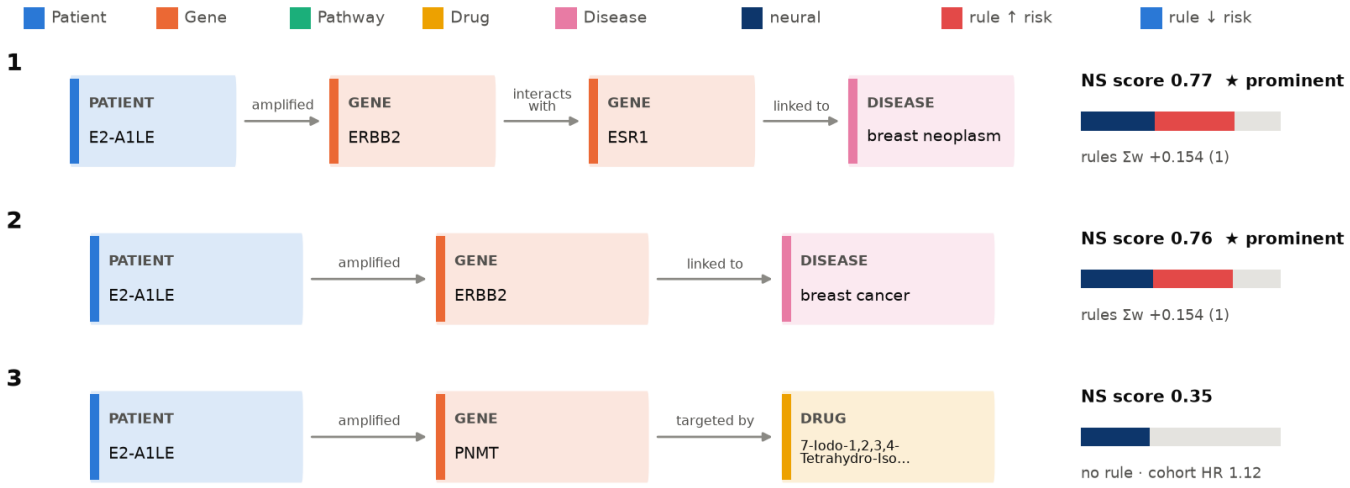
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 71-year-old patient has a subtype-unassigned tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **95th risk percentile** (high tier). The strongest learned formula that holds for them (*NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p)*) raises the progression hazard (weight +0.154 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **breast neoplasm**; the learned formulas it instantiates point to faster progression (Σw +0.154). Progression was recorded at 11.9 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

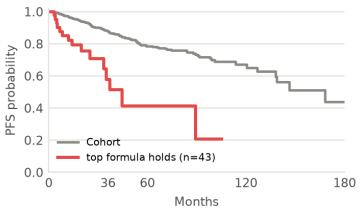


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-E2-A1LE** amplified → **ERBB2** interacts with → **ESR1** linked to → **breast neoplasm**	62	↑ **faster progression** · Σw +0.154 over 1 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.19, n = 797 carriers	**0.77**
2 ★	**TCGA-E2-A1LE** amplified → **ERBB2** linked to → **breast cancer**	60	↑ **faster progression** · Σw +0.154 over 1 formula(s) NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.19, n = 797 carriers	**0.76**
3	**TCGA-E2-A1LE** amplified → **PNMT** targeted by → **7-Iodo-1,2,3,4-Tetrahydro-Isoquinoline**	58	No learned formula fires on this path; its score is neural only. cohort: HR 1.12, n = 91 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

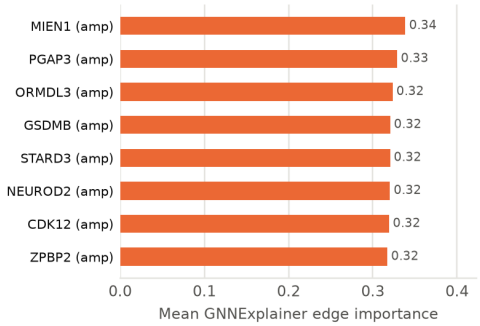
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.154 ↑ risk	NStage(p, N3) ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	mined pair	43 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
MIEN1	High-level amplification	GISTIC ±2	0	0.34
PGAP3	High-level amplification	GISTIC ±2	0	0.33
ORMDL3	High-level amplification	GISTIC ±2	2	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
NEUROD2	High-level amplification	GISTIC ±2	0	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
CACNA1E	High-level amplification	GISTIC ±2	12	0.32



19 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 29 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YG	63	III	LumB	0.66	no progression, followed 21.9 mo
TCGA-A8-A09G	79	III	Her2	0.65	no progression, followed 0.0 mo
TCGA-D8-A3Z5	54	III	missing	0.63	no progression, followed 33.4 mo
TCGA-AO-A0JL	59	III	missing	0.63	no progression, followed 55.3 mo
TCGA-C8-A138	54	III	Her2	0.63	no progression, followed 12.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0I9 · Group A

Report date 31 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

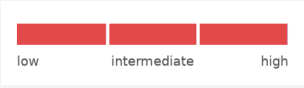
Sample: **TCGA-B6-A0I9-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **2.4** · MSIsensor score: **0.58** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

71% by 36 mo

12 mo 27% · 24 mo 51% · 60 mo 91%

Model-estimated progression time

≈ 14 months

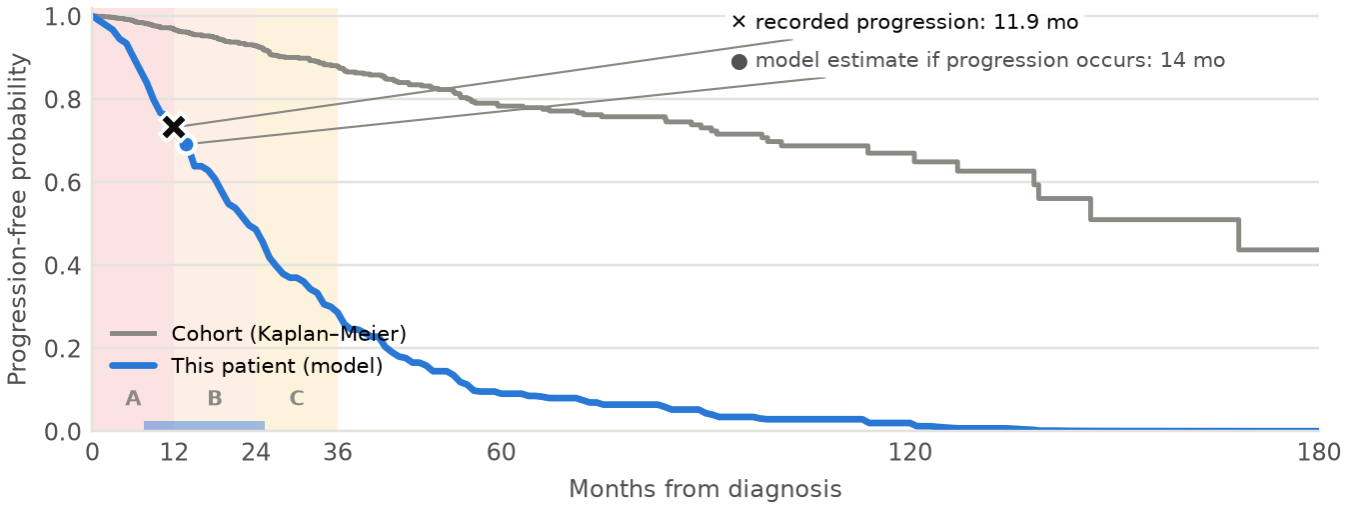
Window 12–24 months · IQR 8–25 mo if progression occurs

Recorded progression

11.9 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 2 mo



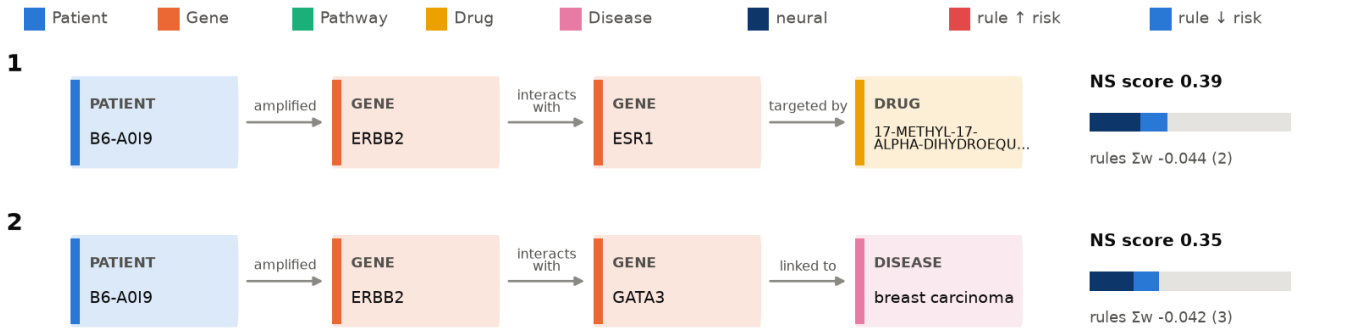
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a Her2 tumour, AJCC stage IV (T3, missing, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The strongest learned formula that holds for them (*Amplified(p, GOLPH3L)* ∧ *Altered(p, g)* ∧ *InPathway(g, 'Neurofascin interactions')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.073 per SD). The highest-scoring knowledge-graph path runs from **ERBB2** to **17-METHYL-17-ALPHA-DIHYDROEQUILENIN**; the learned formulas it instantiates point to slower progression (Σw -0.044). Progression was recorded at 11.9 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

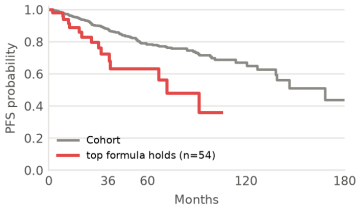


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1	**TCGA-B6-A0I9** amplified → **ERBB2** interacts with → **ESR1** targeted by → **17-METHYL-17-ALPHA-DIHYDROEQUILENIN**	42	↓ **slower progression** · Σw -0.044 over 2 formula(s) Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 2.29, n = 33 carriers	**0.39**
2	**TCGA-B6-A0I9** amplified → **ERBB2** interacts with → **GATA3** linked to → **breast carcinoma**	36	↓ **slower progression** · Σw -0.042 over 3 formula(s) Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) cohort: HR 1.19, n = 780 carriers	**0.35**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

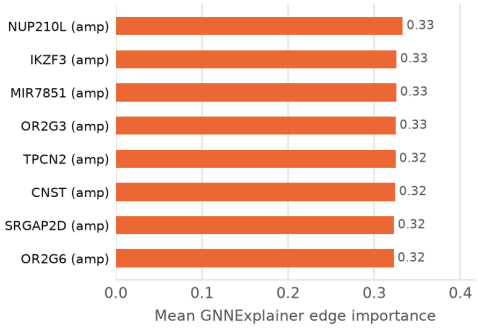
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.073 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	53 pts
-0.067 ↓ risk	Altered(p, g) ∧ InPathway(g, 'Vitamin B2 (riboflavin) metabolism') ∧ Altered(p, g) ∧ InPathway(g, 'Signal transduction by L1') ∧ Altered(p, g) ∧ LinkedTo(g, 'breast ductal adenocarcinoma') ⇒ Progress(p) [KG]	RL-proposed	39 pts
-0.067 ↓ risk	Altered(p, g) ∧ TargetedBy(g, 'Flecainide') ⇒ Progress(p) [KG]	single atom	143 pts
+0.053 ↑ risk	Amplified(p, FAM72C) ∧ Altered(p, g) ∧ InPathway(g, 'Neurofascin interactions') ⇒ Progress(p) [KG]	mined pair	82 pts
+0.027 ↑ risk	Altered(p, g) ∧ InPathway(g, 'PKMTs methylate histone lysines') ∧ Altered(p, g) ∧ InPathway(g, 'RUNX1 and FOXP3 control the development of regulatory T lymphocytes (Tregs)') ⇒ Progress(p) [KG]	RL-proposed	95 pts
+0.026 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'PI3K Cascade') ⇒ Progress(p) [KG]	mined pair	31 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P408Afs*99	13	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
TP53 · driver	Mutation	p.C135Y	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
EGFR · driver	Homozygous deletion	GISTIC ±2	73	0.31
NUP210L	High-level amplification	GISTIC ±2	0	0.33
IKZF3	High-level amplification	GISTIC ±2	0	0.33
MIR7851	High-level amplification	GISTIC ±2	0	0.33
OR2G3	High-level amplification	GISTIC ±2	1	0.33
TPCN2	High-level amplification	GISTIC ±2	1	0.32
CNST	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32



148 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 130 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A12L	67	II	Her2	0.77	no progression, followed 11.9 mo
TCGA-A8-A09C	69	X	LumB	0.76	no progression, followed 1.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.74	no progression, followed 140.9 mo
TCGA-AN-A03X	74	II	LumA	0.74	no progression, followed 0.3 mo
TCGA-C8-A1HL	38	III	LumB	0.72	no progression, followed 10.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EV · Group A

Report date 31 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1EV-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **2.83** · MSIsensor score: **0.41** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

29% by 36 mo

12 mo 7% · 24 mo 17% · 60 mo 50%

Model-estimated progression time

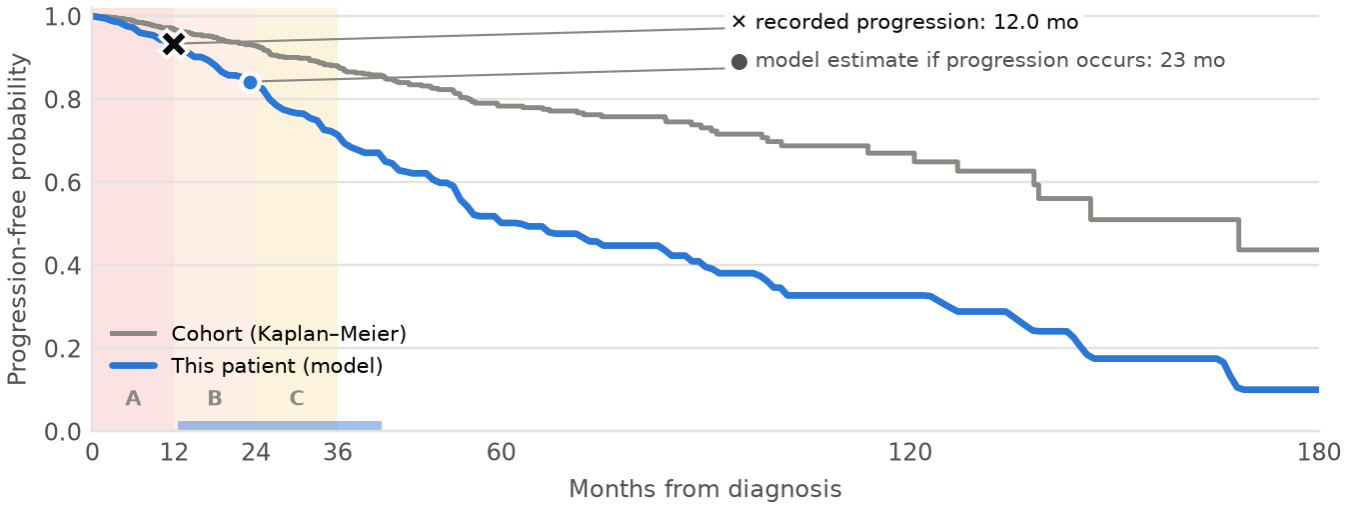
≈ 23 months

Window 12–24 months · IQR 13–42 mo
if progression occurs

Recorded progression

12.0 months

Group A · Early progression
✗ recorded window differs · |error| 11 mo



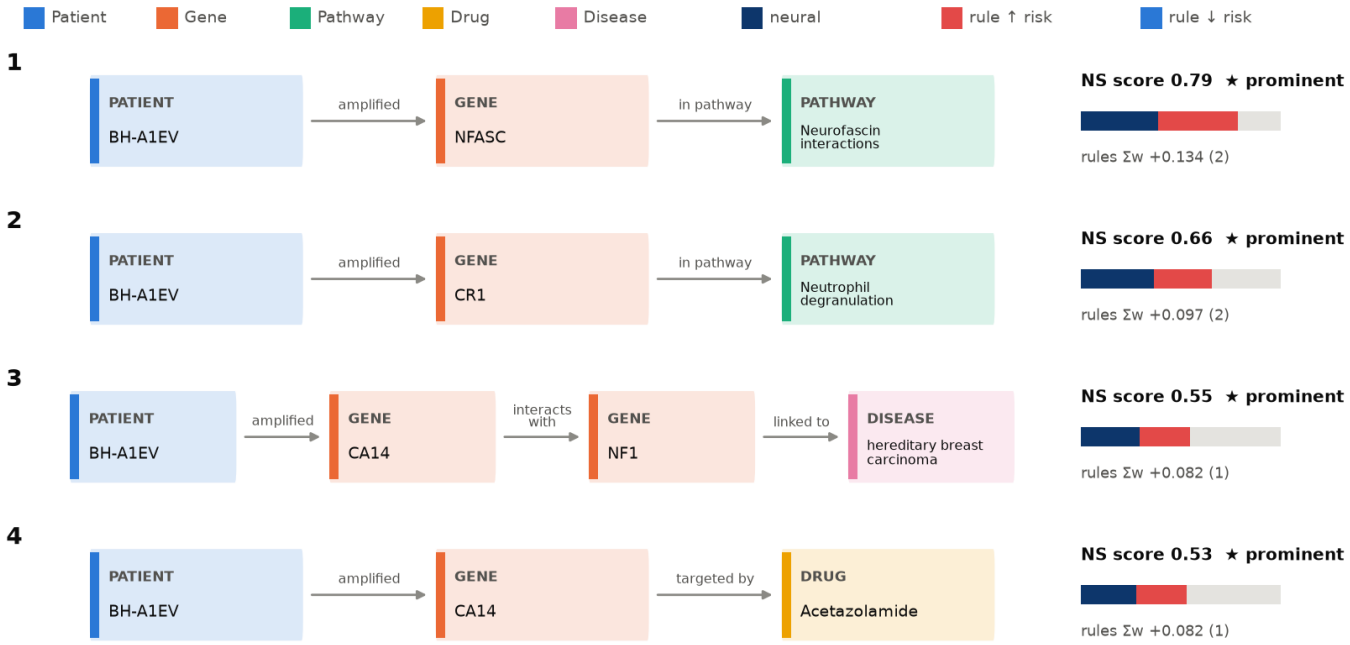
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a Her2 tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The strongest learned formula that holds for them (*Altered(p, g)* ∧ *TargetedBy(g, 'Ellagic acid')* ⇒ *Progress(p)*) raises the progression hazard (weight +0.082 per SD). The highest-scoring knowledge-graph path runs from **NFASC** to **Neurofascin interactions**; the learned formulas it instantiates point to faster progression (Σw +0.134). Progression was recorded at 12.0 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

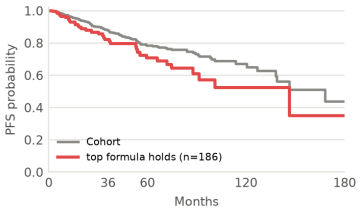


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1EV** amplified → **NFASC** in pathway → **Neurofascin interactions**	64	↑ **faster progression** · Σw +0.134 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.51, n = 173 carriers	**0.79**
2 ★	**TCGA-BH-A1EV** amplified → **CR1** in pathway → **Neutrophil degranulation**	61	↑ **faster progression** · Σw +0.097 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.36, n = 316 carriers	**0.66**
3 ★	**TCGA-BH-A1EV** amplified → **CA14** interacts with → **NF1** linked to → **hereditary breast carcinoma**	49	↑ **faster progression** · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 0.94, n = 792 carriers	**0.55**
4 ★	**TCGA-BH-A1EV** amplified → **CA14** targeted by → **Acetazolamide**	47	↑ **faster progression** · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 1.73, n = 91 carriers	**0.53**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

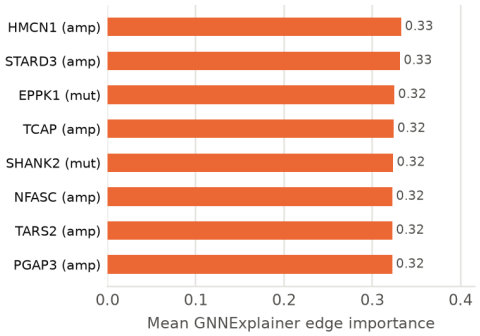
Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.079 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) [KG]	mined pair	40 pts
+0.055 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Interaction between L1 and Ankyrins') ⇒ Progress(p) [KG]	mined pair	27 pts
+0.018 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	mined pair	42 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA2 · driver	High-level amplification	GISTIC ±2	14	0.31
HMCN1	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.33
EPPK1	Mutation	p.L1474=	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
SHANK2	Mutation	p.P1240T	1	0.32
NFASC	High-level amplification	GISTIC ±2	3	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
PGAP3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32



46 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A1HL	38	III	LumB	0.70	no progression, followed 10.4 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.62	no progression, followed 140.9 mo
TCGA-B6-A0WX	40	III	Basal	0.62	progressed at 18.5 mo (group B)
TCGA-A8-A09B	58	III	LumA	0.62	no progression, followed 12.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.